

PropertyKindCalculus Blueprint

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Project

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PropertyKindCalculus is a Lean 4 formalization of René Dybæk's **An Ontology on Property for Physical, Chemical, and Biological Systems** (2009), extended with David Flater's full tracking of kinds of quantities (NIST Technical Note 1943, Appendix C). This blueprint is the design map: it records what is already proved (linked to real declarations) and the **capstone theorems we plan to provide**, with the dependency graph and a status summary at the end.

The whole blueprint is also available as a single paginated document: [download the PDF](#).

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1. What PropertyKindCalculus provides

This project addresses **thirteen requirements** about formalizing *metrology* — the science of measurement — and discharges most of them as machine-checked theorems rather than prose. A substantial part of the library is grounded directly on the published **ISO and IEC 80000** metrology standards — *eleven* of the thirteen parts (every part but the two general ones, 1 *General* and 2 *Mathematics*), catalogued item by item. The table below is generated directly from the formalized catalogues, so its per-part counts and total stay in step with the Lean source:

Part	Title		Quantity kinds
ISO 80000-3	Space and time		42
ISO 80000-4	Mechanics		54
ISO 80000-5	Thermodynamics		54
IEC 80000-6	Electromagnetism		85
ISO 80000-7	Light and radiation		66
ISO 80000-8	Acoustics		18
ISO 80000-9	Physical chemistry and molecular physics		62
ISO 80000-10	Atomic and nuclear physics		125
ISO 80000-11	Characteristic numbers		115
ISO 80000-12	Condensed matter physics		60
IEC 80000-13	Information science and technology		42
Total			723

In plain terms, *rigorous metrology* here means:

- **Telling apart quantities that look identical.** Two measurements in the same units can still be different in nature — volumetric and gravimetric soil moisture are both "just a ratio", yet confusing them corrupts a result. The calculus keeps such quantities distinct *types*, so the mix-up is impossible rather than merely discouraged; agreeing units (or dimensions) are treated as necessary but never sufficient.
- **Rejecting nonsense arithmetic before it runs.** Adding a length to a mass — or even a torque to an energy, which share the same dimension — is a compile-time error, not something caught (or missed) at run time. Quantities that genuinely combine, such as a torque times an angle giving an energy, are sanctioned explicitly; everything else is refused.

- **Knowing which operations even make sense.** What you may do to a measurement depends on its scale: you can rank mineral hardness but not average it; you can subtract Celsius temperatures but only take ratios of thermodynamic ones. These rules are enforced automatically.
- **A faithful hierarchy of kinds.** Width and height are special cases of length — usable where a length is wanted, never the reverse — while a general kind (length) is never confused with one individual measurement (the length of this pencil). A kind can even be tied to the system and component it is *dedicated* to, so "water content of soil" is named in its own right.
- **Units that convert safely and never silently.** A unit is a chosen reference of one specific kind; a conversion round-trips exactly, and there is simply no conversion between units of different kinds.
- **Summing only what may be summed.** The masses of the parts add up to the mass of the whole; the volumes of mixed liquids do not. The calculus tracks which quantities aggregate and refuses to assume it of the rest.
- **Classifications you can trust because they are earned.** Labelling a value "an area" requires a proof that it really is a width times a height; an arbitrary number cannot wear the label.
- **The same model from proof to running code.** One quantity can carry exact real numbers for proving properties and IEEE floating-point for execution — with a theorem guaranteeing the running code matches the proven specification, rounding and all — and a vector quantity is a list of numbers under a *single* shared unit, not a bag of separately-united components.
- **Honest unit classification.** Not even the SI base units are all alike — some (the mole, the candela) are derived in disguise — so the calculus names that third category instead of pretending the base/derived split is clean.

Each of these is stated precisely as one of the thirteen requirements below, and the status table at the end of that section maps every requirement to the checked declaration that discharges it.

2. Requirements

What follows is the fixed set of requirements PropertyKindCalculus is **specified** to meet. The next section, **Why a calculus, not a taxonomy**, argues **why** a calculus is needed where a taxonomy stops, and the chapters after it develop and prove each requirement; this section states them up front, as the axes the design is judged against, with a status table at the end mapping each axis to the declaration that discharges it.

The requirements fall into four groups: how kinds are **structured**, how **operations** on them are gated, how the kind layer is kept **consistent** with the coarser dimension and unit layers, and how values **aggregate** over parts.

2.1. Kind structure (R1, R2, R3)

R1 — Kinds are first-class and discriminate within a dimension. Two properties of the same physical dimension can still be different kinds, and the type system must keep them apart. Volumetric water content (Quantity vwc), gravimetric water content (Quantity gwc), relative permittivity, reflectivity, and emissivity are all dimension-one, yet Quantity vwc and Quantity gwc must be **different types** — not interchangeable values. (QUDT's and SysML v2's QuantityKind is subsumed here as one such kind — a scale-gated leaf — not as the root.) This reaches its limit on ISO 80000-11 *Characteristic numbers*, where **all 115** kinds — every one a dimensionless ratio — share dimension one, so the dimension functor collapses the entire part to a single point and only the kind layer holds its members apart (see the *ISO 80000-11* chapter). The **principled** form of this within-dimension discrimination — naming a kind by the system and component it is *dedicated to*, so volumetric and gravimetric water content differ by their kind-of-property rather than by an identity string — is Dybkær's dedicated kind-of-property (see the *Dedicated kinds-of-property* chapter).

R2 — Specialization is a lattice, with comparability but not identity. Width, Height, and Diameter each specialize Length, and a kind may specialize several parents at once — a lattice, not a tree. Specialization is a preorder (reflexive and transitive) and induces a **one-way** coercion: a Quantity Width may be used where a Quantity Length is wanted, never the reverse, and that up-cast is a visible, deliberate loss of information. Width and Height remain **mutually comparable** — they share the super-kind Length — while staying distinct kinds. This requirement is now realized on the real standard: the ISO 80000-3 length family (width, height, distance, radius, ... of items 3-1.2 ... 3-1.12, all of dimension L) is specified as a specialization lattice over the general length kind, each species individuated **not by fiat but by an explicit measurement principle** (see the *ISO 80000-3* chapter), and the same pattern recurs on Part 4's force family, Part 5's thermodynamic potentials, and IEC 80000-6's AC power family — active, reactive, and apparent power as species of one power kind, carrying three different unit strings (w, var, VA) over one dimension (see the *IEC 80000-6* chapter) — and on ISO 80000-7's radiation trios, where radiant, luminous, and photon flux are one measurand in three modes, the radiant and luminous members even sharing a dimension (see the *ISO 80000-7* chapter). ISO 80000-11 pushes this furthest:

the same name recurs as a *different kind* across its transport-phenomena clauses — the two Froude numbers, the five Stokes numbers, the four Bejan numbers — each sub-suffixed sibling sharing **both** name and dimension (one) with the others, so the measurement principle alone tells them apart (see the *ISO 80000-11* chapter).

R3 — General versus individual is type versus term. A kind is the general notion (a **type**); a particular measured value is an individual (a **term** of that type). Length is a type; the length of this pencil is a term of type Quantity Length. The two are never conflated — the punning OWL permits between a class and its instances is structurally impossible here.

2.2. Operation gating (R4, R5, R6)

R4 — Operations are gated by kind (the additive law). Arithmetic that keeps a quantity within its kind is well-typed; arithmetic across incompatible kinds is a **compile-time type error**, not a runtime check. $\text{Width} + \text{Width} = \text{Width}$ type-checks; $\text{Width} + \text{Height}$ does not — and, decisively, neither does $\text{Torque} + \text{Energy}$, even though torque and energy share a dimension. Matching dimension is not licence to add.

R5 — The interaction algebra is a partial, typed, ternary product. Some kinds combine **across** kinds to yield a third: torque times plane angle is energy, force times length is work. The combination is partial and curated — most pairs combine to **nothing** (fuel-consumption times rainfall is a category error, not a number) — and it carries a division dual, with multiplication and division proved inverse. This multiplicative combination is distinct from R4's additive, same-kind gate.

R6 — Operator availability is gated by the measurement scale, monotonically. Which operations even **exist** between two properties is fixed by their scale type, $\mathrm{nominal} \sqsubset \mathrm{ordinal} \sqsubset \mathrm{interval} \sqsubset \mathrm{ratio}$: a nominal property admits only $=$, a ratio property admits $\times$ and $\div$, and a richer scale licenses every operation a poorer one does. This gate is **orthogonal** to kind — kind says **which** properties may combine, scale says **which operators** are defined at all. This requirement is now realized on the real standard: ISO 80000-5 lists thermodynamic temperature (ratio-scale) and Celsius temperature (interval-scale) at the **same** dimension θ , so a ratio of Celsius temperatures is undefined where a ratio of thermodynamic temperatures is not — the scale type, not the dimension, separates them (see the *ISO 80000-5* chapter). IEC 80000-6 repeats the pattern on electromagnetism: electric potential is gauge-dependent (fixed only up to an additive reference), hence interval-scale, while electric potential difference, of the same dimension v , is ratio-scale (see the *IEC 80000-6* chapter).

2.3. Soundness bridges (R7, R8)

R7 — Dimension certifies coherence; it does not decide legality. Every kind has a dimension, but dimension is a **consistency check**, not the authorization: matching dimensions are necessary yet never sufficient — which is exactly what lets R4 reject $\text{Torque} + \text{Energy}$. Formally the dimension map $\dim :$

$\mathrm{Kind} \rightarrow \mathrm{Dimension}$ is a **forgetful functor** and a **homomorphism**: $\mathrm{KMul} \rightarrow \mathrm{K_1} \times \mathrm{K_2} \times \mathrm{K_3} \rightarrow \mathrm{Dim} \quad \dim k_3 = \dim k_1 \cdot \dim k_2$.

In plain engineering terms: dim is a deliberately **lossy, one-way** translation from the rich kind layer down to the coarse SI-dimension layer. It keeps only the base-quantity exponents (length, mass, time, ...) and **forgets** everything else, so many distinct kinds collapse onto the same dimension and there is no way back — that is what "forgetful" names. Calling it a **homomorphism** says the translation **respects multiplication**: the dimension of a product equals the product of the dimensions (the exponents add), so the kind layer and the dimension layer can never disagree about what multiplying means. It is the discipline of ordinary dimensional analysis, stated and proved as a law rather than performed by hand.

R8 — Units are chosen values of a kind; conversion is a faithful round-trip. A unit is a distinguished value of a kind (a metre is a chosen length), and a measured quantity is a number times a unit. Converting a quantity from one unit to another of the **same** kind and back is the identity — conversion is multiplication by a ratio, defined only within a kind. There is no conversion between units of **different** kinds; that too is a type error, not a runtime check.

2.4. Aggregation (R9)

R9 — Extensive quantities aggregate additively over parts; intensive ones do not. For an **extensive** kind the value over a whole is the sum of the values over its disjoint parts — the mass of an assembly is the sum of its parts' masses. Where this fails it must **not** be assumed: volume on mixing is sub-additive (ethanol and water), and that negation is stated to keep the extensive predicate honest. Tracking which kinds are extensive is the precondition for soundly summing measurements.

2.5. Representation parametricity (R10)

R10 — A quantity value is parametric in its numeric representation type. The magnitude of a scalar quantity is carried at a **representation type** R , a type parameter bounded by whatever algebraic structure a given task demands, so the **same** kind-indexed value can be instantiated at whichever R the task needs — without changing the kind, scale, dimension, interaction, or extensivity layers above it. This is the axis dependent types make not just possible but **necessary**: the carrier the kind sits over is exactly where proof and execution diverge, and a single quantity must serve both. Three **real** representations matter, and they are not interchangeable — and a fourth **complexifies** any of them:

- **Lean's $\mathbb{R}$ — for proofs.** A complete ordered field: the continuous specification against which laws (monotonicity, conversion round-trips, extensivity) are stated and proved.
- **A finite floating-point rounding model — for numerical-accuracy reasoning.** TorchLean's FP32 (NeuralFloat at binary32 precision: round-to-nearest, ties-to-even over $\mathbb{R}$), finite-only, with

`toReal : FP32 → ℝ`). It is **non-computable** — it exists to bound rounding error against the $\mathbb{R}$ spec, not to run.

- **An executable IEEE-754 type — for exact-execution reasoning and code generation.**

TorchLean's IEEE32Exec (a bit-level UInt32 binary32 kernel) that models NaN, infinities, signed zeros, and subnormals, with `toFloat/ofFloat` and `isNaN`. This is the representation that actually runs and the one in which exceptional values (NaN) can be reasoned about — what one might call the "exec" float, distinct from the FP32 rounding **spec**.

- **The complexification $\mathrm{Complex} \setminus \mathbb{R}$ of any of the above — for complex-valued quantities.**

Many physical quantities take a complex value (the relative permittivity of a lossy dielectric $\varepsilon = \varepsilon' + j\varepsilon''$, an impedance, an AC phasor); complex-ness is a property of the **carrier**, not the kind, so $\mathrm{Quantity} \setminus k \setminus (\mathrm{Complex} \setminus \mathbb{R})$ is the **same** kind layer over a complexified carrier. As a functor $\mathbb{R} \mapsto \mathrm{Complex} \setminus \mathbb{R}$ it lifts each of the three above — $\mathrm{Complex} \setminus \mathbb{R}$ to prove, complex binary32 to certify rounding, $\mathrm{Complex} \setminus \mathrm{Float}$ to run. It is a `Carrier` — and a `LawfulCarrier` whenever $\mathbb{R}$ is, since complex addition is componentwise so the additivity laws lift for free — carrying the full field $+, -, \times, \div$; but it is **unordered** (there is no $\leq$ on $\mathbb{C}$), so unlike the real carriers it sits outside the ordinal/interval/ratio carrier tower: complex magnitudes are compared by modulus, not ranked.

Soundness **across** representations is itself a theorem, not an assumption: an exec/spec **refinement** (the executable float result is the rounding of the real result) lets a law proved over $\mathbb{R}$ transfer to the executable run. In plain engineering terms: choose the number type to fit the job — reals to prove, the rounding model to bound error, the IEEE kernel to run and to catch NaN — while the kind, dimension, and unit machinery above is written once and is identical for all three. This refines R1/R3/R4's kind-indexed value into a doubly-indexed `Quantity (k : KindOfProperty) (R : Type)` whose arithmetic is bounded by a `Carrier` typeclass on $\mathbb{R}$. The value layer and its additivity laws are **realized** (sorry-free): `Quantity k R` with same-kind addition, the additivity laws proved once over any lawful carrier, and three carriers — `Int` and $\mathbb{R}$ (lawful, for proof) and `Float` (executable, deliberately not lawful). The exec/spec **refinement** across representations is now **also realized** (sorry-free): the abstract bridge `CarrierRefinement` and its kind-indexed capstone `Quantity.add_refines` — a law over the lawful spec carrier descends to the exec carrier as one rounding step — instantiated in the separately-built Torch library at TorchLean's binary32: FP32 (the rounding **spec**, an **unconditional** refinement of $\mathbb{R}$) and IEEE32Exec (the **executable** kernel, refining $\mathbb{R}$ on the finite/no-overflow path, with overflow surfaced as an explicit side condition rather than silently dropped).

2.6. Value representation: vectors and scalar units (R11)

R11 — Units are scalar; a vector quantity is a numerical array times one scalar unit. Following ISO 80000-2 §18 (scalars, vectors and tensors), a vector (or tensor) quantity is written as a **numerical** vector multiplied by a single unit, and "all units are scalars" — **not** as a collection of per-coordinate quantity values each carrying its own number $\times$ unit; the quantity itself is independent of the coordinate system

while its numerical components are not. This is exactly the R10 carrier taken at a numerical-array type: a vector quantity is `Quantity (k : KindOfProperty) (Fin n → R)` — one kind *k*, the numbers an *n*-vector in the carrier *R*, and one **scalar** `MetrologicalUnit` of the kind for the whole vector — and the additivity laws transfer to the vector carrier by the **same** parametric proof used for scalars (the pointwise `Carrier (Fin n → R)`). It is the ISO 80000-2 §18 **numerical-array** reading of value representation, and it aligns with the VIM's "a quantity is scalar; a vector/tensor is a composite of scalar quantities" — the priority SysML v2 inverts (see **Why a calculus, not a taxonomy**). The fuller **structural** apparatus is a separate axis, still owed below.

2.7. Verified classification: kind-laws that instantiate at the quantity level (R12)

R12 — A classification is a certificate, not an assertion; kind-laws instantiate at the quantity level.

By default a kind is a **tag**: `Quantity k R` records the kind but the magnitude is arbitrary, so a value can be labelled with a kind it has not earned. R12 upgrades a classification to a **certificate** — a proof that a quantity satisfies its kind's **defining relation** (the formalized ISO 80000 **Remark**) to quantities of the kinds it is built from. A length built as speed × time is certified by construction; a bare `<999>` cannot be certified against those factors. The kind becomes a **refinement** (a tag plus its earned proof).

The reason this strengthens the calculus is **instantiation**. Every kind-law is stated as a carrier-parametric universal over quantities with the kind-relation as a premise, so it instantiates to a quantity-level fact by ordinary application — the inter-kind generalization of R10's carrier-parametric laws — and any property of a kind's defining relation transports to **every** quantity certified under it (every certified surface area is ≥ 0 , because the area element is). The pattern comes in three algebraic shapes — **product** (area = length · length, energy = force · length), **quotient** (speed = length / duration, plane angle = arc / radius), and **reciprocal** (frequency = 1 / period, curvature = 1 / radius) — each a kind-law family in the core; analysis-shaped relations such as area as a surface integral $\iint \sqrt{g} \, du \, dv$ follow the same discipline in the layer where their mathematics lives. These are exercised directly on ISO 80000-3's own **Remarks** (see the *ISO 80000-3* chapter), where, for instance, the plane angle is **computed** to be dimension one because it is a ratio of two lengths. R12 unifies the two halves of the standards work: the formalized remark **is** the defining relation, and classification is the witnessed instantiation of it.

2.8. Unit classification (R13)

R13 — Unit classification needs a third category beyond base and derived: scale-spanning units.

The SI sorts units into **base** (kilogram, metre, second, ampere, kelvin, candela, mole) and **derived** (everything built from them). Finkelstein and Whitehead (*Eur. J. Phys.* 46 (2025) 035701) argue this dichotomy is insufficient: of the seven base units only four — the kilogram, metre, second, and ampere — are genuinely **dimensionally independent**, while the kelvin (the dimension of energy, via the Boltzmann constant), the candela (the dimension of power, via the luminous efficacy), and the mole (a

dimensionless number, via the Avogadro constant) are not — yet each is kept a distinct coherent unit because it carries a **human-selected coefficient** sized to span vast scales. They are **scale-spanning units**, a third category. R13 specifies that category, and its point is the unit-layer twin of R1: **whether a unit is scale-spanning is not a function of its dimension alone**. The dimension layer can detect that the candela and the mole are **mechanically reducible** (to power and to one), but it cannot detect the kelvin — this work keeps thermodynamic temperature an independent dimension (ISO 80000-5 needs it), so the kelvin's scale-spanning character is visible only through its coefficient, never through dimensional analysis. The ampere, by contrast, **is** dimensionally independent (it carries the electromagnetic generator), a genuine base unit — so the third category is non-empty and distinct from the first. In plain engineering terms: just as a **kind** carries more than its dimension, a **unit's category** carries more than its dimension; the candela and mole (ISO 80000-7) and the kelvin (ISO 80000-5) are where this lands on the standard, the ampere (IEC 80000-6) the base-unit foil.

2.9. Out of scope (for now)

User-specified base units — making the dimension system parametric in the chosen base set (so a quantity's dimensional decomposition is computed relative to a user's choice of generators, expressing e.g. Gaussian-CGS electromagnetism, natural units, or the four-base Finkelstein system) — is **not specified** here. The controversy over which units are base is real, but in this calculus it is largely answered one layer up: a kind is invariant under the choice of base, and only its dimensional shadow changes. R13 captures the specific base/scale-spanning/derived controversy cheaply over the fixed basis; a parametric change-of-basis layer, if added, would be a separate, additive layer over the shared Dimension, never an edit to it.

Value representation in the structural sense — coordinate frames, tensor variance (the covariant/contravariant split), bound-versus-free vectors, and frame transformations — is **not yet specified** here, beyond the ISO 80000-2 §18 numerical-array reading now specified as R11. This **structural** axis is distinct from R10's **numeric carrier** R and R11's **numerical array**. It is a genuine systems-engineering requirement, but a **separate, orthogonal** axis: its principled home is an index **over** the kind, never a layer the kind hangs beneath (the inversion the **Why a calculus, not a taxonomy** section charges against SysML v2). Stating it as owed keeps the boundary honest.

2.10. The requirements at a glance

Requirement	Concrete test	Specified as	Status
R1 — kind discrimination within a dimension	$vwc \neq gwc$, both dimension one; all 115 ISO 80000-11 characteristic numbers dimension one, distinct kinds	kind-indexed Quantity k R; dimension-1 disambiguation	proved
R2 — specialization lattice + comparability	Width, Height, Diameter specialize Length (ISO 80000-3); weight, static vs kinetic friction force specialize Force (ISO 80000-4); Helmholtz vs Gibbs energy specialize Energy (ISO 80000-5); active vs reactive power specialize Power (IEC 80000-6); radiant vs luminous vs photon flux as radiation-mode trios (ISO 80000-7); the two Froude, five Stokes, four Bejan numbers as sub-suffixed homonyms (ISO 80000-11) — comparable yet distinct, by examination principle	Specializes preorder; MutuallyComparable; examination defining-aspect (Refines, distinct_of_examPrinciple)	proved
R3 — general vs individual (type vs term)	Length a type; this pencil's length a term	KindOfProperty vs Quantity k R	proved
R4 — kind-gated addition	Width+Width ok; Width+Height, Torque+Energy rejected	Quantity.add over Quantity k R (same-kind gate)	proved
R5 — interaction		KMul / KDiv; multiplication–division inverse	proved

Requirement	Concrete test	Specified as	Status
algebra (partial, typed)	torque × angle = energy; fuel × rainfall an error		
R6 — scale- type operator availability (monotone)	nominal: only =; ratio: ×, ÷; thermodynamic temperature (ratio) vs Celsius (interval), same dimension θ (ISO 80000-5); electric potential (interval, gauge-dependent) vs potential difference (ratio), same dimension v (IEC 80000-6)	ScaleType order; operator monotonicity; AllowsRatio on the standard's temperatures	proved
R7 — dimension certifies, not decides	dim many-to-one; necessary ≠ sufficient	dim homomorphism	proved
R8 — units and conversion round-trip	round-trip metre/foot conversion is the identity; cross-kind is a type error	Unit k ; conversion round-trip	planned
R9 — extensive aggregation	mass sums over parts; volume-on-mixing does not	Extensive; additive law + counterexample	proved
R10 — numeric representation parametricity	same value at $\mathbb{R}/\text{Int}$ (proof), Float (executable); FP32/ IEEE32Exec refine $\mathbb{R}$	Carrier-bounded Quantity k R ; parametric additivity laws; CarrierRefinement + Quantity.add_refines	proved (layer + laws + exec/spec refinement)
R11 — scalar units; vector = numerical array × one scalar unit (ISO 80000-2 §18)	a displacement as Quantity k ($\text{Fin } 3 \rightarrow$ $\mathbb{R}$) with one metre; laws transfer	pointwise Carrier ($\text{Fin } n \rightarrow R$); scalar MetrologicalUnit	proved

Requirement	Concrete test	Specified as	Status
R12 — verified classification; kind-laws instantiate at the quantity level	a length built as speed $\times$ time is certified by construction; (999) cannot be certified; every certified surface area is ≥ 0	ProductKind kind-laws; Quantity.IsProduct certificate + smart constructor; defining relations from the ISO 80000 remarks (area = $\ f\ _g$)	proved (product family + area instance)
R13 — scale-spanning units (a third unit category)	candela $\equiv$ power and mole $\equiv 1$ are reducible, the kelvin's reduction is invisible (Θ kept independent), the ampere is a genuine base — so scale-spanning is not a function of dimension (ISO 80000-7 candela/mole, ISO 80000-5 kelvin, IEC 80000-6 ampere; Finkelstein–Whitehead 2025)	UnitCategory; Dimension.MechanicallyReducible; ScaleSpanningUnit; scaleSpanning_not_determined_by_dimension	proved
(out of scope) structural value representation	coordinate frames, tensor variance, transforms	an orthogonal index over the kind	not specified

3. Why a calculus, not a taxonomy

PropertyKindCalculus continues a line of machine-checkable metrology modeling, and inherits its problem statement from exactly where that line stops.

3.1. From QUDV to an OML metrology vocabulary

OMG SysML 1.2 introduced [QUDV](https://www.omgwiki.org/OMGSysML/doku.php?id=sysml-qudv:quantities_units_dimensions_values_qudv) (https://www.omgwiki.org/OMGSysML/doku.php?id=sysml-qudv:quantities_units_dimensions_values_qudv) (Quantities, Units, Dimensions, Values) as a library for declaring quantity kinds, units, and their dimensional factoring; QUDV's foundations were then revised substantially across SysML 1.3 and 1.4.

In parallel, a formalization in OML — leveraging the semantics of OWL2-DL for classification-based reasoning — reworked those foundations, the quantity-kind and unit *stereotypes* carried by SysML, to align them with Dybkær's *An Ontology on Property*. That effort became the [opencaesar metrology vocabulary](https://github.com/opencaesar/metrology-vocabularies) (<https://github.com/opencaesar/metrology-vocabularies>), whose current version combines the updates and motivations of the VIM4 2nd Committee Draft (2CD, 2023-07-31) with Dybkær's ontological grounding. OWL2 (SROIQ) carries the metrology *taxonomy* well: specialization, the general-vs-individual quantity distinction (encoded as concept vs. instance), dimensional factoring, and OWL2-DL/SPARQL consistency checks over units.

Where a description logic stops is visible in that vocabulary's own *dimensional analysis*: computing a derived quantity's dimension from its factors is an arithmetic induction that has to be pushed out of OWL2 into SPARQL 1.1 Update, and even there it is brittle — updates report success while inserting no triples, because "update succeeded" is not a statement about *what was established*. Dimension arithmetic sits right at that boundary; the kind interaction algebra and the algebraic *laws* are past it.

3.2. What a description logic cannot express

A description logic captures the *taxonomy* but is structurally unable to capture the *calculus*. The recurring gap is arithmetic, operations, definedness-conditions, laws, and proofs:

- **Dimension arithmetic** ($\dim(a \cdot b) = \dim a + \dim b$): OWL2 has no datatype arithmetic; in Lean it is an ordinary function with a homomorphism theorem.
- **The interaction algebra** (torque $\times$ angle = energy): role chains are binary, regular, and carry no arithmetic side-conditions; in Lean it is a ternary relation with a dimensional-coherence proof obligation.
- **Scale-type operator gating** (which of $=, <, +, \times$ are **defined**): DL has no operations; in Lean the scale indexes which typeclasses apply, and monotonicity of availability is a proved meta-theorem.

- **Extensivity** ($\mathrm{value}(\text{whole}) = \sum \mathrm{value}(\text{parts})$): no quantified arithmetic in DL; in Lean it is a `forall`-theorem.
- **Units** (a unit is a chosen value of a kind; conversion is a ratio): no arithmetic in DL; in Lean it is a definition with a round-trip theorem.
- **Meta-theorems** (specialization is a preorder; `dim` is a homomorphism): a DL reasoner checks **consistency**, not **laws** — proving the laws is the point.

3.3. SysML v2: a quantity model organized by value representation

OMG's current generation ships a standard-library treatment of quantities and units, examined here at tag [SysML v2, 2026-04](https://github.com/Systems-Modeling/SysML-v2-Release/tree/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units) (its `Quantities`, `MeasurementReferences`, and `ISQ` packages). Read against Dybkær, it is built around a different organizing axis than the one taken here as primary, and the contrast is the cleanest way to say what a **kind-primary** model buys.

3.3.1. The evidence: representation is the root, kind is a leaf

The quantity-value hierarchy is rooted in **value representation**, not in quantity kind. The most general value is `TensorQuantityValue`, a tensor of any order, and the specialization chain runs **toward** the scalar, each more specific type being a lower-order tensor — `VectorQuantityValue` $\rightarrow$ `TensorQuantityValue`, then `ScalarQuantityValue` $\rightarrow$ `VectorQuantityValue` ([SysML v2, 2026-04 — Quantities.sysml, lines 17–46](https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/Quantities.sysml#L17-L46)) (`\text{Scalar} \sqsubseteq \text{Vector} \sqsubseteq \text{Tensor}`). So a scalar **is a** tensor of order 0 and a vector **is a** tensor of order 1. What a value's type primarily records is the apparatus of that representation:

- its tensor order,
- the split of that order into **contravariant** and **covariant** indices — the attributes `contravariantOrder` and `covariantOrder` on `TensorQuantityValue`, constrained to sum to the order ([SysML v2, 2026-04 — Quantities.sysml, lines 33–36](https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/Quantities.sysml#L33-L36))
- whether a vector is bound or free, and
- a matching measurement reference (tensor / vector / scalar) carrying the coordinate frame and its transformations.

Each of these is a statement about **how a quantity is measured, coordinatized, and valued**, not about **what kind of property** it is. The contravariant/covariant split is the sharpest case of this representation-centric view — the one record-field that is **unambiguously** representational, on three escalating counts.

1. **Pure coordinate bookkeeping.** It records only how a value's components transform under a change of frame — nothing about what the quantity **is**. The other three fields each record a facet of the quantity itself:
 - tensor order tracks its mathematical character,
 - bound-versus-free is a genuine physical distinction (and one the library actually uses), and
 - the measurement reference names a dimension and a frame.

The variance split alone carries no information about the quantity kind.

2. **Not quantity-invariant.** Using the metric to raise a covariant (lower) index or lower a contravariant (upper) one moves a unit between the two counts — say from `contravariantOrder 2`, `covariantOrder 0` to a mixed 1 and 1 — while the order, and the physical quantity, stay fixed. That the quantity is unchanged is a fact of **tensor algebra** — the metric's raising/lowering isomorphism — not a SysML v2 statement: the library defines no raise- or lower-index operation, so it would record the two splits as distinct

`TensorQuantityValues` with nothing to mark them as one quantity. A feature that a pure change of representation can move while the quantity is held fixed is thus a property of the representation, not of the quantity's kind. Hold the quantity fixed and the other three fields are pinned with it — order, bound-status, and reference do not change — so the variance split is the one field free to vary while the modeled quantity stays put (worked through just below).

3. **Empirically inert.** The split is declared on the abstract root and named by its constraint, yet assigned by **no** quantity anywhere in the library: even the genuine order-2 kinds — `Cartesian3dStressTensor`, with strain and moment of inertia alongside it — inherit the two slots and leave them unset, fixing only `isBound`, `num`, and `mRef` ([SysML v2, 2026-04 — ISQMechanics.sysml, lines 730–746](https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/ISQMechanics.sysml#L730-L746) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/ISQMechanics.sysml#L730-L746>)). The apparatus is carried at the root whether or not any kind exercises it. By contrast `isBound` **is** used (88 free against 7 bound), so the point is precise rather than blanket: bound-versus-free is live machinery, while the variance split is declared-but-unexercised overhead inherited from the tensor root.

To see why the second count bites — and what the model **loses** by it — follow the library's own machinery. `Cartesian3dStressTensor` stores its value as `num: Real[9]`, nine bare reals, with the variance split left blank (same listing). In an orthonormal Cartesian frame that omission is harmless: under the Euclidean metric the contravariant components σ^{ij} , the mixed $\sigma^i{}_j$, and the covariant σ_{ij} are numerically equal, so it does not matter which the nine numbers are. But `TensorCalculations::transform` is defined to carry a tensor value from a source frame to a target frame through a `CoordinateTransformation` ([SysML v2, 2026-04 — TensorCalculations.sysml, lines 45–49](https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/>

[TensorCalculations.sysml#L45-L49](#)), [MeasurementReferences.sysml, lines 126–135](#) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/MeasurementReferences.sysml#L126-L135>)), and the transformation rule is keyed on the split: a contravariant index transforms by the Jacobian of the coordinate change, a covariant index by its inverse — mutually inverse rules, so the two generally yield different numbers. Move to a frame where they differ — any non-orthonormal frame, which is to say the everyday cylindrical, spherical, or material coordinates of continuum mechanics — and the nine numbers transform by **different** rules depending on a datum the model declared but never filled in.

What is missing, then, is the ability to transform an order-2 (or higher-order) quantity between frames **at all**: the transform contract cannot be met without the split, so for exactly the quantities where it would matter the operation is underdetermined. And the gap is **silent**. The concrete types are pinned to a Cartesian frame, where the omission never shows; the model type-checks and looks complete, and fails only where one would actually rely on it — in the curvilinear frames it was never exercised in. That is precisely a representational hole that validation cannot see: correct on every case anyone tested, wrong on the first case anyone needed.

The lone constraint the library attaches to the split sharpens the reading. `orderSum` requires `contravariantOrder + covariantOrder == order` ([SysML v2, 2026-04 — Quantities.sysml, line 36](#) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/Quantities.sysml#L36>))). Metrologically it is a well-formedness invariant, not a statement about the quantity: it says only that the chosen split must be a valid partition of the quantity's actual order. Its content is that **order is the frame-invariant datum and the split is a convention that must respect it** — raising or lowering an index slides one unit between `covariantOrder` and `contravariantOrder` but holds their sum, the order, fixed. So `orderSum` is exactly what makes **the same quantity, a different split** formally coherent: the order — which carries the quantity's mathematical character — is conserved while the representation changes underneath it. It constrains nothing metrological: not dimension, unit, scale, or kind, only that the coordinate bookkeeping adds up. The one rule SysML v2 writes about variance is thus itself a pure-representation consistency check — the section's thesis in miniature.

The kind-of-quantity sits one level **down**, as a leaf, and the source is decisive on this point: there is no first-class "kind" entity. Length, mass, temperature, and the other base quantities are each declared as a **subtype of the scalar value type** — `attribute def LengthValue :> ScalarQuantityValue` — with the dimension carried only by the corresponding unit's `quantityDimension`, e.g. `LengthUnit` ([SysML v2, 2026-04 — ISQBase.sysml, lines 16–38](#) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/ISQBase.sysml#L16-L38>))), and that `quantityDimension` attribute itself lives on `ScalarMeasurementReference` ([SysML v2, 2026-04 — MeasurementReferences.sysml, lines 82–99](#) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/MeasurementReferences.sysml#L82-L99>))). In other words, "length-the-kind" exists only as (i) a nominal subtype hanging beneath `ScalarQuantityValue` and (ii) a power-product of base quantities on its unit — never as a property whose nature is named in its own right. To reach a kind one must already have committed to a representation branch: the kind is **nested under** the representation. The proportions confirm the weight — across the ISQ library 319 kinds are declared on the scalar branch, against 2 vector, 4 tensor, and 51 fixed-three-vector kinds.

That places the concern precisely. A valued quantity has two independent classification axes:

- **(A) kind-of-property** — its nature, fixed by Dybkær's examination principle, common to all mutually comparable quantities;
- **(B) value representation** — scalar / vector / tensor order, coordinate frame, bound-versus-free — what SysML v2 packages as the measurement reference.

Dybkær's ontology lives entirely on axis (A) and is deliberately **agnostic** to (B): whether a position is recorded as one number, three Cartesian components, or a tensor in a chosen frame does not change **what kind of property** it is. SysML v2 makes axis (B) the root and nests (A) beneath it. The model even flags the departure from its own source: it quotes the VIM3 "quantity" Note 5 — "A quantity as defined here is a scalar. However, a vector or a tensor, the components of which are quantities, is also considered to be a quantity" — then notes that "the rest of [VIM] does not explicitly define how tensor and vector quantities can be or should be supported" ([SysML v2, 2026-04 — MeasurementReferences.sysml, lines 29–31](https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/MeasurementReferences.sysml#L29-L31) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/MeasurementReferences.sysml#L29-L31>)). The tensor-rooted hierarchy is thus SysML's own interpretation, and it **inverts** the VIM's priority — both the quoted VIM3 and the VIM4 2CD (quantity, 1.1 Note 4: "As defined here, quantities are scalar ... vectors, and tensors are considered quantities in a broader sense if their components are (scalar) quantities") make the quantity fundamentally scalar, with vectors and tensors as composites **of** scalar quantities; SysML v2 makes the tensor primary and the scalar the degenerate order-0 case.

3.3.2. How PropertyKindCalculus takes the kind as root instead

PropertyKindCalculus inverts the nesting: axis (A) is the root. A kind-of-property is a first-class value (KindOfProperty), and a quantity is indexed **by its kind** — Quantity (k : KindOfProperty) — so the kind is fixed before, and independently of, any choice of how the value is recorded. Dimension is not the organizing principle but a forgetful functor $\dim : \mathrm{Kind} \rightarrow \mathrm{Dimension}$ that **certifies** coherence without **deciding** what is allowed (developed in the next section). Two consequences fall out directly: two kinds that share a dimension remain distinct types — so the dimension-1 disambiguation is expressible — and an operation across kinds is a type error rather than a silent success. Value representation — the axis SysML v2 puts at the root — is, candidly, **not yet specified** here; its principled home is an **orthogonal** index over the kind, never a layer the kind hangs beneath.

3.3.3. The two designs side by side

Aspect	SysML v2 (2026-04)	PropertyKindCalculus
Root organizing axis	value representation (tensor order, frame)	kind-of-property (Dybækær)
<i>A kind of quantity is...</i>	a leaf subtype of <code>ScalarQuantityValue</code>	a first-class type index, <code>Quantity (k : KindOfProperty)</code>
Role of dimension	the discriminator for commensurability, on the reference	a forgetful functor <code>dim</code> ; certifies, never decides
Two kinds, one dimension (vwc vs gwc)	indistinguishable	distinct by construction
Adding a length to a mass	well-typed (<code>Scalar × Scalar → Scalar</code>)	a type error
Scalar / vector / tensor, frames, transforms	first-class and rich	not yet specified (owed — an orthogonal index)
Algebraic laws	a modelling library; none machine-checked	proved theorems (preorder, homomorphism, monotonicity)

The table is honest in both directions: SysML v2's representation-first design buys real, first-class machinery for coordinate frames, transformations, and stress and strain tensors that systems engineering genuinely needs — machinery PropertyKindCalculus does not yet have. The claim is not that one model is right; it is that the axes are **orthogonal and should not be nested**.

3.3.4. Limitations of the SysML v2 approach

Rooting the hierarchy in representation has three consequences that bear directly on PropertyKindCalculus's problem statement:

1. **A kind cannot be named without first choosing a representation, so one kind fragments.** Position is a single kind-of-property, yet appears as `CartesianPosition3dVector`, `CylindricalPosition3dVector`, `SphericalPosition3dVector`, and `PlanetaryPosition3dVector` — individuated by **coordinate frame**, not by the nature of the property. The cylindrical and spherical forms even carry mixed-dimension components (a length and two angles), so the single documented "quantity dimension L^1 " does not survive its own representations ([SysML v2, 2026-04 — ISQSpaceTime.sysml, lines 304–353](https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/ISQSpaceTime.sysml#L304-L353) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/ISQSpaceTime.sysml#L304-L353>)).
2. **The within-dimension conflation is left untouched.** On the scalar branch the dimension is the only discriminator, so volumetric and gravimetric water content, relative permittivity, reflectivity, and

emissivity remain indistinguishable — exactly the confusion that motivates PropertyKindCalculus. Putting representation at the root does nothing about it; the two are orthogonal.

3. **The calculus is kind- and dimension-blind.** Scalar addition is declared `ScalarQuantityValue × ScalarQuantityValue → ScalarQuantityValue`, so adding a length to a mass is well-typed; nothing gates an operation on agreeing kind or dimension ([SysML v2, 2026-04 — QuantityCalculations.sysml, line 29](https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/QuantityCalculations.sysml#L29) (<https://github.com/Systems-Modeling/SysML-v2-Release/blob/2026-04/sysml.library/Domain%20Libraries/Quantities%20and%20Units/QuantityCalculations.sysml#L29>)).

3.4. How the calculus maps into Lean

Two things in particular resist a DL encoding: the *interaction algebra* (which `kind \times which kind = which kind`, a *partial* relation with arithmetic side-conditions, not a role chain), and the difference between *instantiation* and *subtyping* (that a radius *is* a length, versus that this pencil's length *is an instance of* the length kind — conflated by punning in OWL). In Lean both map cleanly, and the mapping is the design:

1. **Kinds and units are type-level parameters.** `Quantity (k : QuantityKind) (u : Unit) : Type` — a kind and a unit index a *type*, not a value of one universal `Quantity` class. This is the move OWL cannot make.
2. **Specialization ($\sqsubseteq$) induces coercions between types.** It lives on those parameters: `Quantity Radius metre → Quantity Length metre` is sound — a radius *is* a length — and the reverse is not. Up-casting to a parent kind is an explicit, visible loss of information.
3. **Instantiation is term-of-type, not subtyping.** The length of a pencil is a *term* `pencil : Quantity Length metre := {5.2}` — a value of that type, related to its kind by typing/instantiation, never by subtyping. The general/individual distinction *is* the type/term distinction.
4. **The kind algebra is a partial typed algebra, not a group.** PhysLib's `Dimension` is a *total* `CommGroup` — multiply, divide, invert anything. Kinds are not: `Torque × Angle = Work` is meaningful; `Torque + Energy` must be *blocked* even though the dimensions match; `FuelConsumption × Rainfall` is an error (Flater's own example). The meaningful products are encoded as a relation surfaced through typeclass resolution — Flater's "map kinds onto classes + operator overloading," lifted into Lean's elaborator.
5. **Dimension is a soundness functor, not the primary type.** In QUDV and most quantity libraries dimension is the *primary* classifier: two quantities interoperate exactly when their dimensions agree — which is precisely the move that collapses every dimension-one quantity into one indistinguishable type. PropertyKindCalculus inverts the dependency: the *kind* is the primary type-level index, and dimension is recovered as a forgetful functor `\dim : \mathrm{Kind} \to \mathrm{Dimension}`. Its job is *certification*, not decision — the coherence theorem $\mathrm{KMul} \ k_1 \ k_2 \ k_3 \rightarrow \dim \ k_3 = \dim \ k_1 \cdot \dim \ k_2$ proves every product the kind algebra admits is dimensionally consistent — but `\dim` never decides whether an operation is *allowed*; the kind relation does. Agreeing dimensions are necessary, not sufficient.

6. **Specialization ($\sqsubseteq$) is a preorder — a lattice, not a tree.** Flater stresses this (citing Formal Concept Analysis): a kind can specialize several parents. Unit-1 subtyping (Flater §6) is then just the *dimension-1 slice* of that kind lattice — and nothing forces it to be uniform: units of plane angle, for one, could carry their own subtyping reflecting the different measurement principles by which they are realized.
7. **The numeric carrier is a type parameter, bounded by a typeclass (R10).** A scalar quantity is `Quantity (k : KindOfProperty) (R : Type)` with the arithmetic on `R` supplied by a `Carrier` typeclass — the kind fixes **what is measured**, `R` fixes **in what numbers**. Everything above `R` (kind discrimination, scale gating, the dimension functor, the interaction algebra, extensivity) is written once, against the typeclass, and is reused verbatim at every `R` (the additivity laws, for one, are proved a single time over any lawful carrier and hold at $\mathbb{R}$ and `Int` with no per-carrier proof). This is the architecture TorchLean already runs: models are written against a `Context  $\alpha$` typeclass (arithmetic + transcendentals
 - comparison) and instantiated at $\alpha := \mathbb{R}$ for the **spec**, $\alpha := \text{FP32}$ for the finite **rounding** model, and $\alpha := \text{IEEE32Exec}$ for the **executable** IEEE-754 kernel — the **same** source elaborated at three carriers. The kind layer here is the missing index **above** that carrier: TorchLean makes a tensor polymorphic in its scalar type; `PropertyKindCalculus` makes the scalar polymorphic in its **kind**, and the two indices compose. Crucially the carriers are bridged by **refinement theorems** (`toReal (op_exec x) = round (op_real (toReal x))`), so a law proved at $\mathbb{R}$ is not stranded in the proof world — it descends to the float that runs. A description logic has no type parameters and no typeclasses, so this whole axis is unavailable to it; here it is the ordinary discipline of parametric polymorphism.

The trigger was concrete. PhysLib's `Dimension` makes all dimension-one quantities equal, so it cannot distinguish volumetric (L^3/L^3) from gravimetric (M/M) soil moisture, nor either from permittivity, reflectivity, or emissivity. Every dangerous soil-moisture confusion is dimension one. Indexing quantities and units by **kind** is what restores the distinction — this is the dimension-1 disambiguation capstone in the Units chapter.

4. How to read the status

Nodes that link a real declaration with `(lean := "PropertyKindCalculus....")` report their **proved** status straight from the checked, sorry-free core. Nodes tagged `planned` carry an informal statement and a proof sketch only; they show as in-progress goals until formalized. The headline deliverables are tagged `capstone`.

5. The Proved Spine

This chapter records what PropertyKindCalculus already establishes, sorry-free, in its Mathlib-free core. Every node here links to a real declaration with (lean := "PropertyKindCalculus...."), so its *proved* status in the dependency graph is read directly from the checked library rather than asserted in prose.

The point to keep in view: each of these is a law or an operation, not a class membership. A description-logic reasoner (OWL2/SROIQ) can record that width specializes length; it cannot state, let alone prove, that specialization is transitive or that operator availability is monotone in scale. Those are the entries below.

5.1. Foundations — system and object (Dybkaer Ch. 3)

Definition

5.1.1.

uses 0

used by 4

✓ LEAN

A *system* (Dybkaer §3.3) is a demarcated arrangement of elements and their relationships; *object* (§3.3 Note 6) is given as a synonym. Specified abstractly by identity here; the mereological structure that extensivity needs is added in a later module.

Lean code for Definition

5.1.1.

1 definition

structure (1 field) · defined in PropertyKindCalculus/ Foundations.lean

complete

```
structure PropertyKindCalculus.System : Type
```

Proof for Definition 5.1.1

uses 0

A one-field structure with DecidableEq; nothing to prove. The synonym `Object := System` keeps the instance layer reading as "characterizes an object".

5.2. Scale — the operator-based division of ⟨property⟩ (Dybkaer Ch. 12)

Definition

5.2.1.

group

uses 0

used by 7

✓

The four *scale types* of Dybkaer Fig. 12.21, ordered by operator richness: $\mathrm{nominal} \sqsubset \mathrm{ordinal} \sqsubset \mathrm{interval} \sqsubset \mathrm{ratio}$, admitting respectively $\{=, \neq\}$, then $\{<, >\}$, then $\{+, -\}$, then $\{\times, \div\}$.

Lean code for Definition 5.2.1.

1 definition

inductive (4 constructors) · defined in PropertyKindCalculus/Scale.lean

complete

`inductive PropertyKindCalculus.ScaleType : Type`

Proof for Definition 5.2.1 uses 0

A four-constructor inductive with a rank : ScaleType → Nat and the induced ≤. The lemmas le_def, le_refl, le_trans, and le_antisymm establish that ≤ is the expected linear order.

Theorem 5.2.2. group statement">uses 1 by 0

Owner

Tags

Nicolas Rouquette

proved

The scale order is a linear order: reflexive (le_refl), transitive (le_trans), and antisymmetric (le_antisymm) on [the scale types](#).

Lean code for Theorem 5.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Scale.lean

complete

`theorem PropertyKindCalculus.ScaleType.le_antisymm {a b : PropertyKindCalculus.ScaleType} (h1 : a ≤ b) (h2 : b ≤ a) : a = b`

Proof for Theorem 5.2.2 uses 0

Reflexivity and transitivity are inherited from Nat.le on rank. Antisymmetry reduces to Nat.le_antisymm on the ranks followed by a four-by-four case split (cases a <;> cases b <;> simp_all).

Theorem 5.2.3. group statement">uses 1 by 2

Owner

Priority

Tags

Nicolas Rouquette

high

capstone

proved

Operator-availability monotonicity. A richer scale licenses every operation a poorer one does: if $a \leq b$ then *AllowsOrder*, *AllowsDifference*, and *AllowsRatio* each transport from a to b . This is a meta-

theorem OWL2's reasoner cannot even phrase, since it quantifies over which operations are defined for a kind. Builds on [the scale types](#).

Lean code for Theorem 5.2.3.

1 theorem

```
theorem · defined in PropertyKindCalculus/Scale.lean

theorem PropertyKindCalculus.ScaleType.allows_mono
  {a b : PropertyKindCalculus.ScaleType} (h : a ≤ b) :
  (a.AllowsOrder → b.AllowsOrder) ∧
  (a.AllowsDifference → b.AllowsDifference) ∧
  (a.AllowsRatio → b.AllowsRatio)
```

Proof for Theorem 5.2.3

uses 0

Decide all sixteen ordered pairs: cases a <=> cases b <=> simp_all [AllowsOrder, AllowsDifference, AllowsRatio, le_def, rank].

5.3. Kind-of-property and kind-of-quantity (Dybæk Ch. 6, 13)

Definition 5.3.1.

group statement">used by 15

✓

A *kind-of-property* (§6.19) bundles a terminological id, a scale (from [the scale types](#)), and an optional examination principle. A *kind* is a *kind-of-quantity* (§13.3.1) iff its scale has magnitude (is at least ordinal).

Lean code for Definition 5.3.1.

1 definition

```
structure (3 fields) · defined in PropertyKindCalculus/Kind.lean

structure PropertyKindCalculus.KindOfProperty : Type
```

Proof for Definition 5.3.1

uses 0

A structure with DecidableEq. IsQuantity, and the scale divisions IsNominal/IsOrdinal/IsDifferential/IsRational (§13.2), are definitional predicates over the scale field.

Theorem 5.3.2.

group statement">used by 15

Owner	Tags
Nicolas Rouquette	proved

| A nominal kind is, by construction, not a kind-of-quantity: it has no magnitude. Uses [kind-of-property](#).

Lean code for Theorem 5.3.2.

1 theorem

complete

theorem · defined in PropertyKindCalculus/
Kind.lean

```
theorem PropertyKindCalculus.KindOfProperty.nominal_not_quantity
  {k : PropertyKindCalculus.KindOfProperty} (h : k.IsNominal) :
  ¬k.IsQuantity
```

Proof for Theorem 5.3.2

uses 0

| Unfold IsQuantity, IsNominal, and HasMagnitude; the scale is nominal, so scale ≠ nominal is false.

Theorem 5.3.3.

group

statement">us

Owner

Tags

Nicolas Rouquette

proved

| A rational kind is a kind-of-quantity. Uses [kind-of-property](#).

Lean code for Theorem 5.3.3.

1 theorem

complete

theorem · defined in PropertyKindCalculus/
Kind.lean

```
theorem PropertyKindCalculus.KindOfProperty.rational_isQuantity
  {k : PropertyKindCalculus.KindOfProperty} (h : k.IsRational) :
  k.IsQuantity
```

Proof for Theorem 5.3.3

uses 0

| The scale is ratio, so scale ≠ nominal holds; unfold and simplify.

Definition 5.3.4.

group

uses 0

A kind is a *ratio quantity* (VIM4 2CD 1.3) when its scale is ratio: the operators \times, \div are defined, so its quantities have an absolute zero and form ratios.

Lean code for Definition 5.3.4.

1 definition

complete

def · defined in PropertyKindCalculus/
Kind.lean

```
def PropertyKindCalculus.KindOfProperty.IsRational
  (k : PropertyKindCalculus.KindOfProperty) : Prop
```

uses 0

Proof for Definition 5.3.4

```
IsRational k := k.scale = .ratio.
```

Definition 5.3.5.

group

uses 0

A kind is an *interval quantity* (VIM4 2CD 1.4) when its scale is interval: the operators +, - are defined (differences are meaningful), but ratios are not.

Lean code for Definition 5.3.5.

1 definition

```
def · defined in PropertyKindCalculus/
Kind.lean
```

complete

```
def PropertyKindCalculus.KindOfProperty.IsDifferential
  (k : PropertyKindCalculus.KindOfProperty) : Prop
```

Proof for Definition 5.3.5

uses 0

```
IsDifferential k := k.scale = .interval.
```

Definition 5.3.6.

group

uses 0

A kind is an *ordinal quantity* (VIM4 2CD 1.33) when its scale is ordinal: only the order <, > is defined, with neither differences nor ratios.

Lean code for Definition 5.3.6.

1 definition

```
def · defined in PropertyKindCalculus/
Kind.lean
```

complete

```
def PropertyKindCalculus.KindOfProperty.IsOrdinal
  (k : PropertyKindCalculus.KindOfProperty) : Prop
```

Proof for Definition 5.3.6

uses 0

```
IsOrdinal k := k.scale = .ordinal.
```

5.4. Specialization and mutual comparability (OML / Dybkær §6)

Definition 5.4.1.

group

statement" > used by 8

✓

Specialization Specializes E is the reflexive–transitive closure of a direct-parent edge relation E over [kind-of-property](#). Different E give different per-application hierarchies — the extensibility a fixed inductive would deny.

Lean code for Definition 5.4.1.

1 definition

complete

```
inductive (2 constructors, Prop, 3 parameters)
· defined PropertyKindCalculus/
in Specialization.lean

inductive PropertyKindCalculus.Specializes
(E :
PropertyKindCalculus.KindOfProperty →
PropertyKindCalculus.KindOfProperty → Prop) :
PropertyKindCalculus.KindOfProperty →
PropertyKindCalculus.KindOfProperty → Prop
```

Proof for Definition 5.4.1

uses 0

An inductive with `refl` and `step` constructors; `of_edge` lifts a single edge.

Theorem 5.4.2.

group

statement" > uses 1 by 0

Owner	Priority	Tags
Nicolas Rouquette	high	capstone proved

Specialization is a preorder. It is reflexive (the `refl` constructor) and transitive (`Specializes.trans`). Stating — let alone proving — an algebraic law of the taxonomy is exactly what an OWL2 reasoner cannot do; it checks consistency, not laws. Uses [specialization](#).

Lean code for Theorem 5.4.2.

1 theorem

complete

```
theorem · defined PropertyKindCalculus/
in Specialization.lean

theorem PropertyKindCalculus.Specializes.trans
{E :
PropertyKindCalculus.KindOfProperty →
PropertyKindCalculus.KindOfProperty → Prop}
{a b c : PropertyKindCalculus.KindOfProperty}
(h1 : PropertyKindCalculus.Specializes E a b)
```

$(h_2 : \text{PropertyKindCalculus.Specializes } E \ b \ c) :$
 $\text{PropertyKindCalculus.Specializes } E \ a \ c$

Proof for Theorem 5.4.2

uses 0

Transitivity is by induction on the first derivation: `refl` returns the second derivation, `step` re-applies the constructor under the inductive hypothesis.

Definition 5.4.3.

group

statement">uses by 2

Two kinds are *mutually comparable* iff they share a common super-kind ($\exists p, \ a \sqsubseteq p \wedge b \sqsubseteq p$). This is how width and height stay comparable as lengths while remaining distinct sub-kinds. Uses [specialization](#).

Lean code for Definition 5.4.3.

1 definition

def · defined in PropertyKindCalculus/
Specialization.lean

complete

```
def PropertyKindCalculus.MutuallyComparable
  (E :
    PropertyKindCalculus.KindOfProperty →
    PropertyKindCalculus.KindOfProperty → Prop)
  (a b : PropertyKindCalculus.KindOfProperty) : Prop
```

Proof for Definition 5.4.3

uses 0

An existential over a shared upper bound.

Theorem 5.4.4.

group

statement">uses by 0

Owner

Tags

Nicolas Rouquette

proved

Mutual comparability is reflexive (`refl`) and symmetric (`symm`), and any specialization induces it (`of_specializes`). Uses [mutual comparability](#).

Lean code for Theorem 5.4.4.

1 theorem

theorem · defined in PropertyKindCalculus/
Specialization.lean

complete

```
theorem PropertyKindCalculus.MutuallyComparable.symm
  {E :
```

```

PropertyKindCalculus.KindOfProperty →
  PropertyKindCalculus.KindOfProperty → Prop}
{a b : PropertyKindCalculus.KindOfProperty}
(h : PropertyKindCalculus.MutuallyComparable E a b) :
PropertyKindCalculus.MutuallyComparable E b a

```

Proof for Theorem 5.4.4

uses 0

Reflexivity takes the witness a itself; symmetry swaps the two halves of the conjunction.

5.5. Examination — principle, method, procedure (Dybæk Ch. 7, 14)

Definition 5.5.1.

group

uses 0

used by 17

The examination chain has three layers — an *examination principle* (§7.5 / VIM4 2CD 2.4), a *method* based on a principle (VIM4 2CD 2.5), and a *procedure* based on a method (VIM4 2CD 2.7) — gathered into one carrier `ExaminationItem` so that refinement is a homogeneous relation. The forgetful projection `basePrinciple` sends any item down to the principle it rests on.

Lean code for Definition 5.5.1.

1 definition

```

structure (1 field) · defined in PropertyKindCalculus/
                             Examination.lean

```

complete

```

structure PropertyKindCalculus.ExaminationPrinciple : Type

```

Proof for Definition 5.5.1

uses 0

Three small structures with `DecidableEq`, united by an `ExaminationItem` inductive; `basePrinciple` is the projection to the principle layer.

Definition 5.5.2.

group

statement">statement">used by

An *examination method* (Dybæk Ch. 7 / VIM4 2CD 2.5) is a generic description of an examination, based on an [examination principle](#).

Lean code for Definition 5.5.2.

1 definition

```
structure (2 fields) · defined in PropertyKindCalculus/
Examination.lean
```

complete

```
structure PropertyKindCalculus.ExaminationMethod : Type
```

Proof for Definition 5.5.2

uses 0

A structure carrying an identifier and the ExaminationPrinciple it is based on.

Definition 5.5.3.

group

statement" > uses 1 by 0

An *examination procedure* (Dybæk Ch. 7 / VIM4 2CD 2.7) is a detailed description, *based on* an [examination method](#); its principle is the method's principle.

Lean code for Definition 5.5.3.

1 definition

```
structure (2 fields) · defined in PropertyKindCalculus/
Examination.lean
```

complete

```
structure PropertyKindCalculus.ExaminationProcedure : Type
```

Proof for Definition 5.5.3

uses 0

A structure carrying an identifier and the ExaminationMethod it is based on.

Definition 5.5.4.

group

statement" > uses 1 by 2

Refinement *Refines* is the reflexive–transitive closure of the "based-on" edges (a procedure is based on its method, a method on its principle): $\text{procedure} \sqsubseteq \text{method} \sqsubseteq \text{principle}$. It mirrors [specialization](#) over kinds — the same closure construction, one level down at the defining aspect.

Lean code for Definition 5.5.4.

1 definition

```
inductive (2 constructors, Prop, 2 parameters) · defined in PropertyKindCalculus/
Examination.lean
```

complete

```
inductive PropertyKindCalculus.Refines :
  PropertyKindCalculus.ExaminationItem →
  PropertyKindCalculus.ExaminationItem → Prop
```

Proof for Definition 5.5.4

uses 0

An inductive with `refl` and `step` constructors over a `BasedOn` edge relation; `of_basedOn` lifts a single edge.

Theorem 5.5.5.

group

statement" > uses 1 by 0

✓

Owner

Nicolas Rouquette

Tags

proved

Refinement is a preorder. It is reflexive (the `refl` constructor) and transitive (`Refines.trans`) on [the examination chain](#) — the same algebraic law proved for specialization, which an OWL2 reasoner can neither state nor prove.

Lean code for Theorem 5.5.5.

1 theorem

complete

theorem · defined in PropertyKindCalculus/
Examination.lean

```
theorem PropertyKindCalculus.Refines.trans
  {a b c : PropertyKindCalculus.ExaminationItem}
  (h1 : PropertyKindCalculus.Refines a b)
  (h2 : PropertyKindCalculus.Refines b c) :
  PropertyKindCalculus.Refines a c
```

Proof for Theorem 5.5.5

uses 0

Transitivity is by induction on the first derivation, exactly as for `Specializes`: `refl` returns the second derivation, `step` re-applies the constructor under the inductive hypothesis.

Theorem 5.5.6.

group

statement" > uses 1 by 0

✓

Owner

Nicolas Rouquette

Priority

high

Tags

capstone

proved

Refinement preserves the principle. Every examination in one refinement chain rests on the same base principle: $\mathrm{Refines}\ a\ b\ ;\ \Longrightarrow\ \mathrm{basePrinciple}\ a\ =\ \mathrm{basePrinciple}\ b$. The forgetful projection to the principle layer is invariant along refinement — the examination-layer analogue of $\dim$ being a homomorphism, and one provable here in the Mathlib-free core without `PhysLib`. Builds on [refinement](#).

Lean code for Theorem 5.5.6.

1 theorem

theorem · defined in PropertyKindCalculus/ Examination.lean

complete

```
theorem PropertyKindCalculus.Refines.basePrinciple_eq
  {a b : PropertyKindCalculus.ExaminationItem}
  (h : PropertyKindCalculus.Refines a b) :
  a.basePrinciple = b.basePrinciple
```

Proof for Theorem 5.5.6 uses 0

By induction on the refinement derivation: each BasedOn edge preserves the base principle (cases ...
<;> rfl), and the inductive step chains the resulting equalities.

Theorem 5.5.7. group uses 2 statement">used by

Owner

Tags

Nicolas Rouquette

proved

The examination principle is a defining aspect. Two [kinds](#) examined by different principles are distinct kinds — independently of scale and dimension. This is what keeps width and height distinct while both remain rational lengths, [mutually comparable](#) under their shared super-kind Length.

Lean code for Theorem 5.5.7.

1 theorem

theorem · defined in PropertyKindCalculus/ Examination.lean

complete

```
theorem PropertyKindCalculus.KindOfProperty.distinct_of_examPrinciple
  {k1 k2 : PropertyKindCalculus.KindOfProperty}
  (h : k1.examPrinciple ≠ k2.examPrinciple) : k1 ≠ k2
```

Proof for Theorem 5.5.7 uses 0

The kinds differ in their examPrinciple field, so they are unequal by congruence: substitute the assumed kind-equality and derive a contradiction from the field inequality.

5.6. Property value and value scale (Dybækær Ch. 9, 16, 10, 17)

Definition 5.6.1. group statement">uses by 4

A *property value* (§9.15) bundles its [kind-of-property](#), a numeral (the numerical value, §16.10), and the reference the numeral is taken against — a metrological unit for a quantity value, a designation for a nominal one. A value *is a quantity value* (§16.10) iff its kind has magnitude, and two values are *comparable* iff they are of the same kind.

Lean code for Definition 5.6.1.

1 definition

```
structure (3 fields) · defined in PropertyKindCalculus/
                          PropertyValue.lean
```

complete

```
structure PropertyKindCalculus.PropertyValue : Type
```

Proof for Definition 5.6.1

uses 0

A structure with DecidableEq, scale, IsQuantityValue, and Comparable are definitional projections over the kind field.

Theorem 5.6.2.

group

statement">use

Owner

Tags

Nicolas Rouquette

proved

Comparability of values is an equivalence. Two [property values](#) are comparable iff they are of the same kind (§9.15); the relation is reflexive, symmetric, and transitive. Comparable values are moreover governed by the same scale type, so the same operators are defined on both (*scale_eq_of_comparable*). Transitivity is exactly the algebraic law a description logic cannot state.

Lean code for Theorem 5.6.2.

1 theorem

```
theorem · defined in PropertyKindCalculus/
               PropertyValue.lean
```

complete

```
theorem PropertyKindCalculus.PropertyValue.Comparable.trans
  {u v w : PropertyKindCalculus.PropertyValue} (h1 : u.Comparable v)
  (h2 : v.Comparable w) : u.Comparable w
```

Proof for Theorem 5.6.2

uses 0

Comparability unfolds to equality of the kind field, so reflexivity, symmetry, and transitivity are those of Eq; *scale_eq_of_comparable* is *congrArg* on the *scale* projection.

Definition 5.6.3.

group

statement">state

A *property value scale* (§10.14) is the ordered set of possible, mutually comparable values of one [kind](#). It carries that kind and a provenance — *true* (consistent with the property's definition, §10.16.1) or *examined* (obtained by an examination procedure, §10.16.2). Its `scaleType` is the kind's scale, the datum Table 17.4 uses to gate which manipulations a scale admits, and `KindOfProperty.valueScale` gives the canonical true scale of a kind (§9.15 ↔ §10.14).

Lean code for Definition 5.6.3.

1 definition

```
structure (2 fields) · defined in PropertyKindCalculus/ValueScale.lean
```

complete

```
structure PropertyKindCalculus.ValueScale : Type
```

Proof for Definition 5.6.3

uses 0

A structure carrying kind and a `ScaleProvenance`; Admits `v` is `v.kind = s.kind`, and `KindOfProperty.valueScale` builds the canonical true scale.

Theorem 5.6.4.

group

uses 3

used by 0

Owner

Nicolas Rouquette

Priority

high

Tags

capstone

proved

A value scale holds only mutually comparable values. Any two values a [value scale](#) admits are of the scale's kind, hence [comparable](#) (§10.14) — the defining property of a value scale, quantified over its members. The scale type further licenses operations monotonically (`allows_mono_of_le`): a richer value scale supports every manipulation a poorer one does (Table 17.4), inherited from [operator monotonicity](#).

Lean code for Theorem 5.6.4.

1 theorem

```
theorem · defined in PropertyKindCalculus/ValueScale.lean
```

complete

```
theorem PropertyKindCalculus.ValueScale.comparable_of_mem
  {s : PropertyKindCalculus.ValueScale}
  {v w : PropertyKindCalculus.PropertyValue} (hv : s.Admits v)
  (hw : s.Admits w) : v.Comparable w
```

Proof for Theorem 5.6.4

uses 0

Membership unfolds to equality with the scale's kind, so two members' kinds are equal by trans/symm; operator monotonicity is the scale-layer allows_mono re-exported through scaleType.

5.7. Metrological units (Dybæk Ch. 18, §13.3.3)

Definition 5.7.1.

group

uses 2

used by 22

A *metrological unit* (§18.12) carries the [kind](#) it references and a terminological symbol (the unit id, e.g. "m", "kg"). A kind *bears a unit* (BearsUnit, §13.3.3) iff its scale is at least differential, and a unit is *well-formed* iff its kind bears one; `measure u n` builds the [property value](#) " $n \times \text{symbol}$ ".

Lean code for Definition 5.7.1.

1 definition

```
structure (2 fields) · defined in PropertyKindCalculus/Unit.lean
```

complete

```
structure PropertyKindCalculus.MetrologicalUnit : Type
```

Proof for Definition 5.7.1

uses 0

A structure with `DecidableEq`. `BearsUnit` is `scale.AllowsDifference`, `WellFormed` is `kind.BearsUnit`, and `measure` fills the value's numeral and reference fields from `n` and the unit's symbol.

Theorem 5.7.2.

group

statement">state

Owner

Tags

Nicolas Rouquette

proved

Commensurability is an equivalence. Two [units](#) are commensurable iff they reference the same kind (§9.13.4); the relation is reflexive, symmetric, and transitive. A metre and a centimetre are commensurable; a metre and a kilogram are not — incommensurability across kinds is a fact of the types, not a runtime check. Transitivity is exactly the law a description logic cannot state.

Lean code for Theorem 5.7.2.

1 theorem

```
theorem · defined in PropertyKindCalculus/Unit.lean
```

complete

```
theorem PropertyKindCalculus.MetrologicalUnit.Commensurable.trans
  {u1 u2 u3 : PropertyKindCalculus.MetrologicalUnit}
  (h1 : u1.Commensurable u2) (h2 : u2.Commensurable u3) :
  u1.Commensurable u3
```

Proof for Theorem 5.7.2

uses 0

Commensurability unfolds to equality of the kind field, so reflexivity, symmetry, and transitivity are those of Eq.

Theorem

5.7.3.

group

statement">us

Owner

Tags

Nicolas Rouquette

proved

Only a unitary kind bears a unit. A nominal kind has no magnitude, and — the distinctive §9.13.4 fact — an ordinal kind, though rankable, has no reference-times-number magnitude either, so neither bears a metrological unit (`not_wellFormed_of_nominal`, `not_wellFormed_of_ordinal`). Conversely a rational kind does (`rational_bears_unit`), and bearing a unit implies being a kind-of-quantity (`bearsUnit_isQuantity`). This gate is invisible to a dimension-and-scale-only view that treats every magnitude alike. Uses [the metrological unit](#).

Lean code for Theorem

5.7.3.

1 theorem

theorem · defined in

PropertyKindCalculus/Unit.lean

complete

```
theorem PropertyKindCalculus.MetrologicalUnit.not_wellFormed_of_ordinal
  {u : PropertyKindCalculus.MetrologicalUnit} (h : u.kind.IsOrdinal) :
  ¬u.WellFormed
```

Proof for Theorem 5.7.3

uses 0

BearsUnit is scale.AllowsDifference, which is False at nominal and ordinal; substitute the scale from IsNominal/IsOrdinal and the goal is ¬ False. The rational case and bearsUnit_isQuantity are the same four-way case split on the scale.

Theorem

5.7.4.

group

uses 3

used by 0

✓

Owner

Priority

Tags

Nicolas Rouquette

high

capstone

proved

The number-and-reference form is a faithful round-trip (§13.3.3). Measuring a quantity in a unit yields a numeral, and re-applying the unit to that numeral recovers the value: $\mathrm{numeral} \backslash, (\mathrm{measure} \backslash u \ n) = n \quad \text{and} \quad \mathrm{measure} \backslash u \ (\mathrm{numeral} \backslash v) = v \ ; \text{ when } v \text{ is measured in } u$. So quantity / unit = number and number × unit = quantity invert one another — the §13.3.3 "reference quantity multiplied by a number",

checked as a law. Measuring in a well-formed unit moreover yields a [quantity value](#) (`measure_isQuantityValue`), and values measured in [commensurable](#) units are comparable. Builds on [the metrological unit](#).

Lean code for Theorem 5.7.4.

1 theorem

complete

theorem · defined in PropertyKindCalculus/
Unit.lean

```
theorem PropertyKindCalculus.MetrologicalUnit.measure_eq_of_measures
  {u : PropertyKindCalculus.MetrologicalUnit}
  {v : PropertyKindCalculus.PropertyValue} (h : u.Measures v) :
  u.measure v.numeral = v
```

Proof for Theorem 5.7.4

uses 0

`measure_numeral` is `rfl`. For the converse, destructure "measured in `u`" into the kind- and reference-equalities, unfold `measure`, and rewrite the unit's kind and symbol back to the value's fields; structure `eta` closes the goal.

6. Dedicated kinds-of-property

A bare kind-of-property — *length, volume fraction, substance concentration* — is generic: it names *what is measured* without saying *of what*. Dybkær's Chapter 20 closes that gap with the *dedicated kind-of-property*, defined there as "a kind-of-property with given sort of system and any pertinent component". It is the syntax laboratory medicine already uses for a fully-specified measurand — *System – Component ; kind-of-property*, as in "*Plasma — Glucose ; substance concentration*", or, in this project's home domain, "*Soil — Water ; volume fraction*" (volumetric water content).

This is the construct the QUDV → OML/OWL2 lineage cannot express. A description logic can record that a quantity *has* a system and *has* a component, but it cannot make "*Soil — Water ; volume fraction*" and "*Soil — Water ; mass fraction*" provably distinct kinds *while keeping both dedicated to the same system and the same component*. Here that distinctness is a theorem, and it is the *principled* form of the dimension-1 disambiguation: volumetric and gravimetric water content are the **same measurement target** — soil, water — examined as two different kinds-of-property, so they differ **because the kinds-of-property differ**, not because someone chose two different identity strings.

6.1. The dedicated kind (Dybkær Ch. 20)

Definition6.1.1.

groupstatement">uses by 3

A *component* (§20) is the pertinent component of the [system](#) a dedicated kind is about — *water* in a *soil* sample, *glucose* in *plasma*. Specified abstractly by identity, exactly mirroring the system carrier.

Lean code for Definition6.1.1.

1 definition

structure (1 field) · defined in PropertyKindCalculus/ DedicatedKind.leancomplete

structure PropertyKindCalculus.Component : Type

Proof for Definition 6.1.1

uses 0

A one-field structure with DecidableEq; nothing to prove.

Definition6.1.2.

groupuses 3used by 3

A *dedicated kind-of-property* (§20) bundles a [system](#), a [component](#), and a [kind-of-property](#). Its `systematicTerm` renders the three as *System – Component ; kind*, and it *is a dedicated kind-of-quantity* iff its underlying kind has magnitude (§13.3.1). `dedicatedTo` builds one from a bare kind.

Lean code for Definition 6.1.2.

1 definition

structure (3 fields) · defined in PropertyKindCalculus/DedicatedKind.lean

complete

```
structure PropertyKindCalculus.DedicatedKind : Type
```

Proof for Definition 6.1.2 uses 0

A three-field structure with DecidableEq. systematicTerm is string interpolation of the three ids; IsQuantity is the underlying kind's IsQuantity.

Theorem 6.1.3. group uses 2 used by 0

Priority high

Tags

capstone

proved

Same system and component, a different kind-of-property: a distinct dedicated kind. This is the principled volumetric-versus-gravimetric distinctness. Both water contents are "Soil — Water", but volume fraction and mass fraction are different [kinds-of-property](#), so the [dedicated kinds](#) differ — with the system and component held identical, and without appeal to any hand-chosen identity string. This is exactly the discrimination an OWL2 reasoner cannot phrase.

Lean code for Theorem 6.1.3.

1 theorem

theorem · defined in PropertyKindCalculus/DedicatedKind.lean

complete

```
theorem PropertyKindCalculus.DedicatedKind.distinct_of_kind
  {d1 d2 : PropertyKindCalculus.DedicatedKind} (h : d1.kind ≠ d2.kind) :
  d1 ≠ d2
```

Proof for Theorem 6.1.3 uses 0

Assume the two dedicated kinds are equal; then their kind projections are equal by congruence, contradicting the hypothesis (intro he; apply h; rw [he]). The companion laws distinct_of_system and distinct_of_component are the same one-line argument on the other two projections.

Theorem 6.1.4. group uses 4 statement">used by

Tags

proved

Same system and kind, a different component: a distinct dedicated kind. The water content and the air content of one soil sample share their [system](#) and their [kind-of-property](#), differing only in the [component](#) — and that alone makes them distinct [dedicated kinds](#).

Lean code for Theorem 6.1.4.

1 theorem

theorem · defined in PropertyKindCalculus/
DedicatedKind.lean

complete

```
theorem PropertyKindCalculus.DedicatedKind.distinct_of_component
  {d1 d2 : PropertyKindCalculus.DedicatedKind}
  (h : d1.component ≠ d2.component) : d1 ≠ d2
```

Proof for Theorem 6.1.4

uses 0

Congruence on the component projection, exactly as for `distinct_of_kind`.

7. Units and the Dimension-1 Problem

This chapter states the layer that originally motivated PropertyKindCalculus. Its motivating capstone — the dimension-1 disambiguation — is now *proved* (see the *Dimension as a Forgetful Functor* chapter for the forgetful map it rests on); the unit-arithmetic nodes around it remain *planned*, carrying informal statements and proof sketches that appear as in-progress goals in the dependency graph until a (lean := ...) declaration or checked code block is attached.

PhysLib's Dimension (a $\mathbb{Q}$ -exponent free commutative group over the SI base quantities) makes every dimension-one quantity equal. Volumetric water content ($\mathrm{L^3/L^3}$), gravimetric water content ($\mathrm{M/M}$), relative permittivity, reflectivity, and emissivity are all dimension one — so a dimension-only type system cannot tell them apart. In soil-moisture retrieval that is precisely the class of confusion that silently corrupts a result. The kind layer is what restores the distinction.

7.1. Quantities and units (VIM4 2CD, Dybkær §13.3.3)

Definition

7.1.1.

group

uses 0

used by 4

A *numeric carrier* `Carrier R` is the minimal arithmetic a kind-indexed magnitude needs, supplied once per representation type `R` (a zero and an addition; a `LawfulCarrier` adds the additive-monoid laws). In plain engineering terms it is the *number type plug*: everything above it — kind discrimination, scale gating, the dimension functor, the interaction algebra, extensivity — is written against this class and so reused verbatim at every `R`. This is the role TorchLean's `Context α` typeclass plays for tensor element types; the kind index is the layer *above* the carrier that TorchLean does not have.

Lean code for Definition

7.1.1.

1 definition

class (2 methods) · defined in PropertyKindCalculus/Quantity.lean

complete

class PropertyKindCalculus.Carrier (R : Type) : Type

Proof for Definition 7.1.1

uses 0

Realized as the `Carrier / LawfulCarrier` classes. `Carrier R` bundles zero and add; `LawfulCarrier R` extends it with associativity, commutativity, and the unit laws. `Int` and (in the `Dimension` library) $\mathbb{R}$ are lawful carriers; `Float` is a `Carrier` but deliberately *not* lawful (floating-point addition is not associative).

Definition

7.1.2.

group

uses 2

used by 21

A *quantity* `Quantity k R` is a magnitude of a fixed kind `k`, carried at a [representation type](#) `R` (`R10`). It is indexed by `k`, a [kind-of-property](#), so the type system forbids forming or comparing a `Quantity k1 R` with a `Quantity k2 R` when `k1 \neq k2` — even if both kinds are dimension one. This indexing *is* the fix

for the dimension-1 conflation. The second index R is the numeric carrier: the same value type is instantiated at $\mathbb{R}$ and Int to prove, at Float to run today, and (the planned float carriers) at FP32 to bound rounding and at IEEE32Exec to run with NaN — the kind machinery above it written once for all of them.

Lean code for Definition 7.1.2.

1 definition

```
structure (1 field) · defined in PropertyKindCalculus/
                        Quantity.lean
```

complete

```
structure PropertyKindCalculus.Quantity (k :
PropertyKindCalculus.KindOfProperty)
(R : Type) : Type
```

Proof for Definition 7.1.2

uses 2

Realized as structure `Quantity (k : KindOfProperty) (R : Type)` carrying a magnitude in R , with same-kind addition `Quantity.add : Quantity k R → Quantity k R → Quantity k R` ($R4$: the type forces both summands to share the kind). The algebraic operations come from the [carrier](#) typeclass on R ; which operators are even *admissible* is the orthogonal scale gate (the proved operator stratification, [operator monotonicity](#)).

Theorem 7.1.3.

group

uses 2

used by 2

Effort
medium

Tags

capstone

proved

Representation-parametric additivity (R10). The additivity laws on quantities — commutativity, associativity, and the zero unit — $q_1 + q_2 = q_2 + q_1$, $\quad (q_1 + q_2) + q_3 = q_1 + (q_2 + q_3)$, $\quad 0 + q = q$ hold over every lawful carrier, established by a single proof and so available at each R at once: at $\mathbb{R}$ (the proof carrier) and at Int with no R -specific argument. The very carrier where these laws are absent — an executable Float , a [carrier](#) but not a lawful one — is what the *exec/spec refinement bridge* below exists to reconcile. Builds on [the quantity layer](#).

Lean code for Theorem 7.1.3.

1 theorem

```
theorem · defined in PropertyKindCalculus/
                        Quantity.lean
```

complete

```
theorem PropertyKindCalculus.Quantity.laws_parametric
{k : PropertyKindCalculus.KindOfProperty} {R : Type}
```

```

[PropertyKindCalculus.LawfulCarrier R]
(h : PropertyKindCalculus.DifferenceKind k)
(x y z : PropertyKindCalculus.Quantity k R) :
PropertyKindCalculus.Quantity.add h x y =
  PropertyKindCalculus.Quantity.add h y x ∧
PropertyKindCalculus.Quantity.add h
  (PropertyKindCalculus.Quantity.add h x y) z =
PropertyKindCalculus.Quantity.add h x
  (PropertyKindCalculus.Quantity.add h y z) ∧
PropertyKindCalculus.Quantity.add h
  PropertyKindCalculus.Quantity.zero x =
x ∧
PropertyKindCalculus.Quantity.add h x
  PropertyKindCalculus.Quantity.zero =
x

```

Proof for Theorem 7.1.3

uses 0

Proved in `PropertyKindCalculus.Quantity` (axiom-free). Each law unfolds `Quantity.add` to its carrier operation and rewrites by the corresponding `LawfulCarrier` field; `laws_parametric` bundles the four. Specializing to `Int` (core) and $\mathbb{R}$ (in the `Dimension` library) reuses the same proof verbatim, which is the R10 payoff.

Definition 7.1.4.

group

statement">state

A *carrier refinement* `CarrierRefinement E S` specifies how an exec carrier `E` (fast, lossy) stands in for a *spec* carrier `S` (exact): a forgetful map `toSpec : E → S` (the analogue of `TorchLean's toReal`) and a spec-side `round : S → S`, with the single law that forgetting an exec addition equals rounding the spec addition, $\mathrm{toSpec}(x +_E y) = \mathrm{round}(\mathrm{toSpec}(x +_S \mathrm{toSpec}(y)))$. In plain engineering terms it is the *rounding contract* between a slow exact number type and the fast machine number that stands in for it. Builds on the [carrier](#).

Lean code for Definition 7.1.4.

1 definition

class (4 methods) · defined in PropertyKindCalculus/QuantityRefinement.lean

complete

```

class PropertyKindCalculus.CarrierRefinement (E S : Type)
  [PropertyKindCalculus.Carrier E] [PropertyKindCalculus.Carrier S] :
  Type

```

Proof for Definition 7.1.4

uses 0

Realized as the `CarrierRefinement` class (core, axiom-free). TorchLean's FP32 gives an *unconditional* instance over $\mathbb{R}$, and IEEE32Exec refines $\mathbb{R}$ on the finite path — both in the separately built Torch library.

Theorem 7.1.5.

group

uses 2

used by 0

✓

Effort	Priority	Tags
large	high	capstone proved

Exec refines spec across representations (R10). For a kind- k quantity, the exec sum viewed in the spec carrier is the rounding of the spec sum, $\mathrm{toSpec}(x +_E y) = \mathrm{round}(\mathrm{toSpec} \backslash, x +_S \mathrm{toSpec} \backslash, y)$, established over any [carrier refinement](#) and so holding at each at once. A law proved over the lawful spec carrier therefore transfers to the executable run with one rounding step — the bridge that carries the [representation-parametric laws](#) from the lawful $\mathbb{R}$ onto a float that is not lawful, so one kind-indexed value serves both proof and execution.

Lean code for Theorem 7.1.5.

1 theorem

theorem · defined in PropertyKindCalculus/QuantityRefinement.lean

complete

```

theorem PropertyKindCalculus.Quantity.add_refines
  {k : PropertyKindCalculus.KindOfProperty} {E S : Type}
  [PropertyKindCalculus.Carrier E] [PropertyKindCalculus.Carrier S]
  [PropertyKindCalculus.CarrierRefinement E S]
  (h : PropertyKindCalculus.DifferenceKind k)
  (x y : PropertyKindCalculus.Quantity k E) :
  (PropertyKindCalculus.Quantity.add h x y).toSpec =
    PropertyKindCalculus.Quantity.roundBy
      (PropertyKindCalculus.CarrierRefinement.round E)
      (PropertyKindCalculus.Quantity.add h x.toSpec y.toSpec)

```

Proof for Theorem 7.1.5

uses 0

Proved as `Quantity.add_refines` (core, axiom-free): the `CarrierRefinement` bridge law lifted along the kind index. It is instantiated concretely in the separately built Torch library at TorchLean's binary32 — an *unconditional* `CarrierRefinement FP32  $\mathbb{R}$` (the rounding spec has no overflow, so the law holds by computation) and a *conditional* `IEEE32Exec` refinement on the finite/no-overflow path (`Quantity.add_refines_exec`), where overflow is an explicit hypothesis rather than a silent failure. This is TorchLean's `BridgeFP32` pattern ("compute in $\mathbb{R}$, then round").

Definition7.1.6.

groupuses 3used by 0

Vector quantities: numerical array $\times$ one scalar unit (R11, ISO 80000-2 §18). A vector (or tensor) quantity is a *numerical* array carried at one kind with one *scalar unit* — not a per-coordinate collection of number $\times$ unit values; "all units are scalars". This is the *carrier* taken at a function-space type: $\text{Fin } n \rightarrow R$ is a (lawful) carrier whenever R is, pointwise, so a vector quantity is $\text{Quantity } k \ (\text{Fin } n \rightarrow R)$ and the *additivity laws* transfer to it by the same parametric proof. The quantity is coordinate-independent; only its numerical components depend on the frame.

Lean code for Definition7.1.6.

1 definition

def · defined in PropertyKindCalculus/QuantityVector.lean

complete

def PropertyKindCalculus.instLawfulCarrierPi {ι R : Type} [PropertyKindCalculus.LawfulCarrier R] : PropertyKindCalculus.LawfulCarrier (ι → R)

Proof for Definition 7.1.6

uses 0

Realized as the pointwise $\text{Carrier } (\iota \rightarrow R) / \text{LawfulCarrier } (\iota \rightarrow R)$ instances (core). The fuller structural apparatus — coordinate frames, tensor variance, transforms — remains a separate, owed axis.

Definition7.1.7.

groupuses 2statement">used by

A *unit* $\text{Unit } k$ is a distinguished *quantity* of kind k chosen as the reference: measuring expresses any quantity of kind k as $\text{\texttt{number}} \times \text{\texttt{unit}}$. Because Unit is indexed by the same k , "a metre" and "a unit of gravimetric water content" inhabit different types. This is the real-valued refinement of the proved *metrological unit*: it replaces the symbolic numeral-and-reference form with a magnitude in $\mathbb{R}$, so that conversion *ratios* become arithmetic.

Proof for Definition 7.1.7

uses 0

Planned. A $\text{Unit } k := \{ \text{ref} : \text{Quantity } k \}$ (or a chosen nonzero reference), with $\text{measure} : \text{Quantity } k \rightarrow \text{Unit } k \rightarrow \mathbb{R}$ giving the numeric value in that unit.

Theorem7.1.8.

groupstatement">use

Effort	Priority	Tags
medium	high	capstone planned

Unit conversion is a faithful round-trip. For two units u_1, u_2 of the same kind k , converting a quantity from u_1 to u_2 and back is the identity: $\text{convert}_{u_2} \circ \text{convert}_{u_1} = \text{id}$

$u_1 \backslash \text{bigl}(\mathrm{convert}_{u_1 \text{ to } u_2}(q) \backslash \text{bigr}) = q$. Conversion is multiplication by the ratio u_1 / u_2 , defined only within a kind. There is no conversion between units of different kinds — that is a type error, not a runtime check. Builds on [the unit definition](#).

Proof for Theorem 7.1.8

uses 0

Planned. The conversion factor is $r = \mathrm{measure}(u_1, u_2)$ and its inverse r^{-1} ; the round-trip is $r^{-1} \cdot (r \cdot q) = q$, requiring $r \neq 0$ from the chosen-reference nonzeroness. Ratio formation is licensed by the ratio scale.

Theorem 7.1.9.

group

uses 2

statement">used by

Effort	Priority	Tags
small	high	capstone proved

The dimension-1 disambiguation (the motivating capstone). There exist distinct kinds that map to the same dimension. Concretely, for volumetric and gravimetric water content, $\mathrm{vwc} \neq \mathrm{gwc} \quad \text{yet} \quad \dim(\mathrm{vwc}) = \dim(\mathrm{gwc}) = \mathbf{1}$, and likewise for permittivity and reflectivity. So the kind layer is a strict refinement of the dimension layer: $\dim$ is not injective. This is the statement PhysLib — and any dimension-only model, whether OWL2 or a representation-rooted hierarchy like SysML v2's — structurally cannot make. Uses [kind-of-property](#) and [the dimension map](#).

Lean code for Theorem 7.1.9.

1 theorem

theorem · defined in PropertyKindCalculus/Dimension.lean	complete
<pre>theorem PropertyKindCalculus.dim_not_injective : ∃ a b, a.kind ≠ b.kind ∧ a.toDimension = b.toDimension ∧ a.toDimension = 1</pre>	

Proof for Theorem 7.1.9

uses 0

Proved in the Dimension library. `vwc` and `gwc` are declared as dimensioned kinds with distinct kind ids, so `vwc.kind ≠ gwc.kind` is by `decide`; both carry `dim = 1`, so their dimensions agree by `rfl`. Together they witness $\exists a \ b, \ a.\mathrm{kind} \neq b.\mathrm{kind} \wedge \dim a = \dim b \wedge \dim a = 1$.

Definition 7.1.10.

group

uses 0

A decimal SI prefix `SIPrefix` is a name and symbol denoting a power-of-ten factor (e.g. `centi` = 10^{-2}). Applied to a unit it produces a multiple (VIM4 2CD 1.20) or submultiple (1.21) of that unit.

Lean code for Definition 7.1.10.

1 definition

```
structure (3 fields) · defined in PropertyKindCalculus/
UnitPrefix.lean
```

complete

```
structure PropertyKindCalculus.SIPrefix : Type
```

Proof for Definition 7.1.10

uses 0

A structure `SIPrefix` carrying name, symbol, and an integer exponent (the factor is 10^{exponent}); the full SI set (quetta ... quecto) is enumerated as defs.

7.2. Verified classification and instantiable kind-laws (R12)

A quantity `Quantity k R` records its kind `k` as a *tag*: the magnitude is an arbitrary `R`, so a value can be labelled with a kind it has not earned (`{999}` type-checks as an area). R12 upgrades a classification from an *assertion* to a *certificate*: a quantity is classified under a kind by a proof that it satisfies the kind's *defining relation* — the formalized ISO 80000 *Remark* — to quantities of the kinds it is built from. The kind becomes a *refinement* (a tag plus its earned proof), not a label.

The payoff is *instantiation*. Every kind-law is stated as a carrier-parametric universal over quantities with the kind-relation as a premise; "instantiating it at the quantity level" is then ordinary application, and any property of a kind's defining relation transports to *every* quantity certified under it. This is the inter-kind generalization of the carrier-parametric laws of R10.

Definition 7.2.1.

group

uses 0

statement">used by

A *product kind-law* `ProductKind k1 k2 k` records that `k` is the product kind of `k1` and `k2`. In the core it carries the scale precondition — all three are ratio-scale, since only ratio quantities multiply (Dybkaer §13.3.5); the dimension layer refines it with the dimensional equation $k.\text{dim} = k_1.\text{dim} \cdot k_2.\text{dim}$.

Lean code for Definition 7.2.1.

1 definition

```
structure (3 fields) · defined in PropertyKindCalculus/
QuantityClassification.lean
```

complete

```
structure PropertyKindCalculus.ProductKind
(k1 k2 k : PropertyKindCalculus.KindOfProperty) : Prop
```

Proof for Definition 7.2.1

uses 0

Realized as structure `ProductKind` with three `IsRational` fields. A concrete law such as `area = length · length` is the value `(rfl, rfl, rfl)`.

Definition 7.2.2.

group

statement" > uses 1 by 0

✓

The *certificate* `q.IsProduct h a b` states that `q` (of kind `k`) is the product of `a` (of `k1`) and `b` (of `k2`): its magnitude is the product of theirs. A proof of this certifies `q`'s classification, rather than asserting it. It is a separate proposition, so `Quantity` stays a clean tag and certificates are carried only when needed. The smart constructor `Quantity.mul` yields a quantity whose certificate holds by construction.

Lean code for Definition 7.2.2.

1 definition

complete

def · defined in PropertyKindCalculus/QuantityClassification.lean

```
def PropertyKindCalculus.Quantity.IsProduct {R : Type} [Mul R]
  {k1 k2 k : PropertyKindCalculus.KindOfProperty}
  (_h : PropertyKindCalculus.ProductKind k1 k2 k)
  (q : PropertyKindCalculus.Quantity k R)
  (a : PropertyKindCalculus.Quantity k1 R)
  (b : PropertyKindCalculus.Quantity k2 R) : Prop
```

Proof for Definition 7.2.2

uses 0

`Quantity.IsProduct h q a b := q.magnitude = a.magnitude * b.magnitude` over a carrier with multiplication; `Quantity.mul h a b := (a.magnitude * b.magnitude)`, and `mul_isProduct` is `rfl`.

Theorem 7.2.3.

group

uses 0

used by 0

✓

Effort Tags

small

proved

A kind-law that instantiates at the quantity level. A quantity certified as the product of `a` and `b` is unique — and any certified product equals the smart-constructed one (`eq_mul_of_isProduct`, *canonicity*). Supplying concrete quantities gives the quantity-level fact by application. Axiom-free.

Lean code for Theorem 7.2.3.

1 theorem

complete

theorem · defined in PropertyKindCalculus/QuantityClassification.lean

```
theorem PropertyKindCalculus.Quantity.isProduct_unique {R : Type} [Mul R]
  {k1 k2 k : PropertyKindCalculus.KindOfProperty}
  (h : PropertyKindCalculus.ProductKind k1 k2 k)
```

```

{q q' : PropertyKindCalculus.Quantity k R}
{a : PropertyKindCalculus.Quantity k1 R}
{b : PropertyKindCalculus.Quantity k2 R}
(hq : PropertyKindCalculus.Quantity.IsProduct h q a b)
(hq' : PropertyKindCalculus.Quantity.IsProduct h q' a b) : q = q'

```

Proof for Theorem 7.2.3

uses 0

isProduct_unique rewrites both magnitudes to a.magnitude * b.magnitude and closes by structure η; eq_mul_of_isProduct is isProduct_unique against mul_isProduct. Both depend on no axioms.

Definition 7.2.4.

group

uses 0

used by 9

A *quotient kind-law* QuotientKind k₁ k₂ k records that k is the quotient kind $k = k_1 / k_2$ (speed = length / duration; plane angle = arc / radius, where the lengths cancel). Like the product, it carries the ratio-scale precondition in the core and is refined by $k.\text{dim} = k_1.\text{dim} / k_2.\text{dim}$ in the dimension layer. Its certificate Quantity.IsQuotient and smart constructor Quantity.div mirror the product family exactly, over the core Div class.

Lean code for Definition 7.2.4.

1 definition

```

structure (3 fields) · defined in PropertyKindCalculus/
                        QuantityClassification.lean

```

complete

```

structure PropertyKindCalculus.QuotientKind
  (k1 k2 k : PropertyKindCalculus.KindOfProperty) : Prop

```

Proof for Definition 7.2.4

uses 0

structure QuotientKind with three IsRational fields; Quantity.div h a b := {a.magnitude / b.magnitude}, div_isQuotient is rfl, and isQuotient_unique / eq_div_of_isQuotient are the product family's proofs transcribed. Axiom-free.

Definition 7.2.5.

group

uses 0

statement">used by

A *reciprocal kind-law* ReciprocalKind k₁ k records that $k = 1 / k_1$ (frequency = 1 / period; curvature = 1 / radius; repetency = 1 / wavelength) — the unary special case, over the core Inv class, refined by $k.\text{dim} = (k_1.\text{dim})^{-1}$. All three families (product, quotient, reciprocal) stay Mathlib-free, using only toolchain-level arithmetic classes.

Lean code for Definition 7.2.5.

1 definition

```

structure (2 fields) · defined in PropertyKindCalculus/
                        QuantityClassification.lean

```

complete

```
structure PropertyKindCalculus.ReciprocalKind
  ( $k_1$   $k$  : PropertyKindCalculus.KindOfProperty) : Prop
```

Proof for Definition 7.2.5

uses 0

```
structure ReciprocalKind with two IsRational fields; Quantity.recip h a := {a.magnitude-1},
  recip_isReciprocal is rfl, with isReciprocal_unique / eq_recip_of_isReciprocal as before. Axiom-
  free.
```

8. Dimension as a Forgetful Functor

PropertyKindCalculus does not discard PhysLib's `Dimension`; it keeps it as a forgetful functor. `Dimension` answers the coarse question — "are these even commensurable in the SI base quantities?" — and the kind layer answers the fine one. The forgetful map, its multiplicativity, and the dimension-1 disambiguation it makes possible are now *realized*, sorry-free, in the `Dimension` library; the full coherence law — that `\dim` is a homomorphism over the interaction algebra — is realized in the `Interaction` module and stated as the homomorphism capstone below.

Two terms recur below; in plain engineering terms they mean this. A *forgetful functor* is a deliberately lossy, one-way map: `\dim` keeps only a kind's SI base-quantity exponents (length, mass, time, ...) and discards everything else that distinguishes it, so many kinds share one dimension and there is no inverse. A *homomorphism* is a map that respects an operation: here `\dim` respects multiplication — the dimension of a product is the product of the dimensions, the exponents adding — which is exactly the bookkeeping of everyday dimensional analysis, stated and proved here as a law rather than performed by hand.

8.1. The forgetful map (Dybkaer Ch. 19)

Definition8.1.1.

groupuses 2used by 43

The *dimension map* `\dim` sends a *dimensioned kind* — a [kind-of-property](#) paired with a PhysLib `Dimension` ($\mathbb{Q}$ -exponents over length/time/mass/charge/temperature) — to that dimension. It is deliberately many-to-one: that is what makes [the dimension-1 disambiguation](#) true.

Lean code for Definition8.1.1.

1 definition

def · defined in PropertyKindCalculus/Dimension.lean

complete

```
def PropertyKindCalculus.DimensionedKind.toDimension
  (dk : PropertyKindCalculus.DimensionedKind) : Dimension
```

Proof for Definition 8.1.1

uses 0

Realized in the `Dimension` library (`dimension/`). `DimensionedKind` is the structure pairing a kind with its `Dimension`; `toDimension` is the forgetful projection. PhysLib + Mathlib enter only here, as a separate library; the spine stays Mathlib-free.

Theorem8.1.2.

groupuses 2statement">used by

Tags

proved

The forgetful functor preserves products. The dimension of a product of dimensioned kinds is the product of their dimensions, $\dim (a \cdot b) = \dim a \cdot \dim b$, the SI exponents adding — ordinary dimensional bookkeeping, proved as a law. This is the dimension-component of functoriality, stated for the provisional product; the full interaction-algebra version is [the homomorphism capstone](#) below. Builds on [the dimension map](#).

Lean code for Theorem 8.1.2.

1 theorem

theorem · defined in PropertyKindCalculus/Dimension.lean

complete

```
theorem PropertyKindCalculus.DimensionedKind.toDimension_times
  (a b : PropertyKindCalculus.DimensionedKind) :
  (a.times b).toDimension = a.toDimension * b.toDimension
```

Proof for Theorem 8.1.2

uses 0

The provisional product multiplies the dimension components (PhysLib's `Dimension.mul`, i.e. exponent addition), so the equation holds by `rfl`. The unit law $\dim 1 = 1$ holds likewise.

Theorem 8.1.3.

group

uses 2

used by 4

Effort
medium

Priority
high

Tags

capstone

proved

Dimensional coherence — $\dim$ is a homomorphism over the full interaction algebra. Whenever the interaction algebra says k_1 and k_2 combine to k_3 , their dimensions combine the same way: $\mathrm{KMul} \ k_1 \ k_2 \ k_3 \ ; \ ; \Longrightarrow \ ; \dim k_3 = \dim k_1 \cdot \dim k_2$. This is the theorem that licenses forgetting to the dimension layer without losing soundness: a kind-level product is always dimensionally consistent. It upgrades [the proved multiplicativity](#) from the provisional product to Flater's gated [kind product](#).

Lean code for Theorem 8.1.3.

1 theorem

theorem · defined in PropertyKindCalculus/Interaction.lean

complete

```

theorem PropertyKindCalculus.InteractionAlgebra.dim_homomorphism
  (A : PropertyKindCalculus.InteractionAlgebra)
  {a b c : PropertyKindCalculus.DimensionedKind} (h : A.KMul a b c) :
  c.toDimension = a.toDimension * b.toDimension

```

Proof for Theorem 8.1.3

uses 0

Realized in the Interaction module as `InteractionAlgebra.dim_homomorphism`. Coherence is a well-formedness field of every `InteractionAlgebra`, so the homomorphism is its projection: any algebra that can be formed at all already guarantees $\dim k_3 = \dim k_1 \cdot \dim k_2$ on every sanctioned edge. The content sits in the worked algebra, where the obligation is discharged by the Dimension-group computation, case by case.

9. Scale-spanning units — a third unit category

(R13)

The SI sorts its units into two categories — *base* (the kilogram, metre, second, ampere, kelvin, candela, mole) and *derived* (everything built from them). Finkelstein and Whitehead (*Eur. J. Phys.* **46** (2025) 035701) argue this two-way split is *insufficient*. Of the seven base units, only four — the kilogram, metre, second, and ampere — are genuinely *dimensionally independent*, each fixed by a natural constant (h , c , $\Delta\nu_{Cs}$, e). The other three are *not* dimensionally independent of those four: the kelvin has the dimension of *energy* ($M \cdot L^2 \cdot T^{-2}$, via the Boltzmann constant k_B); the candela has the dimension of *power* ($M \cdot L^2 \cdot T^{-3}$, via the luminous efficacy K_{cd}); and the mole is a dimensionless *number* (via the Avogadro constant N_A). Yet each is kept as a distinct, coherent unit because it carries a *human-selected coefficient* — chosen for historical significance and a size that spans vast scales without scientific notation (a temperature difference of 100 K, not 1.38×10^{-21} J per degree of freedom). The authors propose a third category for them: *scale-spanning units*.

R13 specifies that third category. Its point is the unit-layer twin of R1 (a kind is more than its dimension): *whether a unit is scale-spanning is not a function of its dimension alone*. ISO 80000-7 (the *Light and radiation* chapter) is where this lands on the standard — the candela and the mole appear there, and are handled by reduction, not by a new base generator.

9.1. A criterion the dimension layer can apply — and one place it cannot

Definition

9.1.1.

group

uses 0

used by 3

✓

A dimension is *mechanically reducible* when it is expressible in mass, length, and time alone — it carries no charge and no temperature exponent. This is the Finkelstein–Whitehead test for "not dimensionally independent of the four physical base units", specialized to the mechanical generators: such a unit needs no electromagnetic, thermal, luminous, or chemical base of its own.

Lean code for Definition 9.1.1.

1 definition

```
def · defined in PropertyKindCalculus/ScaleSpanning.lean

def PropertyKindCalculus.Dimension.MechanicallyReducible (d : Dimension) : Prop
```

complete

Proof for Definition 9.1.1

uses 0

Dimension.MechanicallyReducible d := d.charge = 0 ∧ d.temperature = 0. Power and energy are reducible (power_mechanicallyReducible, energy_mechanicallyReducible), as is dimension one; electric current is *not* (current_not_mechanicallyReducible — it carries the electromagnetic generator), and neither is thermodynamic temperature in this work's algebra (temperature_not_mechanicallyReducible).

Definition 9.1.2.

group

statement">statement">used by

The proposed three-way classification of SI units: *physical base* (the four dimensionally independent units kg, m, s, A), *scale-spanning* (the kelvin, candela, and mole), and *derived* (every plain product of powers, with no special coefficient). The [scale-spanning units](#) are the new middle category the two-way base/derived split omits.

Lean code for Definition 9.1.2.

1 definition

inductive (3 constructors) · defined in PropertyKindCalculus/
ScaleSpanning.lean

complete

inductive PropertyKindCalculus.UnitCategory : Type

Proof for Definition 9.1.2

uses 0

inductive UnitCategory | physicalBase | scaleSpanning | derived. Realized as a plain enumeration; a scale-spanning unit's category is scaleSpanning by construction.

Definition 9.1.3.

group

statement">used by 4

A *scale-spanning unit* pairs a [metrological unit](#) with its mechanical-dimension reduction and the human-selected coefficient that defines it. The worked set is the candela (reducing to power, via $K_{cd} = 683 \text{ lm/W}$), the mole (to dimension one, via N_A), and the kelvin (to energy, via k_B). The coefficient and its value are *citation locators* — the published constants — not reproduced normative text.

Lean code for Definition 9.1.3.

1 definition

structure (4 fields) · defined in PropertyKindCalculus/
ScaleSpanning.lean

complete

structure PropertyKindCalculus.ScaleSpanningUnit : Type

Proof for Definition 9.1.3

uses 0

structure ScaleSpanningUnit carries the unit, the reducesTo dimension, and the coefficient symbol and value. The three canonical instances candela, mole, kelvin are declared in the ScaleSpanning namespace (so their names do not collide with the standards parts that also declare them).

9.2. The requirement, proved

Theorem	9.2.1.	group	statement" >uses 1 by 0	✓
Effort small	Tags proved			
The candela reduces to power — the dimension layer can see it. Luminous intensity carries the dimension of power, which is mechanically reducible; so the dimension layer correctly detects that the candela is not dimensionally independent. The mole is likewise reducible (to dimension one). Uses mechanical reducibility.				
Lean code for Theorem	9.2.1.		1 theorem	
theorem · defined in	PropertyKindCalculus/ ScaleSpanning.lean		complete	
<pre>theorem PropertyKindCalculus.ScaleSpanning.candela_reduction_reducible : PropertyKindCalculus.Dimension.MechanicallyReducible PropertyKindCalculus.ScaleSpanning.candela.reducesTo</pre>				
Proof for Theorem 9.2.1			uses 0	
candela_reduction_reducible : Dimension.MechanicallyReducible candela.reducesTo, from power_mechanicallyReducible (power.charge = 0, power.temperature = 0, checked in the PhysLib dimension group). The companion mole_reduction_reducible covers dimension one. Axiom-free.				
Theorem	9.2.2.	group	uses 2	used by 0
Effort small	Tags proved			
The kelvin's scale-spanning character is invisible to the dimension layer. Its proposed reduction (energy, via k_B) is mechanically reducible, yet the dimension this work assigns thermodynamic temperature (θ) is not — so no dimensional analysis reveals the kelvin to be scale-spanning. The fact is carried at the unit-classification layer, through the coefficient k_B, exactly as a kind carries more than its dimension. Uses mechanical reducibility.				
Lean code for Theorem	9.2.2.		1 theorem	
theorem · defined in	PropertyKindCalculus/ ScaleSpanning.lean		complete	

```

theorem PropertyKindCalculus.ScaleSpanning.kelvin_reduction_invisible_to_dim
  PropertyKindCalculus.Dimension.MechanicallyReducible
    PropertyKindCalculus.ScaleSpanning.kelvin.reducesTo  $\wedge$ 
       $\neg$ PropertyKindCalculus.Dimension.MechanicallyReducible
        PropertyKindCalculus.Dim.temperature

```

Proof for Theorem 9.2.2

uses 0

kelvin_reduction_invisible_to_dimension : Dimension.MechanicallyReducible kelvin.reducesTo $\wedge$
 $\neg$ Dimension.MechanicallyReducible Dim.temperature, the first conjunct from
 energy_mechanicallyReducible, the second from temperature_not_mechanicallyReducible
 (Dim.temperature.temperature = 1 $\neq$ 0). Axiom-free.

Theorem 9.2.3.

group

statement" >uses 1 by 0

Effort

small

Tags

proved

The ampere is a genuine physical base — not scale-spanning. Electric current is not mechanically reducible: it carries the electromagnetic generator, so it is dimensionally independent. This is the contrast that makes the third category non-empty and distinct from the first — the ampere (IEC 80000-6) belongs to the base set, the candela (ISO 80000-7) does not. Uses [mechanical reducibility](#).

Lean code for Theorem 9.2.3.

1 theorem

theorem · defined in PropertyKindCalculus/
ScaleSpanning.lean

complete

```

theorem PropertyKindCalculus.ScaleSpanning.ampere_is_physicalBase :
   $\neg$ PropertyKindCalculus.Dimension.MechanicallyReducible
    PropertyKindCalculus.Dim.current

```

Proof for Theorem 9.2.3

uses 0

ampere_is_physicalBase : $\neg$ Dimension.MechanicallyReducible Dim.current, from
 Dim.current_charge = 1 $\neq$ 0. Axiom-free.

Theorem 9.2.4.

group

uses 2

used by 0

Effort

small

Priority

high

Tags

capstone

proved

R13 — the base/derived dichotomy is insufficient; scale-spanning is a third category, not a function of dimension. Two scale-spanning units disagree on mechanical reducibility — the candela's dimension (reducible) and the kelvin's actual dimension θ (not reducible) — so membership in the third category cannot be decided from the dimension. The classification is a genuine extra datum, the unit-layer analogue of R1. Uses [the unit categories](#) and [the scale-spanning units](#).

Lean code for Theorem 9.2.4.

1 theorem

theorem · defined in PropertyKindCalculus/ScaleSpanning.lean

complete

```
theorem PropertyKindCalculus.ScaleSpanning.scaleSpanning_not_determined_by_
  ∃ a b,
    PropertyKindCalculus.Dimension.MechanicallyReducible a.reducesTo ∧
    ¬PropertyKindCalculus.Dimension.MechanicallyReducible
      PropertyKindCalculus.Dim.temperature ∧
    b.unit = PropertyKindCalculus.ScaleSpanning.kelvin.unit
```

Proof for Theorem 9.2.4

uses 0

scaleSpanning_not_determined_by_dimension : ∃ a b : ScaleSpanningUnit,
 Dimension.MechanicallyReducible a.reducesTo ∧ ¬ Dimension.MechanicallyReducible
 Dim.temperature ∧ b.unit = kelvin.unit. witnessed by {candela, kelvin, ...}. Axiom-free. This is
 the requirement ISO 80000-7's candela and mole, and ISO 80000-5's kelvin, exercise on the standard.

10. The Interaction Algebra (Flater Appendix C)

This is the contribution beyond the VIM4 2CD and OML, and the reason Flater's note is titled *full tracking of kinds of quantities*. Dimensional analysis says torque and energy share a dimension ($\mathrm{M}\cdot\mathrm{L}^2\cdot\mathrm{T}^{-2}$), yet they are not the same kind and not interchangeable. The interaction algebra records which kinds legitimately combine, with the dimensional product as a coherence side-condition rather than the definition. It is now *realized*, sorry-free, in the Interaction module: the curated product, its division dual, the multiplication–division round-trip, and the worked torque-versus-energy instance.

10.1. The ternary product of kinds (Flater App. C)

Definition10.1.1.

groupuses 2used by 6

KMul $k_1\ k_2\ k_3$ holds when kind k_1 times kind k_2 yields kind k_3 . It is partial and curated: an application asserts the legitimate products of its [kinds](#), each carrying the dimensional coherence obligation discharged by [the homomorphism theorem](#).

Lean code for Definition10.1.1.

1 definition

abbrev · defined in PropertyKindCalculus/Interaction.lean

complete

```
abbrev PropertyKindCalculus.InteractionAlgebra.KMul
  (self : PropertyKindCalculus.InteractionAlgebra) :
    PropertyKindCalculus.DimensionedKind →
      PropertyKindCalculus.DimensionedKind →
        PropertyKindCalculus.DimensionedKind → Prop
```

Proof for Definition 10.1.1

proof">uses 1

Realized as the KMul field of `InteractionAlgebra`: a ternary relation over [dimensioned kinds](#), paired in the same structure with the well-formedness field `kMul_coherent` that requires every edge to satisfy $\dim k_3 = \dim k_1 \cdot \dim k_2$. The structure cannot be formed for an algebra that violates coherence.

Definition10.1.2.

groupstatement">uses 2by 2

KDiv $k_3\ k_2\ k_1$ is the division dual of [the kind product](#): k_3 divided by k_2 yields k_1 . Fuel-consumption $\times$ rainfall is the classic Flater example whose careless handling yields a category error rather than a number.

Lean code for Definition 10.1.2.

1 definition

def · defined in PropertyKindCalculus/Interaction.lean

complete

```
def PropertyKindCalculus.InteractionAlgebra.KDiv
  (A : PropertyKindCalculus.InteractionAlgebra)
  (c b a : PropertyKindCalculus.DimensionedKind) : Prop
```

Proof for Definition 10.1.2 uses 0
Realized as InteractionAlgebra.KDiv, defined as the product read backwards ($\mathrm{KDiv} \ c \ b \ a := \mathrm{KMul} \ a \ b \ c$), so it inherits the same partiality and coherence side-condition.

Theorem 10.1.3. group uses 3 used by 0

Effort
medium

Priority
high

Tags
capstone
proved

Multiplication and division are inverse on kinds. A product and its quotient hold together — the round-trip is an equivalence: $\mathrm{KMul} \ k_1 \ k_2 \ k_3 \ ; \Longleftrightarrow \ \mathrm{KDiv} \ k_3 \ k_2 \ k_1$. Together with [dimensional coherence](#) this is the calculus that OWL2 cannot host: an algebraic law with arithmetic side-conditions, proved rather than consistency-checked. Builds on [the kind product](#) and [the kind quotient](#).

Lean code for Theorem 10.1.3.

1 theorem

theorem · defined in PropertyKindCalculus/Interaction.lean

complete

```
theorem PropertyKindCalculus.InteractionAlgebra.kMul_iff_kDiv
  (A : PropertyKindCalculus.InteractionAlgebra)
  (a b c : PropertyKindCalculus.DimensionedKind) :
  A.KMul a b c  $\leftrightarrow$  A.KDiv c b a
```

Proof for Theorem 10.1.3 uses 0
Immediate from defining KDiv as the converse of KMul: the biconditional is `Iff.rfl`. The round-trip is the symmetry of that definition.

Theorem 10.1.4. group statement">uses 1 used by 0

Effort

small

Tags

proved

Worked instance. Torque times plane angle is energy, and energy is not torque, even though $\dim(\mathrm{torque}) = \dim(\mathrm{energy})$. Formally $\mathrm{KMul} \setminus \mathrm{torque} \setminus \mathrm{angle} \setminus \mathrm{energy}$ holds while $\mathrm{energy} \neq \mathrm{torque}$ — the small example that shows the algebra carrying information dimension alone discards. Uses [the kind product](#).

Lean code for Theorem 10.1.4.

1 theorem

theorem · defined in PropertyKindCalculus/
Interaction.lean

complete

```
theorem PropertyKindCalculus.torque_angle_work :
  PropertyKindCalculus.siMech.KMul PropertyKindCalculus.torqueKind
    PropertyKindCalculus.angleKind PropertyKindCalculus.energyKind ∧
  PropertyKindCalculus.energyKind.kind ≠
    PropertyKindCalculus.torqueKind.kind ∧
  PropertyKindCalculus.energyKind.toDimension =
    PropertyKindCalculus.torqueKind.toDimension
```

Proof for Theorem 10.1.4

uses 0

Proved in the Interaction module. torque, angle (dimension one), and energy are declared as dimensioned kinds, with torque and energy both given dimension force · length; the product edge torque_angle_energy is asserted in the worked siMech algebra and its coherence checked by the Dimension-group computation, while energy.kind ≠ torque.kind is by decide.

11. The Function Calculus and Complex-Valued Carriers (R12)

The product / quotient / reciprocal calculus covers the *algebraic* defining relations of the ISQ (area is length times length, speed is length over duration). Scientific models do two further things the algebra alone does not reach: they apply *functions* to quantities (roots, exponentials, logarithms, the trigonometric functions), and they take *complex values* (the relative permittivity of a lossy dielectric, an impedance, a phasor). This chapter adds the two carrier-level extensions that cover both — the **function calculus** (a carrier *capability*, plus kind-relations for functions) and the **complex carrier** (a carrier *construction*) — and the point of each is that the kind layer is left untouched: only what a magnitude can *do*, or what numbers it lives *in*, changes.

11.1. Functions on quantities

Definition11.1.1.

groupuses 0statement">used by

$\mathrm{MathCarrier} \backslash R$ is the **function operation surface** a kind-indexed magnitude needs — $\exp$, $\log$, $\sin$, $\cos$, the hyperbolic functions, $\sqrt{}$, $|\cdot|$, and the constant π — supplied once per representation type R . It is the function-layer analogue of the numeric carrier `Carrier`, and is by construction the *same* surface as the underlying tensor toolchain's elementary functions, so every carrier that supports them — $\mathbb{R}$, `Float`, the binary32 carriers — is a clean $\mathrm{MathCarrier}$.

Lean code for Definition11.1.1.

1 definition

class (10 methods) · defined in PropertyKindCalculus/QuantityFunction.lean

complete

```
class PropertyKindCalculus.MathCarrier (R : Type) : Type
```

Proof for Definition 11.1.1

uses 0

Realized as the `MathCarrier` class. The richer surface ($\tan$, the inverse trig, $\operatorname{atan2}$, cube root, rounding, rational power) is split off into `MathCarrierExt`, so the executable binary32 carrier — whose function vocabulary is deliberately smaller — is a `MathCarrier` without being a `MathCarrierExt`. The algebraic *identities* ($\sin^2 + \cos^2 = 1$, $\exp(a+b) = \exp a \cdot \exp b$) live in `LawfulMathCarrier`, which holds at $\mathbb{R}$ and *fails* in floating point — the same exec/spec gap the lawful numeric carrier draws for $+$. The `Float` instance is computable (the run carrier); the $\mathbb{R}$ instance is the noncomputable proof carrier in the PhysLib-backed Dimension layer.

Definition11.1.2.

groupstatement">used by 3

$\mathrm{PowerKind} \backslash p \backslash k_1 \backslash k$ records that raising a k_1 -quantity to the rational power p yields a k -quantity ($k = k_1^{\backslash p}$). In the core it carries only the ratio-scale gate (Dybæk §13.3.5 — only ratio quantities have powers); the [dimension](#) layer strengthens it to the exponent-scaling certificate $\mathrm{dim} \ k = (\mathrm{dim} \ k_1)^{\backslash p}$. This is the one function family that is *not* dimension-one-in-out — the dimension-scaling extension of multiplication and division.

Lean code for Definition 11.1.2.

1 definition

```
structure (2 fields) · defined in PropertyKindCalculus/QuantityFunction.lean
```

complete

```
structure PropertyKindCalculus.PowerKind (p : ℚ)
  (k₁ k : PropertyKindCalculus.KindOfProperty) : Prop
```

Proof for Definition 11.1.2

proof" > uses 1

Realized as the `PowerKind` structure (two ratio-scale fields), with the smart constructors [Quantity.sqrt](#) (the half-power), `Quantity.cbrt`, and `Quantity.rpow`. The exponent-scaling refinement is `DimPowerKind`, well-defined because `PhysLib's Dimension` carries $\mathbb{Q}$ exponents (`Pow Dimension ℚ`), making a half-power a genuine group element.

Definition 11.1.3.

group

uses 2

used by 5

Tags

proved

$\mathrm{Quantity.sqrt}$ is the **verified half-power**, the canonical [PowerKind \(1/2\)](#) smart constructor: $\mathrm{sqrt} \{ \backslash, \}$ of a k_1 yields a k , its magnitude $\mathrm{sqrt} \{ a \}$ taken in the carrier. It rides on the base [MathCarrier](#) (not the extended surface), so it runs even on the binary32 executable carrier.

Lean code for Definition 11.1.3.

1 definition

```
def · defined in PropertyKindCalculus/QuantityFunction.lean
```

complete

```
def PropertyKindCalculus.Quantity.sqrt
  {k₁ k : PropertyKindCalculus.KindOfProperty} {R : Type}
  [PropertyKindCalculus.MathCarrier R]
  (_h : PropertyKindCalculus.PowerKind (1 / 2) k₁ k)
  (a : PropertyKindCalculus.Quantity k₁ R) :
  PropertyKindCalculus.Quantity k R
```

Proof for Definition 11.1.3

uses 0

Realized as `Quantity.sqrt`, taking a `PowerKind (1/2) k₁ k` and returning `(\mathrm{MathCarrier.sqrt} \ a.\mathrm{magnitude})`; the projection `sqrt_magnitude` is `rfl`, so an erased computation is exactly the carrier's `\sqrt{\,}`.

Theorem 11.1.4.

group

uses 2

used by 0

✓

Effort

Tags

small

proved

*The square root of an area is a length — checked, not annotated. The half-power of the area dimension L^2 is the length dimension L : $\dim(\sqrt{\mathrm{area}}) = (\dim \mathrm{area})^{1/2} = \mathrm{L}$. The certificate [Quantity.sqrt](#) carries, here discharged in the *PhysLib Dimension group*. Builds on [the power law](#).*

Lean code for Theorem 11.1.4.

1 theorem

theorem · defined in PropertyKindCalculus/Function.lean

complete

```
theorem PropertyKindCalculus.sqrt_area_coherent :
  PropertyKindCalculus.DimPowerKind (1 / 2) PropertyKindCalculus.areaK
  PropertyKindCalculus.lengthKind
```

Proof for Theorem 11.1.4

uses 0

Proved in the `Function` module: `areaK` is declared with dimension L^2 and the goal `lengthKind.dim = areaK.dim ^ (1/2 : ℚ)` is closed component-wise by `norm_num` over the rational exponents of the `Dimension group`.

Definition 11.1.5.

group

uses 3

used by 0

✓

$\mathrm{TranscendentalKind} \setminus k_1 \setminus k$ licenses the smart constructors [for](#) `\exp`, `\log`, the hyperbolic and trigonometric functions. The core records the ratio-scale gate; the [dimension](#) layer strengthens it to "argument and result are dimension one" (Families C), and — for the trigonometric functions — to a *curated algebra* (the unary analogue of [the interaction algebra](#)) that keeps `\sin` on plane angles only. That a plane angle is dimension one yet a distinct *kind* is exactly why `\sin` of an angle is fine while `\sin` of a reflectivity is the category error a dimension-only system cannot forbid.

Lean code for Definition 11.1.5.

1 definition

structure (2 fields) · defined in PropertyKindCalculus/QuantityFunction.lean

complete

```

structure PropertyKindCalculus.TranscendentalKind
  (k₁ k : PropertyKindCalculus.KindOfProperty) : Prop

```

Proof for Definition 11.1.5

uses 0

Realized as TranscendentalKind with the constructors Quantity.exp / log / sin / cos / tanh (base surface) and tan / asin / acos / atan / atan2 (extended surface). The dimensional strengthening is DimTranscendental (tanh_lossTangent_coherent is the worked edge: the conductive-loss \tanh argument is dimensionless and so is its value), and the trigonometric curation is the Dimension-layer trig algebra.

11.2. Complex-valued carriers

Definition 11.2.1.

group

uses 0

used by 0

Tags

proved

$\mathrm{Complex} \setminus R$ is the **complexification of a carrier** — the functor $R \mapsto \mathrm{Complex} \setminus R$ that, given the real arithmetic of R , equips $(\mathrm{re}, \mathrm{im})$ with the complex arithmetic. A complex-valued quantity is $\mathrm{Quantity} \setminus k \setminus (\mathrm{Complex} \setminus R)$; the carrier varies $(\mathbb{R} \text{ to prove, } \text{Float to run})$ while the kind k does not.

This is a **fourth** point on the representation axis (R10): the three scalar carriers $(\mathbb{R} \text{ to prove, a binary32 rounding spec to certify, } \text{Float to run})$ are all real, and $\mathrm{Complex} \setminus R$ lifts each of them — $\mathrm{Complex} \setminus \mathbb{R} \text{ to prove, } \mathrm{Complex} \setminus (\text{binary32})$ to certify rounding, $\mathrm{Complex} \setminus \text{Float to run}$. It is a Carrier, and a LawfulCarrier whenever R is (complex addition is componentwise, so the additivity laws lift for free); it carries a full field $+, -, \times, \div$ but is *unordered*, so — unlike the real carriers — it sits outside the ordinal/interval/ratio carrier tower: complex magnitudes are compared by modulus, not ranked.

Lean code for Definition 11.2.1.

1 definition

```

structure (2 fields) · defined in PropertyKindCalculus/
                        Complex.lean

```

complete

```

structure PropertyKindCalculus.Complex (R : Type) : Type

```

Proof for Definition 11.2.1

uses 0

Realized as Complex R with the toolchain instances (+, -, $\times$, $\div$, negation, zero, one), a Carrier instance, and $|z|^2 / |z|$. Why parametric rather than Mathlib's $\mathbb{C}$? For the same reason the

carrier is a parameter at all: $\mathbb{C}$ is *noncomputable* (its parts are $\mathbb{R}$, which does not reduce — no $\mathbb{C}$ -valued program runs or compiles, yet the executable carrier must be Float-backed), and it is *hardwired to* $\mathbb{R}$ (there is no " $\mathbb{C}$ over Float"), whereas the representation axis needs complex over Float, over a binary32 rounding spec, and over $\mathbb{R}$ from one definition. As with the reals — where $\mathbb{R}$ is *one instance* of the carrier, not the universal one — Mathlib's $\mathbb{C}$ is the natural *proof-carrier target*: a ring isomorphism $\mathrm{Complex} \setminus \mathbb{R} \simeq \mathbb{C}$ (its home the Dimension layer) transports any property of complex quantities at $\mathbb{R}$ to $\mathbb{C}$ and borrows its analysis library, while the run side stays on $\mathrm{Complex} \setminus \mathrm{Float}$.

Definition 11.2.2.

group

uses 0

proof">used by 1 ✓

Tags

proved

The **principal complex square root**. The algebraic operations are total and single-valued, so they are plain instances; the square root is the exception — it has a branch cut along the negative real axis. Selecting the principal branch needs exactly one sign decision (the sign of the imaginary part, with $\operatorname{sign} 0 = +$), isolated as the one-field capability NegTest so the branch-free complex algebra needs only the toolchain.

Lean code for Definition 11.2.2.

1 definition

def · defined in PropertyKindCalculus/Complex.lean

complete

```
def PropertyKindCalculus.Complex.sqrt {R : Type} [One R] [Add R] [Sub R]
  [Mul R] [Div R] [Neg R] [PropertyKindCalculus.MathCarrier R]
  [PropertyKindCalculus.Complex.NegTest R]
  (z : PropertyKindCalculus.Complex R) : PropertyKindCalculus.Complex R
```

Proof for Definition 11.2.2

proof">uses 1

Realized as Complex.sqrt: $\sqrt{z} = \sqrt{\frac{|z| + \operatorname{re}}{2}} \setminus + \setminus;$
 $\operatorname{sign}(\operatorname{im}) \setminus, \sqrt{\frac{|z| - \operatorname{re}}{2}} \setminus, j$, the real and imaginary magnitudes two real [square roots](#) and the imaginary *sign* the one NegTest decision (Float deciding it computably, $\mathbb{R}$ classically). Complex is deliberately *not* made a full MathCarrier: a complex $\exp/\log/\sin$ suite would drag in a quadrant-aware angle and an unprovable lawfulness story the metrology does not need — the one transcendental the electromagnetic forward model uses is given directly and honestly as its own branch-tracked constructor.

Theorem 11.2.3.

group

uses 2

used by 0

✓

Effort

small

Tags

proved

The kind-tracked complex square root. The complex analogue of [Quantity.sqrt](#), licensed by the same [PowerKind \(1/2\)](#): the complex square root of a (dimension-one) relative permittivity is a (dimension-one) refractive index — a checked half-power crossing of kinds, now over a complex carrier. The complexification leaves the kind layer untouched, so this is the only new kind-level constructor the complex numbers require.

Lean code for Theorem 11.2.3.

1 definition

def · defined in PropertyKindCalculus/Complex.lean

complete

```
def PropertyKindCalculus.Quantity.csqrt
  {k₁ k : PropertyKindCalculus.KindOfProperty} {R : Type} [One R]
  [Add R] [Sub R] [Mul R] [Div R] [Neg R]
  [PropertyKindCalculus.MathCarrier R]
  [PropertyKindCalculus.Complex.NegTest R]
  (_h : PropertyKindCalculus.PowerKind (1 / 2) k₁ k)
  (a :
    PropertyKindCalculus.Quantity k₁ (PropertyKindCalculus.Complex R)) :
    PropertyKindCalculus.Quantity k (PropertyKindCalculus.Complex R)
```

Proof for Theorem 11.2.3

proof" > uses 1

Realized as `Quantity.csqrt`, the smart constructor wrapping [the principal complex square root](#) under the same `PowerKind (1/2)` certificate as the real half-power; named `csqrt` because the complex root is a genuinely distinct operation (it has a branch cut) even though its kind discipline is identical.

12. Extensivity

Dybckær §13.5 distinguishes *extensive* kinds (whose value over a whole is the sum over disjoint parts) from kinds where that fails. This is a quantified arithmetic law over the mereology of systems — again something a description logic cannot state. It is now *realized*, sorry-free, in the Mathlib-free Extensivity module: the decomposition mereology, the Extensive predicate, the induction that lifts a single split to the whole tree, and the volume-on-mixing counterexample.

12.1. Extensive and conditionally-extensive kinds (Dybckær §13.5)

Definition 12.1.1. group statement">uses by 2

A kind is *extensive* when, for any partition of a [system](#) into disjoint parts, the quantity of the whole equals the sum over the parts. This requires the mereological structure deferred from the foundations chapter.

Lean code for Definition 12.1.1.

1 definition

```
structure (2 fields) · defined PropertyKindCalculus/
in Extensivity.lean
complete

structure PropertyKindCalculus.Extensive (k :
PropertyKindCalculus.KindOfProperty)
(m : PropertyKindCalculus.Measurement) : Prop
```

Proof for Definition 12.1.1 uses 0

Realized as Extensive k m, a predicate over a kind k and a measurement m (the property value observed on each node of a Decomposition): it asserts that every part is measured as a value of kind k, and that the numeral on a disjoint union is the sum of the numerals on its two parts.

Theorem 12.1.2. group statement">uses by 0

Effort medium

Tags

capstone

proved

Extensive aggregation. For an extensive kind k and a system decomposed into parts s_1, \dots, s_n, \mathrm{value}_k\Bigl(\bigsqcup_i s_i\Bigr) = \sum_i \mathrm{value}_k(s_i). The

\forall\text{forall-quantified additivity law itself} — not a single instance — is the deliverable, and it is exactly what OWL2 has no vocabulary for. Builds on [extensive kinds](#).

Lean code for Theorem 12.1.2.

1 theorem

theorem · defined in PropertyKindCalculus/Extensivity.lean

complete

```
theorem PropertyKindCalculus.extensive_additive
  {k : PropertyKindCalculus.KindOfProperty}
  {m : PropertyKindCalculus.Measurement}
  (h : PropertyKindCalculus.Extensive k m)
  (d : PropertyKindCalculus.Decomposition) :
  (m d).numeral = PropertyKindCalculus.leafSum m d
```

Proof for Theorem 12.1.2

uses 0

Proved in the Extensivity module by induction on the Decomposition: at a leaf the law is reflexivity; at a union the single-split additive field plus the two induction hypotheses give the numeral of the whole as the sum of the leaf sums.

Theorem 12.1.3.

group

statement" > uses 1 by 0

Effort

Tags

small

proved

Counterexample — volume on mixing is not extensive. Volume fails the additivity law for a water/ethanol mixture: the volume of the mixture is strictly less than the sum of the component volumes. Stating the negation keeps the extensive predicate honest. Uses [extensive kinds](#).

Lean code for Theorem 12.1.3.

1 theorem

theorem · defined in PropertyKindCalculus/Extensivity.lean

complete

```
theorem PropertyKindCalculus.mixing_subadditive :
  (PropertyKindCalculus.volMix PropertyKindCalculus.mixture).numeral <
    (PropertyKindCalculus.volMix
      PropertyKindCalculus.waterPart).numeral +
    (PropertyKindCalculus.volMix
      PropertyKindCalculus.ethanolPart).numeral ^
```

$\neg \text{PropertyKindCalculus.Extensive}$ $\text{PropertyKindCalculus.volume}$
 $\text{PropertyKindCalculus.volMix}$

Proof for Theorem 12.1.3

uses 0

Proved in the Extensivity module. 50 mL of water and 50 mL of ethanol measure 96 mL when mixed;
 the strict inequality $96 < 50 + 50$ is by `decide`, and the same `volMix` measurement witnesses `\neg \`
`\mathrm{Extensive} \` by refuting its additive field on that two-part decomposition.

13. Write Once, Correctly

The chapters so far catalogue what PropertyKindCalculus *provides*. This one is about *method*: the order in which an application author reaches for those pieces when specifying a real model, and why the order is forced rather than a matter of taste. The running example is `PropertyKindCalculus.Examples.MinWriteOnce` — a small, self-contained soil-moisture retrieval in the project's own `Examples` library — read here only to *infer the recipe*. It is a deliberately simplified subset, keeping just the salient structure each step needs (a few kinds, a typed constant table, a dedicated kind, one row of a Cholesky factor) and omitting the full retrieval. Every line of it is a *checked fact*, built in the `Examples` library under `CI`, so the illustrations below are machine-verified; PropertyKindCalculus supplies the four primitives named below and the example composes them in this order.

The discipline has a slogan: *write once correctly, go fast automatically*. A quantity is authored once, carrying its kind, over an abstract numeric carrier; the kind layer is checked by the elaborator and then *erases* — projected away by `.magnitude` — to a bare floating-point kernel, bit for bit. Correctness lives in the types and is paid for at compile time, while the number that runs is the same one a hand-written kernel would compute. Getting the *authoring order* right is what makes that erasure sound: each layer must be in place before the one above it can type-check.

13.1. The four steps

13.1.1. Step 1 — Start from the measurement principle: declare the kinds

Definition13.1.1.1.groupuses 3used by 0X

Before any number, name *what each quantity is* by the examination that produces it. A [kind-of-property](#) is the common defining aspect of mutually comparable properties (Dybæk §6.19); two otherwise-comparable kinds are individuated by their [examination principle](#) — the defining aspect of the procedure that measures them, so that [a different principle forces a different kind](#). This is the layer a description logic partly has: it can record a class, but not the operator algebra or the laws the later steps add.

In the example this is the catalogue of bare kinds — the radar backscatter σ^0 , the optical $ndvi$, the surface reflectivity, and the Mironov dielectric fit-constant kinds — each declared as a `def of KindOfProperty`, named by the instrument or principle that examines it. That a different examination principle forces a different kind is itself checked there: monostatic and bistatic reflectivity share name, scale, and dimension yet are distinct kinds. A kind is *general* (a type); *this pixel's* backscatter is *individual* (a term, `Quantity backscatter  $\alpha$`). Fix the kinds first: everything above is indexed by them.

13.1.2. Step 2 — Pair each kind with its dimension

Definition13.1.2.1.groupuses 2used by 0X

Give each kind its dimension, and let dimension do only what it can. The [dimension map](#) is a forgetful functor — it keeps a kind's SI base-quantity exponents and discards the rest — so it answers only the coarse question, *are these even commensurable?* It [respects products](#), which is ordinary dimensional bookkeeping proved as a law, but it *does not decide* identity. This step matters precisely because the quantities that carry the retrieval are almost all dimension one — a volume fraction, a reflectivity, an index. Dimension equates them all; the kind layer keeps them apart. Pairing a kind with its dimension here is what lets a later product be checked for dimensional consistency *and* kind consistency at once.

In the example, volumetric and gravimetric water content (`vwc`, `gwc`) are both dimension one yet distinct kinds — a units library alone would silently accept one for the other, which is exactly the failure this layer removes. (The literal `dim`-functor collapse is the Mathlib-backed dimension layer, shown in the *Units* and *Dimension* chapters; here the point is made one level up, at the kind.)

13.1.3. Step 3 — Build kind-differentiated quantities; gate the arithmetic

Definition **13.1.3.1.** group uses 2 used by 0 x

Now write the model's data and forward maps as records of typed quantities, and let the [interaction algebra](#) gate the arithmetic: a product is formed only where the kinds sanction it (`KMul` for products, `KDiv for quotients`), so a wrong pairing is a compile error, not a wrong number. Same-kind addition is the gate on `+`; products and quotients change kind by the algebra. Authored over an abstract carrier α , the whole record re-represents at the proof carrier ($\mathbb{R}$), the rounding spec (`binary32`), and the executable (`Float`) by one map, preserving every kind.

The example's `MiniMironovCoeffs` α is the pattern: each Mironov-2009 fit constant is a field of its own kind — a dry refractive index `ndA0 : Quantity refractiveIndexKind  $\alpha$` , a relaxation time `taub0 : Quantity relaxTimeKind  $\alpha$` , a conductivity `sigbF0 : Quantity conductivityKind  $\alpha$` — so the elaborator refuses to pass a relaxation time where a conductivity is expected, even though all three erase to the same carrier. Its `MiniMironovCoeffs.map` re-represents the whole table at any carrier (the constants moved from `Int` to executable `Float` by one map), each constant keeping its kind.

13.1.4. Step 4 — Differentiate shared kinds: dedicate, or index by field

Definition **13.1.4.1.** group uses 3 used by 0 x

Finally, decide what to do about quantities that share a kind. There are two cases, and they go on *different axes*.

When two quantities *should* be different kinds because their measurement target differs — same quantity, different system or component — make them distinct with a [dedicated kind](#): holding the [component](#) (or system) apart [is enough to force distinct kinds](#), without inventing identity strings. In the example the bound-water and free-water static permittivities (`boundStaticPerm`, `freeStaticPerm`) are one relative-permittivity kind dedicated to two components, hence provably distinct.

When the algebra *forces* a genuinely shared kind, the leftover distinction is *positional*, not metrological, and belongs on a second axis carried by *named record fields* — not arrays. One row of the Cholesky factor — the example's `CholRowQ α` — is row-homogeneous: three fields `l20 l21 l22`, all Quantity reflectivity α , because $A = L L^{\mathsf{T}}$ makes every $L_{ik} L_{jk}$ land at the same kind regardless of the column — a kind that separated them would make the factorization's own row sums ill-typed. The remaining "which slot" distinction is held by the distinct field names and checked off the kind axis by an erasure `rfl`: the theorem `cholRowQ_magnitudes` proves the kinded row's `.magnitudes` are *definitionally* the bare-Float kernel's outputs, so a same-kind slot swap still type-checks but breaks parity. A `Fin 3`-indexed array would push that positional bookkeeping onto the type and discard the named slots; the record keeps both axes separable — kind by the elaborator, position by parity-to-reference.

13.2. The payoff

Authored in this order, the model is *one source*. The kinds make every operation in the forward physics and the normal-equations solve a checked equation; the dimension layer certifies commensurability without conflating dimension-one quantities; the interaction algebra gates the arithmetic; and dedication plus named fields carry the distinctions the kind axis cannot. Then `.magnitude` erases the whole overlay and the compiled kernel is the bare Float program — proved, by a definitional `rfl` for every input, to be the same numbers. *Write once correctly, go fast automatically.*

The worked example is `PropertyKindCalculus.Examples.MiniWriteOnce`, build-gated by the project's own CI, so every illustration above is a checked fact rather than a claim. The recipe scales to a full retrieval: the same four steps, composed in the same order, carry a complete SMAP–NISAR soil-moisture model — and because each step's facts are real declarations, a downstream *verifiable ATBD* can cite them by name with their proved-or-sorry status, keeping the document and the code in lockstep. `PropertyKindCalculus` supplies the four primitives this chapter names; a model composes them in this order.

14. ISO/IEC 80000 — The Standards Catalogue

The quantity-kinds and units specified here are grounded against the international standard series ISO/IEC 80000, *Quantities and units*. That series is **licensed and copyrighted**: no normative content from it is reproduced in this work. What is recorded is the *citable identity* of each part — body, number, part, edition, and publication date — so that every catalogued quantity-kind or unit can name its exact source. This chapter specifies that references layer; the chapters that follow specify one part's quantity-kinds and units at a time, each added as coverage of that part becomes available.

14.1. The references layer (citation metadata only)

Definition

14.1.1.

group

uses 0

state

A *standard reference* `StandardRef` is the bibliographic identity of one part: its publishing body (ISO or IEC), the series number (80000), the part number, the edition designation as printed, the publication date, and the part's English title. It carries **no** normative content — only what is needed to *cite* the source. Its `cite` renders the conventional form, e.g. IEC 80000-6, Edition 2.0, 2022-11.

Lean code for Definition

14.1.1.

1 definition

structure (6 fields) · defined in PropertyKindCalculus/Iso80000/References.lean

complete

```
structure PropertyKindCalculus.Iso80000.StandardRef : Type
```

Proof for Definition 14.1.1

uses 0

Realized as structure `StandardRef` (Mathlib-free) with a `StandardBody` (ISO/IEC) field and the printed edition, date, and title as strings, plus designation (ISO 80000-3) and `cite` (the full reference). The values are read off each licensed part's cover page; nothing normative is transcribed.

Definition

14.1.2.

group

statement">us

The *catalogue* is the list of the twelve catalogued parts in part order, each a [standard reference](#). It is the single place this work names *which editions* of ISO/IEC 80000 the catalogued quantity-kinds and units are grounded against, so the grounding is auditable rather than implicit.

Lean code for Definition

14.1.2.

1 definition

def · defined in PropertyKindCalculus/Iso80000/References.lean

complete

```
def PropertyKindCalculus.Iso80000.catalogue :  
  List PropertyKindCalculus.Iso80000.StandardRef
```

Proof for Definition 14.1.2

uses 0

Realized as catalogue : List StandardRef, the twelve definitions iso80000_1 ... iso80000_12 (with iec80000_6 the one IEC-published part). The edition is "Second edition" for every part except part 6's "Edition 2.0"; the dates are the cover dates of the licensed copies.

15. ISO 80000-3 — Space and Time

The first part whose quantity-kinds and units are specified in full is ISO 80000-3, *Space and time*. Every item — all of 3-1.1 ... 3-26.3 — is catalogued: length, the width/height/distance/radius family, area, volume, the angles, duration, the frequencies, velocity and speed, acceleration, the wave quantities, and the rest, each carrying its exact source as data: the part, the printed item designation, the principal quantity symbol, and the coherent SI unit symbol. Only **citation locators** are recorded; no normative content (definitions, remarks) from the licensed standard is reproduced. The defining *mathematics* of selected remarks is formalized in its own right, in the sibling modules linked below.

Part 3 turns out to be an unusually good proving ground for the calculus, because the standard itself is full of distinctions that a dimension-only or representation-only model cannot make: a dozen different length quantities, several dimensionless angles, and three families of quantities that share a dimension while remaining distinct kinds.

15.1. Quantity-kinds and units of ISO 80000-3

Definition

15.1.1.

group

statement">uses 1 by 0

✓

A catalogued kind pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "3-3"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream rather than kept in a comment.

Lean code for Definition

15.1.1.

1 definition

structure (5 fields) · defined in PropertyKindCalculus/ISO80000/Catalogue.lean

complete

```
structure PropertyKindCalculus.ISO80000.CataloguedKind : Type
```

Proof for Definition 15.1.1

uses 0

Realized as structure CataloguedKind over a DimensionedKind, with cite rendering e.g. ISO 80000-3, Second edition, 2019-10 item 3-3. The catalogue lists all 42 items in item order — length (3-1.1), area (3-3), volume (3-4), duration (3-9), speed (3-10.2), frequency (3-17.1), ... — each paired with its coherent SI unit symbol.

Theorem

15.1.2.

group

statement">uses 1 by 0

✓

Effort

Tags

small

proved

Area is L^2 (item 3-3). The catalogued area kind carries length-exponent two in PhysLib's dimension group — the standard's dimensional statement reproduced as a checked computation, not an annotation. Uses [the dimension map](#).

Lean code for Theorem 15.1.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part3.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.area_dim_length :
  PropertyKindCalculus.Iso80000.Part3.area.dim.length = 2
```

Proof for Theorem 15.1.2

uses 0

Proved as area_dim_length : area.dim.length = 2 in the Iso80000 library, discharged by Dim.area_length from the Dimension layer. The companion volume_dim_length (item 3-4) gives length-exponent three.

Theorem 15.1.3.

group

statement" > uses 1 by 0

Effort

Tags

small

proved

Speed is $L \cdot T^{-1}$ (item 3-10.2). The catalogued speed kind's dimension equals length divided by time — a checked computation in the dimension group. Uses [the dimension map](#).

Lean code for Theorem 15.1.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part3.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.speed_dim :
  PropertyKindCalculus.Iso80000.Part3.speed.dim =
    PropertyKindCalculus.Dim.length / PropertyKindCalculus.Dim.time
```

Proof for Theorem 15.1.3

uses 0

Proved as speed_dim : speed.dim = Dim.length / Dim.time, discharged by Dim.speed_eq. The catalogue also records curvature_dim_length = -1 (item 3-2, L^{-1}), frequency_dim_time = -1 (item 3-17.1, T^{-1}), and acceleration_dim_time = -2 (item 3-11, $L \cdot T^{-2}$).

Theorem 15.1.4.

group

statement" > uses 1 by 0

Effort	Tags
small	proved

The metre is a well-formed unit (item 3-1.1). Length is a ratio-scale kind, so it bears a [unit](#) (Dybkaer §13.3.5); the metre is one such chosen reference.

Lean code for Theorem 15.1.4.

1 theorem

theorem · defined in	PropertyKindCalculus/Iso80000/Part3.lean	complete
<pre>theorem PropertyKindCalculus.Iso80000.Part3.metre_wellFormed : PropertyKindCalculus.Iso80000.Part3.metre.WellFormed</pre>		

Proof for Theorem 15.1.4

uses 0

Proved as `metre_wellFormed : metre.WellFormed` via `KindOfProperty.rational_bears_unit`.

Theorem 15.1.5.

group

statement" > uses 1 by 0

Effort	Tags
small	proved

A metre and a second are not commensurable. Length and duration are distinct kinds, so their units cannot be compared by ratio — a type-level fact, not a runtime check. Uses [metrological units](#).

Lean code for Theorem 15.1.5.

1 theorem

theorem · defined in	PropertyKindCalculus/Iso80000/Part3.lean	complete
<pre>theorem PropertyKindCalculus.Iso80000.Part3.metre_second_not_commensurable ¬PropertyKindCalculus.Iso80000.Part3.metre.Commensurable PropertyKindCalculus.Iso80000.Part3.second</pre>		

Proof for Theorem 15.1.5

uses 0

Proved as `metre_second_not_commensurable : ¬ metre.Commensurable second` by unfolding `Commensurable` to the underlying kinds and `decide` (length's kind id differs from duration's).

15.2. The length family: distinguishing kinds by measurement principle

ISO 80000-3 lists width (3-1.2), height (3-1.3), thickness (3-1.4), diameter (3-1.5), radius (3-1.6), path length (3-1.7), distance (3-1.8), and more as *separate items* — yet every one of them has dimension L and ratio scale. The standard distinguishes them only in *prose*: a width is a transverse extent, a distance is a shortest path, and so on. A dimension-only or representation-only model cannot capture the difference at all; it sees one type, "a real number of metres", for all of them.

This is exactly requirement *R2*, and Part 3 is where it earns its keep on the real standard. Item 3-1.1, *length*, is the broad genus. Each later item is specified as a *species* of length, distinguished **not by fiat** (not merely by a different name) but by an explicit *measurement principle* — Dybkær's examination principle (§7.5), a genuine defining aspect. The species are then provably distinct kinds, while remaining *mutually comparable* as lengths; and the specialization edges mirror the standard's own "defined in terms of" links (a diameter is a width of a circle, a radius is half a diameter, a distance is a shortest path length).

Definition15.2.1.

groupuses 2used by 0

A *length species* is a ratio-scale [quantity](#)-kind of dimension L carrying an [examination principle](#) — this work's own terse descriptor of the measurement principle that distinguishes it (transverse extent, shortest path, centre-to-rim, ...), never the standard's normative definition. Width, height, thickness, diameter, radius, path length, distance, radial distance, the position and displacement vectors, the radius of curvature, and the wavelength are all species of the general length kind (item 3-1.1).

Lean code for Definition15.2.1.

1 definition

```
def · defined in PropertyKindCalculus/Iso80000/Part3.lean

def PropertyKindCalculus.Iso80000.Part3.lengthSpecies (id : String)
  (p : PropertyKindCalculus.ExaminationPrinciple) :
  PropertyKindCalculus.DimensionedKind
```

Proof for Definition 15.2.1

uses 0

lengthSpecies id p := { kind := { id, scale := .ratio, examPrinciple := some p.id }, dim := Dim.length }. The general length carries no examination principle; the direct-parent edges of the family are collected in an Edge relation mirroring the standard's definitional links (thickness and diameter via width, radius via diameter, distance via path length, ...).

Theorem15.2.2.

groupstatement">uses 1by 0

Effort

small

Tags

proved

Width and distance are distinct kinds — by measurement principle, not by fiat. A width (transverse extent) and a distance (shortest path) are different kinds because they are examined by different principles, proved through `distinct_of_examPrinciple` (§7.5) rather than by appealing to their `id` strings. Same dimension L , same scale, different kind. Uses [the examination principle](#).

Lean code for Theorem 15.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso800000/Part3.lean

complete

```
theorem PropertyKindCalculus.Iso800000.Part3.width_ne_distance :
  PropertyKindCalculus.Iso800000.Part3.width.kind ≠
    PropertyKindCalculus.Iso800000.Part3.distance.kind
```

Proof for Theorem 15.2.2

uses 0

`width_ne_distance` : `width.kind` $\neq$ `distance.kind`, from `KindOfProperty.distinct_of_examPrinciple` applied to the differing examination principles `transverse-extent` and `shortest-path` (a decide on the `Option String` links). The companion `width_examinedBy` checks the kind-to-principle link. Axiom-free.

Theorem 15.2.3.

group

statement"uses 1 by 0

Effort

small

Tags

proved

A radius specializes length — transitively. Through the chain `radius` $\sqsubseteq$ `diameter` $\sqsubseteq$ `width` $\sqsubseteq$ `length`, a radius is a length: specialization is the reflexive-transitive closure of the family's edges, a genuine preorder (a lattice, not just direct edges). Uses [specialization](#).

Lean code for Theorem 15.2.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso800000/Part3.lean

complete

```
theorem PropertyKindCalculus.Iso800000.Part3.radius_specializes_length :
  PropertyKindCalculus.Specializes
    PropertyKindCalculus.Iso800000.Part3.Edge
```

```
PropertyKindCalculus.Iso80000.Part3.radius.kind
PropertyKindCalculus.Iso80000.Part3.length.kind
```

Proof for Theorem 15.2.3

uses 0

radius_specializes_length, built from Specializes.of_edge ON radius_diameter, diameter_width, width_length composed by Specializes.trans. Axiom-free.

Theorem 15.2.4.

group

statement" > uses 1 by 0

Effort

small

Tags

proved

Comparability is preserved. Width and distance, though distinct kinds, remain mutually comparable — they share the super-kind length, so combining them is possible but only via an explicit up-cast, never silently. Uses [specialization](#).

Lean code for Theorem 15.2.4.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part3.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.width_distance_comparable :
  PropertyKindCalculus.MutuallyComparable
    PropertyKindCalculus.Iso80000.Part3.Edge
    PropertyKindCalculus.Iso80000.Part3.width.kind
    PropertyKindCalculus.Iso80000.Part3.distance.kind
```

Proof for Theorem 15.2.4

uses 0

width_distance_comparable : MutuallyComparable Edge width.kind distance.kind, with witness length.kind, width reaching it by one edge and distance by two (distance $\sqsubseteq$ path length $\sqsubseteq$ length). Axiom-free.

15.3. Dimension does not classify; the kind does

Part 3 alone is dense with *dimension collisions* — distinct kinds that the dimension functor cannot tell apart. Plane angle, rotational displacement, phase angle, solid angle, rotation, and the logarithmic decrement are *all* dimension one. Frequency, rotational frequency, angular frequency, angular velocity, and the damping coefficient are *all* T^{-1} . Velocity and speed are both $L \cdot T^{-1}$. The dimension is the same; the kind is not — and so the units differ too, a radian is not a steradian and a hertz is not a radian-per-second, though each pair shares a dimension.

Effort

small

Tags

proved

The dimension-1 disambiguation, on the standard. There exist distinct ISO 80000-3 kinds with the same dimension one — plane angle and solid angle witness it. No dimension-only type system can separate them; the kind layer does. This generalizes the calculus's `dim_not_injective` capstone to the published standard. Uses [the dimension map](#).

Lean code for Theorem15.3.1.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part3.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.iso80000_3_dim_one_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem15.3.1

uses0

```
iso80000_3_dim_one_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim ∧
a.dim = 1, witnessed by ⟨planeAngle, solidAngle, ...⟩ with planeAngle_ne_solidAngle (a decide) and
reflexivity. Axiom-free.
```

Effort

small

Tags

proved

A radian is not a steradian, though both are dimension one. The SI units of plane and solid angle are not commensurable, because their kinds differ — a type-level fact the dimension cannot see. The hertz and the radian-per-second are separated the same way (both T^{-1}). Uses [metrological units](#).

Lean code for Theorem15.3.2.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part3.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.radian_steradian_not_commensura
¬PropertyKindCalculus.Iso80000.Part3.radian.Commensurable
PropertyKindCalculus.Iso80000.Part3.steradian
```

Proof for Theorem 15.3.2

uses 0

radian_steradian_not_commensurable : $\neg$ radian.Commensurable steradian by unfolding to the underlying kinds and decide; hertz_radianPerSecond_not_commensurable is the T^{-1} companion.

Theorem 15.3.3.

group

statement">uses 1 by 0

Effort

Tags

small

proved

Dimension is not a classifier even away from dimension one. Distinct ISO 80000-3 kinds also share dimensionful dimensions — frequency and angular frequency are both T^{-1} , velocity and speed both $L \cdot T^{-1}$. Uses [the dimension map](#).

Lean code for Theorem 15.3.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part3.lean

complete

theorem PropertyKindCalculus.Iso80000.Part3.iso80000_3_dim_collision :
 $\exists a\ b, a.kind \neq b.kind \wedge a.dim = b.dim$

Proof for Theorem 15.3.3

uses 0

iso80000_3_dim_collision, witnessed by (frequency, angularFrequency, ...) with frequency_ne_angularFrequency (a decide) and reflexivity; velocity_ne_speed is the $L \cdot T^{-1}$ companion. Axiom-free.

15.4. Standard prefixes: the centimetre as *centi* · metre

A unit is rarely used alone — a length is reported in millimetres, kilometres, or centimetres, all *the same kind* as the metre but differing by a decimal factor. The VIM4 records this directly: §1.19 lists the SI prefixes (a factor 10^n with a name and a symbol), §1.20 and §1.21 define a *multiple* and a *submultiple* of a unit, and §1.22 the *conversion factor* between two units of the same kind. So the centimetre is not an independent unit named "cm": it is the *centi* submultiple of the metre, and its symbol and its 10^{-2} factor should *follow* from that, not be asserted by hand.

Definition 15.4.1.

group

statement">uses 0

A *prefixed unit* is an SIPrefix (a decimal factor 10^n with its name and symbol, the VIM4 §1.19 table) applied to a base [unit](#). It projects to a metrological unit of the *same kind* with the prefix symbol prepended ("c" ++ "m" = "cm"), and exposes its §1.22 conversion factor as the exponent and its §1.20/ §1.21 status as a multiple or submultiple.

Lean code for Definition 15.4.1.

1 definition

```
structure (2 fields) · defined in PropertyKindCalculus/UnitPrefix.lean
complete

structure PropertyKindCalculus.PrefixedUnit : Type
```

Proof for Definition 15.4.1 uses 0

Realized as structure SIPrefix (name, symbol, exponent : Int) with the full §1.19 table, and structure PrefixedUnit (siPrefix, base) with toUnit (the projection), conversionExponent, IsMultiple, and IsSubmultiple. The projection preserves the kind, so toUnit_wellFormed and commensurable_base follow by reflexivity.

Theorem 15.4.2. group statement">use

Effort

Tags

small

proved

The centimetre is the centi submultiple of the metre, factor 10^{-2} . Its symbol "cm" and its conversion exponent -2 are checked computations from the prefix and the base, and it stays commensurable with the metre. Uses [metrological units](#).

Lean code for Theorem 15.4.2.

1 theorem

```
theorem · defined in PropertyKindCalculus/Iso800000/Part3.lean
complete

theorem PropertyKindCalculus.Iso800000.Part3.centimetre_conversionExponent :
PropertyKindCalculus.Iso800000.Part3.centimetrePrefixed.conversionExponent
=
-2
```

Proof for Theorem 15.4.2 uses 0

centimetre is (metre.withPrefix SIPrefix.centi).toUnit. The facts centimetre_symbol = "cm", centimetre_conversionExponent = -2, centimetre_isSubmultiple, and centimetre_wellFormed are discharged by rfl / decide; metre_centimetre_commensurable holds because the prefix leaves the kind unchanged.

15.5. Capturing a Remark's mathematics: the surface element of an area (item 3-3)

Each ISO 80000 item carries a *Remarks* field, and those remarks often hold the quantity's defining mathematics. Item 3-3 (area) is the example: its remark gives the *surface element* of a surface in terms of the Gaussian coordinates u, v and the determinant g of the metric tensor (ISO 80000-2). Rather than transcribe the licensed text or reduce it to a dimensional annotation, this work *formalizes the relation* — as real differential-geometric objects with proved properties — so the mathematics behind the kind is itself checked.

Definition 15.5.1.

group

uses 0

used by 0

✓

The *first fundamental form* (the metric tensor of ISO 80000-2) at a point is the 2×2 Gram matrix $\langle t_i, t_j \rangle$ of the surface's two tangent vectors. It is symmetric and positive-semidefinite, so its determinant g is non-negative.

Lean code for Definition 15.5.1.

1 definition

def · defined in PropertyKindCalculus/Iso80000/Part3/AreaElement.lean

complete

```
def PropertyKindCalculus.Iso80000.Part3.AreaElement.firstFundamentalForm.
{u_1}
{E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace ℝ E]
(t : Fin 2 → E) : Matrix (Fin 2) (Fin 2) ℝ
```

Proof for Definition 15.5.1

uses 0

Realized as `firstFundamentalForm t := Matrix.gram ℝ t` over Mathlib's Gram-matrix library; symmetry is `isHermitian_gram` and `metricDet_nonneg` is the determinant of a positive-semidefinite matrix.

Definition 15.5.2.

group

statement" > uses 1 by 0

✓

The *surface element* $dA = \sqrt{g}$ is the area-element density at a point, and the *area* of a surface patch is its integral over the parameter region ($A = \iint \sqrt{g} \, du \, dv$). Uses [the dimension map](#) for the L^2 reading of the resulting kind.

Lean code for Definition 15.5.2.

1 definition

def · defined in PropertyKindCalculus/Iso80000/Part3/AreaElement.lean

complete

```
def PropertyKindCalculus.Iso80000.Part3.AreaElement.areaElement.{u_1}
  {E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace ℝ E]
  (t : Fin 2 → E) : ℝ
```

Proof for Definition 15.5.2

uses 0

areaElement t := Real.sqrt (metricDet t) and surfaceArea t s := $\int x \text{ in } s, \text{ areaElement } (t \ x)$.
surfaceArea_const shows a uniformly-parametrized patch has area element $\times$ (area of the parameter region).

Theorem 15.5.3.

group

uses 0

used by 0

Effort

small

Tags

proved

The element squares back to the determinant: $(dA)^2 = g$. The $\sqrt{}$ in $dA = \sqrt{g}$ is faithful — the element is exactly the square root of the determinant of the metric tensor.

Lean code for Theorem 15.5.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part3/AreaElement.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.AreaElement.areaElement_sq.
  {u_1}
  {E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace ℝ E]
  (t : Fin 2 → E) :
  PropertyKindCalculus.Iso80000.Part3.AreaElement.areaElement t ^ 2 =
    PropertyKindCalculus.Iso80000.Part3.AreaElement.metricDet t
```

Proof for Theorem 15.5.3

uses 0

areaElement_sq : areaElement t ^ 2 = metricDet t, from Real.sq_sqrt and the non-negativity of g.

Theorem 15.5.4.

group

uses 0

used by 0

Effort

medium

Tags

proved

A flat patch recovers its Euclidean area. For a surface lying in the plane the element equals $|\det A|$, the change-of-variables area density of the parametrization — so the $\sqrt{g}$ reading computes the actual area. This is the correctness anchor that pins the convention the prose remark leaves implicit.

Lean code for Theorem 15.5.4.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part3/AreaElement.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.AreaElement.areaElement_eq_abs_
(t : Fin 2 → EuclideanSpace ℝ (Fin 2)) :
  PropertyKindCalculus.Iso80000.Part3.AreaElement.areaElement t =
    |(Matrix.of fun i j => (t j).ofLp i).det|
```

Proof for Theorem 15.5.4 uses 0

areaElement_eq_abs_det : areaElement t = |(Matrix.of fun i j => t j i).det|, via
gram_eq_conjTranspose_mul (so g = (det A)²) and Real.sqrt_sq_eq_abs.

Theorem 15.5.5. group uses 0 used by 0

Effort

Tags

small

proved

Regularity. The element is strictly positive exactly when the two tangent vectors are linearly independent — a non-degenerate (regular) point. A vanishing element marks a degenerate parametrization.

Lean code for Theorem 15.5.5.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part3/AreaElement.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.AreaElement.areaElement_pos_iff
{u_1}
{E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace ℝ E]
(t : Fin 2 → E) :
  0 < PropertyKindCalculus.Iso80000.Part3.AreaElement.areaElement t ↔
    LinearIndependent ℝ t
```

Proof for Theorem 15.5.5 uses 0

areaElement_pos_iff : 0 < areaElement t ↔ LinearIndependent ℝ t, from Real.sqrt_pos and
det_gram_ne_zero_iff_linearIndependent.

15.6. The volume element of a region (item 3-4)

The volume item carries the analogous remark — the *volume element* $dV = \sqrt{g}$ for the 3×3 metric tensor of a three-parameter region. It is the three-dimensional analogue of the area construction, the same Gram-determinant raised one rank, and it is formalized the same way.

Definition 15.6.1. group uses 0
The *volume element* $dV = \sqrt{g}$ is the volume-element density of a three-parameter region, and the *volume* is its integral over the parameter region ($V = \iiint \sqrt{g} \, du \, dv \, dw$). It squares back to g , is positive exactly at regular points, and (the correctness anchor) equals $|\det A|$ for a flat region — so it integrates to the actual Euclidean volume.

Lean code for Definition 15.6.1. 1 definition

```
def · defined in PropertyKindCalculus/Iso80000/Part3/VolumeElement.lean
complete

def PropertyKindCalculus.Iso80000.Part3.VolumeElement.volumeElement.{u_1}
  {E : Type u_1} [NormedAddCommGroup E] [InnerProductSpace ℝ E]
  (t : Fin 3 → E) : ℝ
```

Proof for Definition 15.6.1 uses 0
volumeElement t := Real.sqrt (metricDet t) over the 3×3 metricTensor t := Matrix.gram ℝ t for t : Fin 3 → E; regionVolume t s := ∫ x in s, volumeElement (t x) with regionVolume_const the constant-frame case. The proofs mirror AreaElement at one higher rank.

Theorem 15.6.2. group uses 0

Effort	Tags
medium	proved

A flat region recovers its Euclidean volume. For a region in Euclidean 3-space the element equals $|\det A|$, the change-of-variables volume density — the three-dimensional correctness anchor, confirming the $\sqrt{g}$ reading against the volume it must reproduce.

Lean code for Theorem 15.6.2. 1 theorem

```
theorem · defined in PropertyKindCalculus/Iso80000/Part3/VolumeElement.lean
complete
```

```

theorem PropertyKindCalculus.Iso80000.Part3.VolumeElement.volumeElement_eq_
  (t : Fin 3 → EuclideanSpace ℝ (Fin 3)) :
  PropertyKindCalculus.Iso80000.Part3.VolumeElement.volumeElement t =
    |(Matrix.of fun i j => (t j).ofLp i).det|

```

Proof for Theorem 15.6.2

uses 0

volumeElement_eq_abs_det : volumeElement t = |(Matrix.of fun i j => t j i).det| for t : Fin 3 → EuclideanSpace ℝ (Fin 3), via gram_eq_conjTranspose_mul and Real.sqrt_sq_eq_abs. The companion volumeElement_sq gives $(dV)^2 = g$.

15.7. The algebraic Remarks as kind-laws

Many Part-3 remarks state a quantity's defining relation *algebraically* — as a quotient or a reciprocal of other quantities. Curvature is $1/\rho$ (item 3-2); repetency is $1/\lambda$ (3-20); frequency is $1/T$ (3-17.1); speed is ds/dt , path length per duration (3-10.2); a plane angle is s/r , arc length per radius (3-5). Each is formalized as an R12 kind-law over the catalogued kinds, using the quotient and reciprocal families.

Theorem 15.7.1.

group

statement">us

Effort

Tags

small

proved

A plane angle is dimension one because it is a ratio of two lengths. From the remark $\alpha = s/r$ (arc over radius), the lengths cancel and the dimension is computed to be one — not stipulated. Yet plane angle remains a distinct kind from every other dimension-one quantity (the collision capstone above). Uses [the dimension map](#).

Lean code for Theorem 15.7.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part3/DefiningRelations.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part3.DefiningRelations.planeAngle_dim
  PropertyKindCalculus.Iso80000.Part3.planeAngle.dim =
    PropertyKindCalculus.Iso80000.Part3.pathLength.dim /
    PropertyKindCalculus.Iso80000.Part3.radius.dim

```

Proof for Theorem 15.7.1

uses 0

planeAngle_dim_from_arc_over_radius : planeAngle.dim = pathLength.dim / radius.dim, by
 reducing both sides to Dim.length / Dim.length and the lemma length_div_length ($a / a = 1$, by ext
 and simp).

Theorem 15.7.2.

group

statement">us

Effort

small

Tags

proved

Verified construction at the quantity level. A frequency built as the reciprocal of a period duration carries its classification certificate by construction, over the $\mathbb{R}$ carrier — and canonicity makes that certificate determine the quantity. Uses [the reciprocal kind-law](#).

Lean code for Theorem 15.7.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part3/DefiningRelations.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part3.DefiningRelations.frequencyOf_i
  (T :
    PropertyKindCalculus.Quantity
    PropertyKindCalculus.Iso80000.Part3.periodDuration.kind  $\mathbb{R}$ ) :
    PropertyKindCalculus.Quantity.IsReciprocal

PropertyKindCalculus.Iso80000.Part3.DefiningRelations.frequency_recip_peri
odDuration
  (PropertyKindCalculus.Iso80000.Part3.DefiningRelations.frequencyOf
    T)
  T
```

Proof for Theorem 15.7.2

uses 0

frequencyOf T := Quantity.recip frequency_recip_periodDuration T, and
 frequencyOf_isReciprocal is rfl; frequency_certificate_canonical is eq_recip_of_isReciprocal.
 The kind-law frequency_recip_periodDuration is (rfl, rfl) (both kinds ratio-scale). The same
 families give speedOf (speed = path length / duration) and planeAngleOf (plane angle = arc / radius).

15.8. Why it matters: classification and metrological consistency at the quantity level

Formalizing a remark's mathematics is not decoration. It earns its keep by turning the quantity-kind into a working *classifier* and by moving metrological checking down to the *quantity level*, where the type system enforces it.

Classifying quantities. The surface-area formula does not return a bare real number: its output is, by construction, a magnitude of the *one* catalogued kind area. Any quantity a downstream model computes this way — the area of a field, a footprint, a cross-section — is automatically *classified* as the same kind, dimension L^2 , unit the square metre. The kind is the genus under which every such quantity falls, and the defining relation is what places it there. So formalizing the remark gives users a principled answer to "what is this quantity?" — it is a value of area, because the relation that produced it is area's defining relation.

Verifying metrological consistency. Because the kind carries its dimension and its own scalar unit, the consistency checks happen at the quantity level and cannot be skipped. Two areas combine — addition is defined on the area carrier — but an area and a length are *different types*, so adding them does not type-check: the dimensionless numeric conflation a spreadsheet would silently allow is rejected. An area is not measurable in metres, because the square metre and the metre are not commensurable, a fact of the types rather than a runtime guard. And the dimensional invariant L^2 travels with every value of the kind. The remark's mathematics, the kind, and these consistency guarantees are thus one chain: the relation anchors the kind, the kind classifies the quantity, and the classification *is* the metrological check.

These claims are exercised as checked facts in the `DimensionExamples` library: a surface area built from the analytic element and classified as an area quantity, the L^2 invariant, the area-plus-area combination, and the rejected area-versus-length and square-metre-versus-metre conflations.

This is the **verified instantiation** requirement (R12): a classification is a certificate that a quantity satisfies its kind's defining relation, not a bare tag. For area, the defining relation is the surface element above, and it yields two concrete certificate forms.

Theorem15.8.1.

group

Effortsmall

Tags

proved

A rectangle's area is verified-classified by construction. *rectangleArea w h* is built as the product of two lengths (*area = length · length*), so its area-classification certificate holds by construction.

Lean code for Theorem 15.8.1.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part3/AreaClassification.lean

complete

theorem PropertyKindCalculus.Iso80000.Part3.AreaClassification.rectangleArea
 (w h :
 PropertyKindCalculus.Quantity
 PropertyKindCalculus.Iso80000.Part3.length.kind $\mathbb{R}$) :
 PropertyKindCalculus.Quantity.IsProduct

PropertyKindCalculus.Iso80000.Part3.AreaClassification.area_is_length_time
 s_length
 (PropertyKindCalculus.Iso80000.Part3.AreaClassification.rectangleArea
 w h)
 w h

Proof for Theorem 15.8.1

uses 0

rectangleArea w h := Quantity.mul area_is_length_times_length w h , and
rectangleArea_isProduct is rfl — the certificate of the constructed product.

Theorem 15.8.2.

group

uses 0

Effort

Tags

medium

proved

Property transport. Every certified surface area is non-negative — the surface element is ≥ 0 (a property of area's defining relation), instantiated at the quantity level through the certificate.

Lean code for Theorem 15.8.2.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part3/AreaClassification.lean

complete

theorem PropertyKindCalculus.Iso80000.Part3.AreaClassification.certified_su
 { q :
 PropertyKindCalculus.Quantity
 PropertyKindCalculus.Iso80000.Part3.area.kind $\mathbb{R}$ }
 { t : $\mathbb{R} \times \mathbb{R} \rightarrow \text{Fin } 2 \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } 2\text{)}$ } { s : Set ($\mathbb{R} \times \mathbb{R}$)}
 (h_s : MeasurableSet s)
 (h :

PropertyKindCalculus.Iso80000.Part3.AreaClassification.IsSurfaceArea
 $q \ t \ s) :$
 $0 \leq q.magnitude$

Proof for Theorem 15.8.2

uses 0

certified_surfaceArea_nonneg rewrites the certificate $q.magnitude = \text{surfaceArea } t \ s$ and applies $\text{surfaceArea_nonneg}$ (the integral of a non-negative element, by $\text{setIntegral_nonneg}$ and $\text{areaElement_nonneg}$).

15.9. Item index — ISO 80000-3

Every catalogued item of ISO 80000-3, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLib dimension this work assigns it. Each item number links to the formalized result it participates in — the length-family lattice, a dimension collision, a formalized remark, or the catalogue itself. Symbols and unit strings are **citation locators**; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
3-1.1	length	l	m	L
3-1.2	width	b	m	L
3-1.3	height	h	m	L
3-1.4	thickness	d	m	L
3-1.5	diameter	d	m	L
3-1.6	radius	r	m	L
3-1.7	path length	s	m	L
3-1.8	distance	d	m	L
3-1.9	radial distance	r_Q	m	L
3-1.10	position vector	r	m	L
3-1.11	displacement	Δr	m	L
3-1.12	radius of curvature	ρ	m	L
3-2	curvature	κ	m^{-1}	L^{-1}
3-3	area	A	m^2	L^2
3-4	volume	V	m^3	L^3
3-5	plane angle	α	rad	1
3-6	rotational displacement	φ	rad	1
3-7	phase angle	φ	rad	1
3-8	solid angle	Ω	sr	1
3-9	duration	t	s	T
3-10.1	velocity	v	m/s	$L \cdot T^{-1}$
3-10.2	speed	v	m/s	$L \cdot T^{-1}$
3-11	acceleration	a	m/s^2	$L \cdot T^{-2}$
3-12	angular velocity	ω	rad/s	T^{-1}
3-13	angular acceleration	α	rad/s^2	T^{-2}
3-14	period duration	T	s	T
3-15	time constant	τ	s	T
3-16	rotation	N	1	1

Item	Quantity	Symbol	Unit	Dimension
3-17.1	frequency	f	Hz	T^{-1}
3-17.2	rotational frequency	n	s^{-1}	T^{-1}
3-18	angular frequency	ω	rad/s	T^{-1}
3-19	wavelength	λ	m	L
3-20	repetency	σ	m^{-1}	L^{-1}
3-21	wave vector	k	m^{-1}	L^{-1}
3-22	angular repetency	k	m^{-1}	L^{-1}
3-23.1	phase velocity	c	m/s	$L \cdot T^{-1}$
3-23.2	group velocity	c_g	m/s	$L \cdot T^{-1}$
3-24	damping coefficient	δ	s^{-1}	T^{-1}
3-25	logarithmic decrement	Λ	1	1
3-26.1	attenuation	α	m^{-1}	L^{-1}
3-26.2	phase coefficient	β	rad/m	L^{-1}
3-26.3	propagation coefficient	γ	m^{-1}	L^{-1}

16. ISO 80000-4 — Mechanics

The second part specified in full is ISO 80000-4, *Mechanics*. Every item — all of 4-1 ... 4-32 — is catalogued: mass and the densities, momentum, the force family, impulse and the moments, pressure and stress, the strains and elastic moduli, the viscosities, power and the energies, the flows, and action — each carrying its exact source as data: the part, the printed item designation, the principal quantity symbol, and the coherent SI unit symbol. Only **citation locators** are recorded; no normative content (definitions, remarks) from the licensed standard is reproduced. The defining *mathematics* of selected remarks is formalized in the sibling module `Part4.DefiningRelations`.

Mechanics is where the two theses of the calculus bite hardest. The standard's own quantities force the issues: a force family that is a specialization lattice; the textbook torque-versus-energy dimension collision (both $M \cdot L^2 \cdot T^{-2}$, the newton metre reserved for one and the joule for the other); and a web of defining relations that build mechanical quantities out of Part-3 *space-and-time* quantities — momentum from mass and velocity, pressure from force and area.

16.1. Quantity-kinds and units of ISO 80000-4

Definition16.1.1.

groupstatement">uses1by0

A catalogued kind pairs a dimensioned quantity-kind with its exact source in the series — the part, the printed item designation (e.g. "4-9.1"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream rather than kept in a comment.

Lean code for Definition16.1.1.

1 definition

structure (5 fields) · defined inPropertyKindCalculus/Iso80000/Catalogue.lean

complete

structure PropertyKindCalculus.Iso80000.CataloguedKind : Type

Proof for Definition 16.1.1

uses0

Realized as structure CataloguedKind over a DimensionedKind, with cite rendering e.g. ISO 80000-4, Second edition, 2019-08 item 4-9.1. The catalogue lists all 54 items in item order — mass (4-1), momentum (4-8), force (4-9.1), torque (4-12.2), pressure (4-14.1), power (4-27), action (4-32), ... — each paired with its coherent SI unit symbol.

Theorem16.1.2.

groupstatement">uses1by0

Effort	Tags
small	proved

Force is $M \cdot L \cdot T^{-2}$ (item 4-9.1). The catalogued force kind carries time-exponent minus two in PhysLib's dimension group — the standard's dimensional statement reproduced as a checked computation, not an annotation. Uses [the dimension map](#).

Lean code for Theorem 16.1.2.

1 theorem

theorem	defined in	PropertyKindCalculus/Iso800000/Part4.lean	complete
theorem PropertyKindCalculus.Iso800000.Part4.force_dim_time : PropertyKindCalculus.Iso800000.Part4.force.dim.time = -2			

Proof for Theorem 16.1.2

uses 0

Proved as force_dim_time : force.dim.time = -2 in the Iso800000 library, discharged by norm_num over the PhysLib dimension lemmas. The companions energy_dim_length = 2 (item 4-28), pressure_dim_length = -1 (item 4-14.1), and power_dim_time = -3 (item 4-27) record the rest of the mechanical algebra.

Theorem 16.1.3.

group

statement" > uses 1 by 0

Effort	Tags
small	proved

A kilogram and a newton are not commensurable. Mass and force are distinct kinds, so their units cannot be compared by ratio — a type-level fact, not a runtime check. Uses [metrological units](#).

Lean code for Theorem 16.1.3.

1 theorem

theorem	defined in	PropertyKindCalculus/Iso800000/Part4.lean	complete
theorem PropertyKindCalculus.Iso800000.Part4.kilogram_newton_not_commensurable : ¬PropertyKindCalculus.Iso800000.Part4.kilogram.Commensurable PropertyKindCalculus.Iso800000.Part4.newton			

Proof for Theorem 16.1.3

uses 0

Proved as `kilogram_newton_not_commensurable` : $\neg$ `kilogram.Commensurable newton` by unfolding `Commensurable` to the underlying kinds and `decide`. The kilogram and the newton are each `WellFormed` (mass and force are ratio-scale).

16.2. The force family: distinguishing kinds by measurement principle

ISO 80000-4 lists weight (4-9.2), static friction force (4-9.3), kinetic friction force (4-9.4), rolling resistance (4-9.5), and drag force (4-9.6) as *separate items* — yet every one has dimension $M \cdot L \cdot T^{-2}$ and ratio scale. The standard distinguishes them only in *prose*: a weight acts in a gravitational field; a static friction force resists motion *before* a body slides; a kinetic friction force resists motion *while* it slides. A dimension-only or representation-only model cannot capture the difference; it sees one type, "a real number of newtons", for all of them.

This is requirement *R2* again, now on the force family — the exact pattern Part 3 set on the length family. Item 4-9.1, *force*, is the broad genus; each later item is specified as a *species* of force, distinguished **not by fiat** but by an explicit *measurement principle* (Dybæk's examination principle, §7.5). The species are then provably distinct kinds while remaining *mutually comparable* as forces.

Definition 16.2.1.

group

uses 2

used by 0

✓

A *force species* is a ratio-scale [quantity](#)-kind of dimension $M \cdot L \cdot T^{-2}$ carrying an [examination principle](#) — this work's own terse descriptor of the measurement principle that distinguishes it (gravitational-field action, pre-slip resistance, sliding resistance, ...), never the standard's normative definition. Weight, the static and kinetic friction forces, the rolling resistance, and the drag force are all species of the general force kind (item 4-9.1).

Lean code for Definition 16.2.1.

1 definition

def · defined in `PropertyKindCalculus/Iso80000/Part4.lean`

complete

```
def PropertyKindCalculus.Iso80000.Part4.forceSpecies (id : String)
  (p : PropertyKindCalculus.ExaminationPrinciple) :
  PropertyKindCalculus.DimensionedKind
```

Proof for Definition 16.2.1

uses 0

`forceSpecies id p` := { `kind` := { `id`, `scale` := `.ratio`, `examPrinciple` := `some p.id` }, `dim` := `MDim.force` }. The general force carries no examination principle; the direct-parent edges of the family are collected in an `Edge` relation, each species an edge to `force` (mirroring the standard's "force (item 4-9.1) ... acting/resisting ...").

Effort

small

Tags

proved

Static and kinetic friction forces are distinct kinds — by measurement principle, not by fiat. They share dimension $M \cdot L \cdot T^{-2}$ and are both "friction forces", yet are different kinds because they are examined under different conditions (resistance before sliding vs during sliding), proved through `distinct_of_examPrinciple` (§7.5). This is the force-family analogue of `width ≠ distance`. Uses [the examination principle](#).

Lean code for Theorem16.2.2.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part4.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part4.staticFriction_ne_kineticFriction :
  PropertyKindCalculus.Iso80000.Part4.staticFrictionForce.kind ≠
    PropertyKindCalculus.Iso80000.Part4.kineticFrictionForce.kind
```

Proof for Theorem16.2.2

uses0

`staticFriction_ne_kineticFriction` : `staticFrictionForce.kind ≠ kineticFrictionForce.kind`, from `KindOfProperty.distinct_of_examPrinciple` applied to the differing principles `pre-slip-resistance` and `sliding-resistance` (a decide on the `Option String` links). The companions `weight_ne_dragForce` and `weight_ne_force` separate the other species, and `weight_examinedBy` checks the kind-to-principle link. Axiom-free.

Effort

small

Tags

proved

Comparability is preserved. Weight and the drag force, though distinct kinds, remain mutually comparable — they share the super-kind force, so combining them is possible but only via an explicit up-cast, never silently. The companion `weight_specializes_force` records that a weight specializes force. Uses [specialization](#).

Lean code for Theorem16.2.3.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part4.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part4.weight_dragForce_comparable :
  PropertyKindCalculus.MutuallyComparable
    PropertyKindCalculus.Iso80000.Part4.Edge
    PropertyKindCalculus.Iso80000.Part4.weight.kind
    PropertyKindCalculus.Iso80000.Part4.dragForce.kind

```

Proof for Theorem 16.2.3

uses 0

weight_dragForce_comparable : MutuallyComparable Edge weight.kind dragForce.kind, with witness force.kind, each species reaching it by one edge. Axiom-free.

16.3. Dimension does not classify; the kind does — the torque/energy case

Part 4 holds the textbook example of a dimension collision. *Torque* (moment of force, 4-12.1/4-12.2) and *energy* (work, 4-28) are *both* $M \cdot L^2 \cdot T^{-2}$ — yet they are distinct kinds, measured in distinct units: the newton metre is reserved for torque, the joule for energy, precisely because they are different kinds. And Part 4 is dense with such collisions: momentum and impulse ($M \cdot L \cdot T^{-1}$); angular momentum, angular impulse, and action ($M \cdot L^2 \cdot T^{-1}$); pressure, stress, and the three elastic moduli ($M \cdot L^{-1} \cdot T^{-2}$); and a whole dimension-one family — the relative densities, the strains, the Poisson number, the friction factors, the drag coefficient, and the efficiency.

Theorem 16.3.1.

group

statement" > uses by 0

✓

Effort

Tags

small

proved

Torque is not energy, though both are $M \cdot L^2 \cdot T^{-2}$. There exist distinct ISO 80000-4 kinds with the same dimension — torque and energy witness it. No dimension-only type system can separate them; the kind layer does. This is the textbook case that a dimension does not classify a quantity. Uses [the dimension map](#).

Lean code for Theorem 16.3.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part4.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part4.iso80000_4_dim_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim

```

Proof for Theorem 16.3.1

uses 0

iso80000_4_dim_collision : $\exists a\ b : \text{DimensionedKind}, a.\text{kind} \neq b.\text{kind} \wedge a.\text{dim} = b.\text{dim}$,
witnessed by (torque, mechanicalEnergy, ...) with torque_ne_energy (a decide) and reflexivity. The
companion momentum_ne_impulse is the $M \cdot L \cdot T^{-1}$ case. Axiom-free.

Theorem 16.3.2.

group

statement">uses 1 by 0

✓

Effort

Tags

small

proved

A newton metre is not a joule, though both are $M \cdot L^2 \cdot T^{-2}$. The SI units of torque and energy are not commensurable, because their kinds differ — the unit-level form of the torque/energy collision, a type-level fact the dimension cannot see. Uses [metrological units](#).

Lean code for Theorem 16.3.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part4.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part4.newtonMetre_joule_not_commensur
  ¬PropertyKindCalculus.Iso80000.Part4.newtonMetre.Commensurable
    PropertyKindCalculus.Iso80000.Part4.joule
```

Proof for Theorem 16.3.2

uses 0

newtonMetre_joule_not_commensurable : $\neg$ newtonMetre.Commensurable joule by unfolding to the
underlying kinds and decide (torque's kind differs from the energy kind).

Theorem 16.3.3.

group

statement">uses 1 by 0

✓

Effort

Tags

small

proved

The dimension-1 disambiguation, on the standard. There exist distinct ISO 80000-4 kinds with the same dimension one — efficiency and the relative mass density witness it, alongside the strains, the Poisson number, and the friction factors. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem 16.3.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part4.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part4.iso80000_4_dim_one_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 16.3.3

uses 0

```
iso80000_4_dim_one_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim ∧
a.dim = 1, witnessed by {efficiency, relativeMassDensity, ...} with
efficiency_ne_relativeMassDensity (a decide) and reflexivity. Axiom-free.
```

16.4. The algebraic Remarks as kind-laws: mechanics built from space and time

Many Part-4 remarks state a quantity's defining relation *algebraically* — and, in mechanics, the other quantities are often *space-and-time* quantities from Part 3. Momentum is mass × velocity (4-8); pressure is force / area (4-14.1); mass density is mass / volume (4-2); specific volume is 1/ρ (4-3); kinematic viscosity is η/ρ (4-25); efficiency is output power / input power (4-29); a modulus of elasticity is stress / strain (4-19.1). Each is formalized as an R12 kind-law over the catalogued kinds, several composing a Part-4 kind out of Part-3 kinds.

Theorem 16.4.1.

group

statement">us

Effort

Tags

small

proved

Momentum is mass × velocity — a verified, cross-part construction. A momentum built as the product of a mass (Part 4) and a velocity (Part 3) carries its product certificate by construction, over the $\mathbb{R}$ carrier, and canonicity makes the certificate determine the quantity. Uses [the product kind-law](#).

Lean code for Theorem 16.4.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part4/DefiningRelations.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part4.DefiningRelations.momentumOf_is
(m :
  PropertyKindCalculus.Quantity
  PropertyKindCalculus.Iso80000.Part4.mass.kind ℝ)
(v :
  PropertyKindCalculus.Quantity
  PropertyKindCalculus.Iso80000.Part3.velocity.kind ℝ) :
PropertyKindCalculus.Quantity.IsProduct
```

```
PropertyKindCalculus.Iso80000.Part4.DefiningRelations.momentum_prod_mass_v
elocity
  (PropertyKindCalculus.Iso80000.Part4.DefiningRelations.momentumOf m
    v)
m v
```

Proof for Theorem 16.4.1

uses 0

momentumOf m v := Quantity.mul momentum_prod_mass_velocity m v, and momentumOf_isProduct is rfl; momentum_certificate_canonical is eq_mul_of_isProduct. The kind-law momentum_prod_mass_velocity : ProductKind mass.kind Part3.velocity.kind momentum.kind is (rfl, rfl, rfl), and momentum_dim_from_mass_velocity checks $M \cdot L \cdot T^{-1}$ follows from the relation. The same families give pressureOf (pressure = force / area) and efficiencyOf.

Theorem 16.4.2.

group

statement">us

Effort

Tags

small

proved

Efficiency is dimension one because it is a ratio of two powers. From the remark $\eta = P_{out}/P_{in}$, the powers cancel and the dimension is computed to be one — not stipulated, the mechanics analogue of the plane angle being a ratio of two lengths. Yet efficiency remains a distinct kind from every other dimension-one quantity (the collision capstone above). Uses [the dimension map](#).

Lean code for Theorem 16.4.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part4/DefiningRelations.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part4.DefiningRelations.efficiency_di
PropertyKindCalculus.Iso80000.Part4.efficiency.dim =
  PropertyKindCalculus.Iso80000.Part4.power.dim /
  PropertyKindCalculus.Iso80000.Part4.power.dim
```

Proof for Theorem 16.4.2

uses 0

efficiency_dim_from_power_ratio : efficiency.dim = power.dim / power.dim, by reducing both sides via div_self' (a / a = 1). The kind-law efficiency_quot_power_power is (rfl, rfl, rfl) and efficiencyOf builds a certified efficiency from two powers.

16.5. Item index — ISO 80000-4

Every catalogued item of ISO 80000-4, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLip dimension this work assigns it. Each item number links to the formalized result it participates in — the force-family lattice, the torque/energy collision, a cross-part defining relation, or the catalogue itself. Symbols and unit strings are **citation locators**; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
4-1	mass	m	kg	M
4-2	mass density	ρ	kg/m ³	M · L ⁻³
4-3	specific volume	v	m ³ /kg	M ⁻¹ · L ³
4-4	relative mass density	d	1	1
4-5	surface mass density	ρ_A	kg/m ²	M · L ⁻²
4-6	linear mass density	ρ_l	kg/m	M · L ⁻¹
4-7	moment of inertia	J	kg · m ²	M · L ²
4-8	momentum	p	kg · m/s	M · L · T ⁻¹
4-9.1	force	F	N	M · L · T ⁻²
4-9.2	weight	F _g	N	M · L · T ⁻²
4-9.3	static friction force	F _s	N	M · L · T ⁻²
4-9.4	kinetic friction force	F _k	N	M · L · T ⁻²
4-9.5	rolling resistance	F _{rr}	N	M · L · T ⁻²
4-9.6	drag force	F _D	N	M · L · T ⁻²
4-10	impulse	I	N · s	M · L · T ⁻¹
4-11	angular momentum	L	kg · m ² /s	M · L ² · T ⁻¹
4-12.1	moment of force	M	N · m	M · L ² · T ⁻²
4-12.2	torque	T	N · m	M · L ² · T ⁻²
4-13	angular impulse	H	N · m · s	M · L ² · T ⁻¹
4-14.1	pressure	p	Pa	M · L ⁻¹ · T ⁻²
4-14.2	gauge pressure	p _e	Pa	M · L ⁻¹ · T ⁻²
4-15	stress	σ	Pa	M · L ⁻¹ · T ⁻²
4-16.1	normal stress	σ_n	Pa	M · L ⁻¹ · T ⁻²
4-16.2	shear stress	τ	Pa	M · L ⁻¹ · T ⁻²
4-17.1	strain	ϵ	1	1
4-17.2	relative linear strain	ϵ	1	1
4-17.3	shear strain	γ	1	1
4-17.4	relative volume strain	θ	1	1

Item	Quantity	Symbol	Unit	Dimension
4-18	Poisson number	μ	1	1
4-19.1	modulus of elasticity	E	Pa	$M \cdot L^{-1} \cdot T^{-2}$
4-19.2	modulus of rigidity	G	Pa	$M \cdot L^{-1} \cdot T^{-2}$
4-19.3	modulus of compression	K	Pa	$M \cdot L^{-1} \cdot T^{-2}$
4-20	compressibility	κ	Pa^{-1}	$M^{-1} \cdot L \cdot T^2$
4-21.1	second axial moment of area	I_a	m ⁴	L ⁴
4-21.2	second polar moment of area	I_p	m ⁴	L ⁴
4-22	section modulus	Z	m ³	L ³
4-23.1	static friction factor	μ_s	1	1
4-23.2	kinetic friction factor	μ	1	1
4-23.3	rolling resistance factor	C_rr	1	1
4-23.4	drag coefficient	C_D	1	1
4-24	dynamic viscosity	η	Pa · s	$M \cdot L^{-1} \cdot T^{-1}$
4-25	kinematic viscosity	ν	m ² /s	$L^2 \cdot T^{-1}$
4-26	surface tension	γ	N/m	$M \cdot T^{-2}$
4-27	power	P	W	$M \cdot L^2 \cdot T^{-3}$
4-28.1	potential energy	V	J	$M \cdot L^2 \cdot T^{-2}$
4-28.2	kinetic energy	T	J	$M \cdot L^2 \cdot T^{-2}$
4-28.3	mechanical energy	E	J	$M \cdot L^2 \cdot T^{-2}$
4-28.4	mechanical work	W	J	$M \cdot L^2 \cdot T^{-2}$
4-29	efficiency	η	1	1
4-30.1	mass flow	j_m	kg/(m ² · s)	$M \cdot L^{-2} \cdot T^{-1}$
4-30.2	mass flow rate	q_m	kg/s	$M \cdot T^{-1}$
4-30.3	mass change rate	q_m	kg/s	$M \cdot T^{-1}$
4-31	volume flow rate	q_V	m ³ /s	$L^3 \cdot T^{-1}$
4-32	action	S	J · s	$M \cdot L^2 \cdot T^{-1}$

17. ISO 80000-5 — Thermodynamics

The third part specified in full is ISO 80000-5, *Thermodynamics*. Every item — all of 5-1 ... 5-36, including every sub-suffixed item (5-3.1 ... 5-3.3, 5-5.1/5-5.2, 5-6.1/5-6.2, 5-10.1/5-10.2, 5-16.1 ... 5-16.4, 5-17.1/5-17.2, 5-20.1 ... 5-20.5, 5-21.1 ... 5-21.5, 5-25.1/5-25.2) — is catalogued: the temperatures, the expansion and pressure coefficients, heat and the heat-transfer quantities, the heat capacities and entropy, the energy family (the thermodynamic potentials), the specific energies, the Massieu and Planck functions, efficiency, and the humidity quantities — each carrying its exact source as data: the part, the printed item designation, the principal quantity symbol, and the coherent SI unit symbol. Only **citation locators** are recorded; no normative content (definitions, remarks) from the licensed standard is reproduced. The defining *mathematics* of selected remarks is formalized in the sibling module `Part5.DefiningRelations`.

Thermodynamics brings the one axis Parts 3 and 4 could not show, and sharpens the two the calculus already pressed. It introduces the temperature base quantity θ , and with it the *scale-type* distinction — thermodynamic temperature is ratio-scale, but Celsius temperature, of the very same dimension, is only interval-scale (requirement *R6*, here on the standard). It holds the textbook *entropy-versus-heat-capacity* dimension collision (both J/K, down to the same unit string). And item 5-20 lays out the thermodynamic potentials — internal energy, enthalpy, the Helmholtz and Gibbs energies — as a specialization lattice over energy, mirroring Part 3's length family and Part 4's force family.

17.1. Quantity-kinds and units of ISO 80000-5

Definition17.1.1.

groupstatement">uses1by0

A *catalogued kind* pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "5-20.3"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream rather than kept in a comment.

Lean code for Definition17.1.1.

1 definition

structure (5 fields) · defined inPropertyKindCalculus/Iso80000/Catalogue.lean

complete

```
structure PropertyKindCalculus.Iso80000.CataloguedKind : Type
```

Proof for Definition 17.1.1

uses0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with cite rendering e.g. ISO 80000-5, Second edition, 2019-08 item 5-18. The catalogue lists all 54 items in item order — thermodynamic

temperature (5-1), heat (5-6.1), entropy (5-18), the thermodynamic potentials (5-20.1 ... 5-20.5), the dew-point temperature (5-36), ... — each paired with its coherent SI unit symbol.

Theorem 17.1.2.

group

statement" >uses 1 by 0

Effort

Tags

small

proved

Entropy is $M \cdot L^2 \cdot T^{-2} \cdot \theta^{-1}$ (item 5-18). The catalogued entropy kind carries temperature-exponent minus one in PhysLib's dimension group — the standard's dimensional statement reproduced as a checked computation over the temperature generator θ , not an annotation. Uses [the dimension map](#).

Lean code for Theorem 17.1.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part5.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part5.entropy_dim_temperature :
  PropertyKindCalculus.Iso80000.Part5.entropy.dim.temperature = -1
```

Proof for Theorem 17.1.2

uses 0

Proved as entropy_dim_temperature : entropy.dim.temperature = -1 in the Iso80000 library, discharged by norm_num over the PhysLib dimension lemmas for the temperature generator. The companions heatCapacity_dim_temperature = -1, heat_dim_length = 2 (item 5-6.1), and linearExpansionCoefficient_dim_temperature = -1 (item 5-3.1) record the rest of the thermodynamic algebra.

Theorem 17.1.3.

group

statement" >uses 1 by 0

Effort

Tags

small

proved

A kelvin and a joule are not commensurable. Thermodynamic temperature and energy are distinct kinds, so their units cannot be compared by ratio — a type-level fact, not a runtime check. Uses [metrological units](#).

Lean code for Theorem 17.1.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part5.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part5.kelvin_joule_not_commensurable
  ¬PropertyKindCalculus.Iso80000.Part5.kelvin.Commensurable
    PropertyKindCalculus.Iso80000.Part5.joule

```

Proof for Theorem 17.1.3
uses 0

Proved as `kelvin_joule_not_commensurable` : `¬ kelvin.Commensurable joule` by unfolding `Commensurable` to the underlying kinds and `decide`. The kelvin and the joule are each `WellFormed` (thermodynamic temperature and energy are ratio-scale).

17.2. Temperature: one dimension, two scales

ISO 80000-5 lists *thermodynamic temperature* (5-1) and *Celsius temperature* (5-2) as separate items of the *same* dimension θ . But they are not the same kind of quantity. Thermodynamic temperature has an absolute zero, so it is *ratio-scale*: ratios of temperatures are meaningful, and the kelvin admits $\times$ and $\div$. Celsius temperature is measured from the arbitrary zero of the ice point ($t = T - T_0$), so it is *interval-scale*: only its *differences* are meaningful, and a ratio of Celsius temperatures is not a defined operation. A dimension-only model sees one type, "a real number of degrees", for both.

This is requirement *R6* — the scale type fixes which operations are even *defined* — now on the real standard, the axis Part 3 and Part 4 (every quantity ratio-scale) could not exercise. The two temperatures are distinct kinds separated *neither* by dimension *nor* by an examination principle, but by their *scale type* alone.

Theorem **17.2.1.** group **statement">us**

Effort	Tags
small	proved

Celsius temperature does not admit $\times, \div$. It is interval-scale (an arbitrary zero at the ice point), so a ratio of Celsius temperatures is not a meaningful operation — the scale layer refuses it where the dimension would not. Thermodynamic temperature, being ratio-scale, does admit it. Uses [the scale type](#).

Lean code for Theorem 17.2.1.
1 theorem

theorem · defined in <code>PropertyKindCalculus/Iso80000/Part5.lean</code>	complete
---	---------------------

```
theorem PropertyKindCalculus.Iso80000.Part5.celsiusTemperature_not_allowsRa
  ¬PropertyKindCalculus.Iso80000.Part5.celsiusTemperature.kind.scale.AllowsRa
```

Proof for Theorem 17.2.1

uses 0

celsiusTemperature_not_allowsRatio : $\neg$ ScaleType.AllowsRatio celsiusTemperature.kind.scale, with the companion thermodynamicTemperature_allowsRatio for the ratio case. The degree Celsius is nonetheless WellFormed (degreeCelsius_wellFormed, via the differential scale), and thermodynamicTemperature_ne_celsiusTemperature (a decide) records that the two are distinct kinds.

Theorem 17.2.2.

group

statement">us

Effort

Tags

small

proved

The scale disambiguation, on the standard. There exist distinct ISO 80000-5 kinds with the same dimension θ that are separated by scale type alone — thermodynamic temperature (ratio) and Celsius temperature (interval) witness it. Neither the dimension nor an examination principle tells them apart; the scale layer does. This is requirement R6 on the standard. Uses [the scale type](#).

Lean code for Theorem 17.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part5.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part5.iso80000_5_scale_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.kind.scale ≠ b.kind.scale
```

Proof for Theorem 17.2.2

uses 0

iso80000_5_scale_collision : $\exists a b : \text{DimensionedKind}, a.\text{kind} \neq b.\text{kind} \wedge a.\text{dim} = b.\text{dim} \wedge a.\text{kind}.\text{scale} \neq b.\text{kind}.\text{scale}$, witnessed by {thermodynamicTemperature, celsiusTemperature, ...} with reflexivity on the dimension and a decide on the differing scales. Axiom-free.

17.3. The energy family: distinguishing kinds by defining construction

ISO 80000-5 item 5-20 lists *energy* (5-20.1) as a genus and the *thermodynamic potentials* — internal energy (5-20.2), enthalpy (5-20.3), the Helmholtz energy (5-20.4), and the Gibbs energy (5-20.5) — as items beneath it. Every one has dimension $M \cdot L^2 \cdot T^{-2}$ and ratio scale. The standard distinguishes them

only in *prose*, by which thermodynamic variables are natural to the potential: internal energy is a state function of S, V, N ; the enthalpy is $H = U + pV$; the Helmholtz energy is $A = U - TS$; the Gibbs energy is $G = H - TS$. A dimension-only model sees one type, "a real number of joules", for all of them — and cannot even separate them from a Part-4 torque.

This is requirement *R2* again, now on the thermodynamic potentials — the pattern Part 3 set on the length family and Part 4 on the force family. Item 5-20.1 is the broad genus; each later item is a *species* of energy, distinguished **not by fiat** but by an explicit defining construction (Dybkaer's examination principle, §7.5). The species are then provably distinct kinds while remaining *mutually comparable* as energies.

Definition **17.3.1.** **group** **uses 2** **used by 0**

An *energy species* is a ratio-scale [quantity](#)-kind of dimension $M \cdot L^2 \cdot T^{-2}$ carrying an [examination principle](#) — this work's own terse descriptor of the defining construction that distinguishes it (natural in S - V - N , in S - p , in T - V , in T - p), never the standard's normative definition. Internal energy, the enthalpy, and the Helmholtz and Gibbs energies are all species of the general energy kind (item 5-20.1).

Lean code for Definition **17.3.1.** **1 definition**

def · defined in PropertyKindCalculus/Iso800000/Part5.lean

complete

```
def PropertyKindCalculus.Iso800000.Part5.energySpecies (id : String)
  (p : PropertyKindCalculus.ExaminationPrinciple) :
  PropertyKindCalculus.DimensionedKind
```

Proof for Definition 17.3.1 **uses 0**

energySpecies id p := { kind := { id, scale := .ratio, examPrinciple := some p.id }, dim := TDim.energy }. The general energy carries no examination principle; the direct-parent edges of the family are collected in an Edge relation, each species an edge to energy (mirroring the standard's "energy (item 5-20.1) ... given by ...").

Theorem **17.3.2.** **group** **statement"** **uses 1** **by 0**

Effort
small

Tags
proved

The Helmholtz and Gibbs energies are distinct kinds — by defining construction, not by fiat.
*They share dimension $M \cdot L^2 \cdot T^{-2}$ and are both "free energies", yet are different kinds because they are defined by different natural variables ($A = U - TS$, natural in T, V ; $G = H - TS$, natural in T, p), proved through *distinct_of_examPrinciple* (§7.5). This is the thermodynamic-potential analogue of width $\neq$ distance and static $\neq$ kinetic friction force. Uses [the examination principle](#).*

Lean code for Theorem 17.3.2.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part5.lean

complete

theorem PropertyKindCalculus.Iso80000.Part5.helmholtzEnergy_ne_gibbsEnergy
PropertyKindCalculus.Iso80000.Part5.helmholtzEnergy.kind ≠
PropertyKindCalculus.Iso80000.Part5.gibbsEnergy.kind

Proof for Theorem 17.3.2 uses 0

helmholtzEnergy_ne_gibbsEnergy : helmholtzEnergy.kind ≠ gibbsEnergy.kind, from
KindOfProperty.distinct_of_examPrinciple applied to the differing principles natural-T-V and
natural-T-p (a decide on the Option String links). The companions internalEnergy_ne_enthalpy and
internalEnergy_ne_energy separate the other species, and internalEnergy_examinedBy checks the
kind-to-principle link. Axiom-free.

Theorem 17.3.3. group statement">uses 1 by 0

Effort

Tags

small

proved

Comparability is preserved. The Helmholtz and Gibbs energies, though distinct kinds, remain mutually comparable — they share the super-kind energy, so combining them is possible but only via an explicit up-cast, never silently. The companion internalEnergy_specializes_energy records that internal energy specializes energy. Uses [specialization](#).

Lean code for Theorem 17.3.3.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part5.lean

complete

theorem PropertyKindCalculus.Iso80000.Part5.helmholtzEnergy_gibbsEnergy_com
PropertyKindCalculus.MutuallyComparable
PropertyKindCalculus.Iso80000.Part5.Edge
PropertyKindCalculus.Iso80000.Part5.helmholtzEnergy.kind
PropertyKindCalculus.Iso80000.Part5.gibbsEnergy.kind

Proof for Theorem 17.3.3 uses 0

helmholtzEnergy_gibbsEnergy_comparable : MutuallyComparable Edge helmholtzEnergy.kind
gibbsEnergy.kind, with witness energy.kind, each species reaching it by one edge. Axiom-free.

17.4. Dimension does not classify; the kind does — the entropy/heat-capacity case

Part 5 holds another textbook dimension collision. *Entropy* (5-18) and *heat capacity* (5-15) are both $M \cdot L^2 \cdot T^{-2} \cdot \theta^{-1}$ — the joule per kelvin — yet they are distinct kinds, and the collision is unusually sharp: their units carry not only the same dimension but the *same symbol*, "J/K", and are still not commensurable, because their kinds differ. The Massieu (5-22) and Planck (5-23) functions join them at J/K; specific entropy (5-19), specific heat capacity (5-16.1), and the specific gas constant (5-26) collide the same way at J/(kg·K); and the humidity ratios and fractions (5-29 ... 5-35), with the ratio of specific heats and the efficiencies, fill out the dimension-one family.

Theorem17.4.1groupstatement"uses 0

Effortsmall

Tags

proved

Entropy is not heat capacity, though both are $M \cdot L^2 \cdot T^{-2} \cdot \theta^{-1}$. There exist distinct ISO 80000-5 kinds with the same dimension — entropy and heat capacity witness it. No dimension-only type system can separate them; the kind layer does. This is the textbook thermodynamic case that a dimension does not classify a quantity. Uses [the dimension map](#).

Lean code for Theorem17.4.1

1 theorem

theorem · defined inPropertyKindCalculus/Iso80000/Part5.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part5.iso80000_5_dim_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim
```

Proof for Theorem 17.4.1uses 0

iso80000_5_dim_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim, witnessed by ⟨entropy, heatCapacity, ...⟩ with entropy_ne_heatCapacity (a decide) and reflexivity. The companion specificEntropy_ne_specificHeatCapacity is the J/(kg·K) case. Axiom-free.

Theorem17.4.2groupstatement"uses 0

Effortsmall

Tags

proved

The entropy "J/K" is not the heat-capacity "J/K". The units of entropy and heat capacity carry the same symbol and the same dimension $M \cdot L^2 \cdot T^{-2} \cdot \theta^{-1}$, yet are not commensurable — the sharpest form of the rule that neither a dimension nor a unit string determines a kind. Uses [metrological units](#).

Lean code for Theorem 17.4.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part5.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part5.entropy_heatCapacity_unit_not_c
  ¬PropertyKindCalculus.Iso80000.Part5.joulePerKelvinEntropy.Commensurable
    PropertyKindCalculus.Iso80000.Part5.joulePerKelvinHeatCapacity
```

Proof for Theorem 17.4.2

uses 0

entropy_heatCapacity_unit_not_commensurable : ¬ joulePerKelvinEntropy.Commensurable
joulePerKelvinHeatCapacity by unfolding to the underlying kinds and decide (entropy's kind differs from the heat-capacity kind, although both units are written "J/K").

Theorem 17.4.3.

group

statement" > uses 1 by 0

Effort

Tags

small

proved

The dimension-1 disambiguation, on the standard. There exist distinct ISO 80000-5 kinds with the same dimension one — the ratio of specific heat capacities and the efficiency witness it, alongside the isentropic exponent and the humidity ratios and fractions. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem 17.4.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part5.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part5.iso80000_5_dim_one_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 17.4.3

uses 0

iso80000_5_dim_one_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim ∧
a.dim = 1, witnessed by ⟨ratioOfSpecificHeatCapacities, efficiency, ...⟩ with
ratioOfSpecificHeatCapacities_ne_efficiency (a decide) and reflexivity. Axiom-free.

17.5. The algebraic Remarks as kind-laws: thermodynamics built from the earlier parts

Many Part-5 remarks state a quantity's defining relation *algebraically* — and, in thermodynamics, the other quantities are often from *earlier parts* (mass from ISO 80000-4, area from ISO 80000-3) or from temperature, the base quantity this part introduces. Heat capacity is added heat per temperature (5-15); specific heat capacity is heat capacity / mass (5-16.1); specific entropy is entropy / mass (5-19); the density of heat flow rate is heat flow rate / area (5-8); thermal conductance is $1/R$ (5-13); the ratio of specific heats is c_p/c_v (5-17.1). Each is formalized as an R12 kind-law over the catalogued kinds, several composing a Part-5 kind out of an earlier part's kinds.

Theorem

17.5.1.

group

statement">us

Effort small	Tags proved	
------------------------------------	-----------------------------------	--

Specific heat capacity is heat capacity / mass — a verified, cross-part construction. A specific heat capacity built as the quotient of a heat capacity (Part 5) by a mass (Part 4) carries its quotient certificate by construction, over the $\mathbb{R}$ carrier, and canonicity makes the certificate determine the quantity. Uses [the quotient kind-law](#).

Lean code for Theorem 17.5.1.

1 theorem

```
theorem · defined in PropertyKindCalculus/Iso80000/Part5/DefiningRelations.lean

theorem PropertyKindCalculus.Iso80000.Part5.DefiningRelations.specificHeatC
  (c :
    PropertyKindCalculus.Quantity
    PropertyKindCalculus.Iso80000.Part5.heatCapacity.kind ℝ)
  (m :
    PropertyKindCalculus.Quantity
    PropertyKindCalculus.Iso80000.Part4.mass.kind ℝ) :
  PropertyKindCalculus.Quantity.IsQuotient

PropertyKindCalculus.Iso80000.Part5.DefiningRelations.specificHeatCapacity
_quot_heatCapacity_mass

(PropertyKindCalculus.Iso80000.Part5.DefiningRelations.specificHeatCapacity
  c m)
  c m
```

Proof for Theorem 17.5.1

uses 0

specificHeatCapacityOf c m := Quantity.div specificHeatCapacity_quot_heatCapacity_mass c m, and specificHeatCapacityOf_isQuotient is rfl; specificHeatCapacity_certificate_canonical is eq_div_of_isQuotient. The kind-law specificHeatCapacity_quot_heatCapacity_mass : QuotientKind heatCapacity.kind Part4.mass.kind specificHeatCapacity.kind is {rfl, rfl, rfl}, and specificHeatCapacity_dim_from_heatCapacity_mass checks $L^2 \cdot T^{-2} \cdot \theta^{-1}$ follows from the relation. The same families give densityOfHeatFlowRateOf (= heat flow rate / area) and thermalConductanceOf (= the reciprocal of thermal resistance).

Theorem 17.5.2.

group statement">use

Effort	Tags
small	proved

The ratio of specific heats is dimension one because it is a ratio of two specific heat capacities. From the remark $\gamma = cp/cv$, the specific heat capacities cancel and the dimension is computed to be one — not stipulated, the thermodynamic analogue of the plane angle being a ratio of two lengths and the efficiency a ratio of two powers. Yet the ratio remains a distinct kind from every other dimension-one quantity (the collision capstone above). Uses [the dimension map](#).

```
theorem · defined in PropertyKindCalculus/Iso80000/Part5/
  DefiningRelations.lean
```

complete

```
theorem PropertyKindCalculus.Iso80000.Part5.DefiningRelations.ratio_dim_from
  PropertyKindCalculus.Iso80000.Part5.ratioOfSpecificHeatCapacities.dim =

  PropertyKindCalculus.Iso80000.Part5.specificHeatCapacityConstantPressure.dim /
  PropertyKindCalculus.Iso80000.Part5.specificHeatCapacityConstantVolume.dim
```

Proof for Theorem 17.5.2

uses 0

```
ratio_dim_from_specific_heats : ratioOfSpecificHeatCapacities.dim =
  specificHeatCapacityConstantPressure.dim / specificHeatCapacityConstantVolume.dim, by
  reducing both sides via div_self' (a / a = 1). The kind-law
  ratioOfSpecificHeatCapacities_quot_cp_cV is {rfl, rfl, rfl}.
```

17.6. Item index — ISO 80000-5

Every catalogued item of ISO 80000-5, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLib dimension this work assigns it. Each item number links to the formalized result it participates in — the scale-type distinction, the energy-family lattice, the entropy/heat-capacity collision, a cross-part defining relation, or the catalogue itself. Symbols and unit strings are **citation locators**; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
5-1	thermodynamic temperature	T	K	θ
5-2	Celsius temperature	t	$^{\circ}\text{C}$	θ
5-3.1	linear expansion coefficient	α_l	K^{-1}	θ^{-1}
5-3.2	cubic expansion coefficient	α_V	K^{-1}	θ^{-1}
5-3.3	relative pressure coefficient	α_p	K^{-1}	θ^{-1}
5-4	pressure coefficient	β	Pa/K	$\text{M} \cdot \text{L}^{-1} \cdot \text{T}^{-2} \cdot \theta^{-1}$
5-5.1	isothermal compressibility	χ_T	Pa^{-1}	$\text{M}^{-1} \cdot \text{L} \cdot \text{T}^2$
5-5.2	isentropic compressibility	χ_S	Pa^{-1}	$\text{M}^{-1} \cdot \text{L} \cdot \text{T}^2$
5-6.1	heat	Q	J	$\text{M} \cdot \text{L}^2 \cdot \text{T}^{-2}$
5-6.2	latent heat	Q	J	$\text{M} \cdot \text{L}^2 \cdot \text{T}^{-2}$
5-7	heat flow rate	Φ	W	$\text{M} \cdot \text{L}^2 \cdot \text{T}^{-3}$
5-8	density of heat flow rate	q	W/m^2	$\text{M} \cdot \text{T}^{-3}$
5-9	thermal conductivity	λ	$\text{W}/(\text{m} \cdot \text{K})$	$\text{M} \cdot \text{L} \cdot \text{T}^{-3} \cdot \theta^{-1}$
5-10.1	coefficient of heat transfer	K	$\text{W}/(\text{m}^2 \cdot \text{K})$	$\text{M} \cdot \text{T}^{-3} \cdot \theta^{-1}$
5-10.2	surface coefficient of heat transfer	h	$\text{W}/(\text{m}^2 \cdot \text{K})$	$\text{M} \cdot \text{T}^{-3} \cdot \theta^{-1}$
5-11	thermal insulance	M	$\text{m}^2 \cdot \text{K/W}$	$\text{M}^{-1} \cdot \text{T}^3 \cdot \theta$
5-12	thermal resistance	R	K/W	$\text{M}^{-1} \cdot \text{L}^{-2} \cdot \text{T}^3 \cdot \theta$
5-13	thermal conductance	G	W/K	$\text{M} \cdot \text{L}^2 \cdot \text{T}^{-3} \cdot \theta^{-1}$
5-14	thermal diffusivity	a	m^2/s	$\text{L}^2 \cdot \text{T}^{-1}$
5-15	heat capacity	C	J/K	$\text{M} \cdot \text{L}^2 \cdot \text{T}^{-2} \cdot \theta^{-1}$
5-16.1	specific heat capacity	c	$\text{J}/(\text{kg} \cdot \text{K})$	$\text{L}^2 \cdot \text{T}^{-2} \cdot \theta^{-1}$
5-16.2	specific heat capacity at constant pressure	c_p	$\text{J}/(\text{kg} \cdot \text{K})$	$\text{L}^2 \cdot \text{T}^{-2} \cdot \theta^{-1}$
5-16.3	specific heat capacity at constant volume	c_V	$\text{J}/(\text{kg} \cdot \text{K})$	$\text{L}^2 \cdot \text{T}^{-2} \cdot \theta^{-1}$
5-16.4	specific heat capacity at saturated vapour pressure	c_{sat}	$\text{J}/(\text{kg} \cdot \text{K})$	$\text{L}^2 \cdot \text{T}^{-2} \cdot \theta^{-1}$
5-17.1	ratio of specific heat capacities	γ	1	1
5-17.2	isentropic exponent	κ	1	1
5-18	entropy	S	J/K	$\text{M} \cdot \text{L}^2 \cdot \text{T}^{-2} \cdot \theta^{-1}$
5-19	specific entropy	s	$\text{J}/(\text{kg} \cdot \text{K})$	$\text{L}^2 \cdot \text{T}^{-2} \cdot \theta^{-1}$

Item	Quantity	Symbol	Unit	Dimension
5-20.1	energy	E	J	$M \cdot L^2 \cdot T^{-2}$
5-20.2	internal energy	U	J	$M \cdot L^2 \cdot T^{-2}$
5-20.3	enthalpy	H	J	$M \cdot L^2 \cdot T^{-2}$
5-20.4	Helmholtz energy	A	J	$M \cdot L^2 \cdot T^{-2}$
5-20.5	Gibbs energy	G	J	$M \cdot L^2 \cdot T^{-2}$
5-21.1	specific energy	e	J/kg	$L^2 \cdot T^{-2}$
5-21.2	specific internal energy	u	J/kg	$L^2 \cdot T^{-2}$
5-21.3	specific enthalpy	h	J/kg	$L^2 \cdot T^{-2}$
5-21.4	specific Helmholtz energy	a	J/kg	$L^2 \cdot T^{-2}$
5-21.5	specific Gibbs energy	g	J/kg	$L^2 \cdot T^{-2}$
5-22	Massieu function	J	J/K	$M \cdot L^2 \cdot T^{-2} \cdot \theta^{-1}$
5-23	Planck function	Y	J/K	$M \cdot L^2 \cdot T^{-2} \cdot \theta^{-1}$
5-24	Joule-Thomson coefficient	μ_{JT}	K/Pa	$M^{-1} \cdot L \cdot T^2 \cdot \theta$
5-25.1	efficiency	η	1	1
5-25.2	maximum efficiency	η_{max}	1	1
5-26	specific gas constant	R_s	J/(kg·K)	$L^2 \cdot T^{-2} \cdot \theta^{-1}$
5-27	mass concentration of water	w	kg/m ³	$M \cdot L^{-3}$
5-28	mass concentration of water vapour	v	kg/m ³	$M \cdot L^{-3}$
5-29	mass ratio of water to dry matter	u	1	1
5-30	mass ratio of water vapour to dry gas	r	1	1
5-31	mass fraction of water	w_H2O	1	1
5-32	mass fraction of dry matter	w_d	1	1
5-33	relative humidity	ϕ	1	1
5-34	relative mass concentration of vapour	ϕ	1	1
5-35	relative mass ratio of vapour	ψ	1	1
5-36	dew-point temperature	T_d	K	θ

18. IEC 80000-6 — Electromagnetism

The fourth part specified in full is IEC 80000-6, *Electromagnetism* — the one *IEC*-published part of the series. Every item — all of 6-1 ... 6-62, including every sub-suffixed item (6-2.1/6-2.2, 6-11.1 ... 6-11.4, 6-14.1/6-14.2, 6-19.1/6-19.2, 6-22.1 ... 6-22.4, 6-26.1/6-26.2, 6-35.1/6-35.2, 6-37.1 ... 6-37.3, 6-41.1/6-41.2, 6-42.1/6-42.2, 6-51.1 ... 6-51.5, 6-52.1 ... 6-52.5) — is catalogued: electric current and charge, the charge and current distributions, the electric field and potential, the capacitances and permittivities, the magnetic field, flux, and potentials, the magnetic circuit, conductivity and resistivity, impedance and admittance, and the AC power quantities — each carrying its exact source as data: the part, the printed item designation, the principal quantity symbol, and the coherent SI unit symbol. Only *citation locators* are recorded; no normative content (definitions, remarks) from the licensed standard is reproduced. The defining *mathematics* of selected remarks is formalized in the sibling module `Part6.DefiningRelations`.

Electromagnetism brings the last SI base quantity the earlier parts did not exercise — *electric current* — and sharpens the two axes the calculus already pressed. The base axis enters as the ampere; PhysLib's `Dimension` takes electric *charge* `C` as its generator, so electric current appears as $C \cdot T^{-1}$ (coulomb per second). The *scale-type* distinction recurs — electric potential is gauge-dependent, fixed only up to an arbitrary additive reference, hence interval-scale, while electric potential difference, of the very same dimension `V`, is ratio-scale (requirement *R6*). And the AC power quantities give the sharpest *dimension does not classify* case yet: active, reactive, and apparent power are all $M \cdot L^2 \cdot T^{-3}$, yet the standard spends *three* different unit strings on that one dimension — the watt, the var, and the volt-ampere — and arranges them as a specialization lattice over power (requirement *R2*).

18.1. Quantity-kinds and units of IEC 80000-6

Definition18.1.1.

groupstatement">uses1by0

A *catalogued kind* pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "6-51.1"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream rather than kept in a comment.

Lean code for Definition18.1.1.

1 definition

```
structure (5 fields) · defined PropertyKindCalculus/Iso80000/
in Catalogue.lean
complete

structure PropertyKindCalculus.Iso80000.CataloguedKind : Type
```

Proof for Definition18.1.1uses0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with cite rendering e.g. IEC 80000-6, Edition 2.0, 2022-11 item 6-51.1 (the one IEC-published part of the catalogue). The catalogue lists all 85 items in item order — electric current (6-1), electric charge (6-2.1), magnetic flux (6-22.1), the impedance family (6-51.1 ... 6-51.5), the AC power quantities (6-56 ... 6-61), active energy (6-62), ... — each paired with its coherent SI unit symbol.

Theorem 18.1.2.

group

statement" > uses 1 by 0

Effort

Tags

small

proved

A capacitance carries the charge generator squared (item 6-13). The catalogued capacitance kind has charge-exponent two in PhysLib's dimension group — the farad's $C^2 \cdot M^{-1} \cdot L^{-2} \cdot T^2$ reproduced as a checked computation over the new electric-current base axis, not an annotation. Uses [the dimension map](#).

Lean code for Theorem 18.1.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part6.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part6.capacitance_dim_charge :
  PropertyKindCalculus.Iso80000.Part6.capacitance.dim.charge = 2
```

Proof for Theorem 18.1.2

uses 0

Proved as `capacitance_dim_charge : capacitance.dim.charge = 2` in the `Iso80000` library, discharged by `norm_num` over the PhysLib dimension lemmas for the charge generator. The companions `electricCurrent_dim_charge = 1` and `electricCurrent_dim_time = -1` (item 6-1, the ampere as $C \cdot T^{-1}$), `resistance_dim_charge = -2` (item 6-46, the ohm), and `magneticFlux_dim_charge = -1` (item 6-22.1, the weber) record the rest of the electromagnetic algebra.

Theorem 18.1.3.

group

statement" > uses 1 by 0

Effort

Tags

small

proved

An ampere and a volt are not commensurable. Electric current and electric potential are distinct kinds, so their units cannot be compared by ratio — a type-level fact, not a runtime check. Uses [metrological units](#).

Lean code for Theorem 18.1.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part6.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part6.ampere_voltPotential_not_commen
  ¬PropertyKindCalculus.Iso80000.Part6.ampere.Commensurable
    PropertyKindCalculus.Iso80000.Part6.voltPotential
```

Proof for Theorem 18.1.3

uses 0

Proved as ampere_voltPotential_not_commensurable : ¬ ampere.Commensurable voltPotential by unfolding Commensurable to the underlying kinds and decide. The ampere and the ohm are each WellFormed (electric current and resistance are ratio-scale).

18.2. Electric potential: one dimension, two scales

IEC 80000-6 lists *electric potential* (6-11.1) and *electric potential difference* (6-11.2) as separate items of the *same* dimension V. But they are not the same kind of quantity. Electric potential is fixed only up to an arbitrary additive reference — the choice of where "zero potential" sits (the ground, infinity) is a gauge, so the standard notes the potential "is not unique". Only its *differences* are physically meaningful, so it is *interval-scale*: a ratio of two electric potentials is not a defined operation. Electric potential difference is a genuine difference, so it is *ratio-scale*. A dimension-only model sees one type, "a real number of volts", for both.

This is requirement *R6* — the scale type fixes which operations are even *defined* — recurring on electromagnetism exactly as it appeared on thermodynamics. The gauge-dependent electric potential is electromagnetism's Celsius temperature: same dimension as its companion, separated from it *neither* by dimension *nor* by an examination principle, but by *scale type* alone.

Theorem 18.2.1.

group statement">use

Effort Tags

small proved

Electric potential does not admit $\times, \div$. It is *interval-scale* (gauge freedom — fixed only up to an arbitrary additive reference), so a ratio of electric potentials is not a meaningful operation — the scale layer refuses it where the dimension would not. Electric potential difference, being *ratio-scale*, does admit it. Uses [the scale type](#).

Lean code for Theorem 18.2.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part6.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part6.electricPotential_not_allowsRatio :  
  ¬PropertyKindCalculus.Iso80000.Part6.electricPotential.kind.scale.AllowsRatio
```

Proof for Theorem 18.2.1 uses 0

electricPotential_not_allowsRatio : $\neg$ ScaleType.AllowsRatio electricPotential.kind.scale, with the companion electricPotentialDifference_allowsRatio for the ratio case. The volt of electric potential is nonetheless WellFormed (voltPotential_wellFormed, via the differential scale), and voltPotential_voltDifference_not_commensurable records that the two volts — same symbol, same dimension — are not commensurable, because the kinds differ in scale.

Theorem 18.2.2. group statement">us

Effort small

Tags proved

***The scale disambiguation, on the standard.** There exist distinct IEC 80000-6 kinds with the same dimension v that are separated by scale type alone — electric potential (interval, gauge-dependent) and electric potential difference (ratio) witness it. Neither the dimension nor an examination principle tells them apart; the scale layer does. This is requirement R6 on the standard, the electromagnetic counterpart of thermodynamic versus Celsius temperature. Uses [the scale type](#).*

Lean code for Theorem 18.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part6.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part6.iec80000_6_scale_collision :  
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.kind.scale ≠ b.kind.scale
```

Proof for Theorem 18.2.2 uses 0

iec80000_6_scale_collision : $\exists a b : \text{DimensionedKind}, a.\text{kind} \neq b.\text{kind} \wedge a.\text{dim} = b.\text{dim} \wedge a.\text{kind}.\text{scale} \neq b.\text{kind}.\text{scale}$, witnessed by (electricPotential, electricPotentialDifference, ...) with reflexivity on the dimension and a decide on the differing scales. Axiom-free.

18.3. The AC power family: distinguishing kinds by defining construction

IEC 80000-6 lists *power* (6-45) as a genus and the AC power quantities — active power (6-56), apparent power (6-57), complex power (6-59), reactive power (6-60), and non-active power (6-61) — as items of one dimension $M \cdot L^2 \cdot T^{-3}$. The standard distinguishes them by *which component of the periodic process each isolates*: active power is the time-averaged real (resistive) component; reactive power the imaginary part of the complex power; apparent power the product of RMS voltage and current; and so on. Tellingly, it spends *three different unit strings* on this single dimension — the watt for active power, the var for reactive power, the volt-ampere for apparent power — precisely to keep the kinds apart in print. A dimension-only model sees one type, "a real number of watts", for all of them.

This is requirement *R2* again, now on the AC power quantities — the pattern Part 3 set on the length family, Part 4 on the force family, and Part 5 on the thermodynamic potentials. Item 6-45 is the broad genus; each AC quantity is a *species* of power, distinguished **not by fiat** but by an explicit defining construction (Dybæk's examination principle, §7.5). The species are then provably distinct kinds while remaining *mutually comparable* as powers — comparable enough to combine into the complex power $P + jQ$.

Definition 18.3.1.

group

uses 2

used by 0

✓

A *power species* is a ratio-scale [quantity](#)-kind of dimension $M \cdot L^2 \cdot T^{-3}$ carrying an [examination principle](#) — this work's own terse descriptor of the defining construction that distinguishes it (the time-averaged real part, the RMS product, the complex sum $P + jQ$, the imaginary part, the residual $\sqrt{(S^2 - P^2)}$), never the standard's normative definition. Active, apparent, complex, reactive, and non-active power are all species of the general power kind (item 6-45).

Lean code for Definition 18.3.1.

1 definition

def · defined in PropertyKindCalculus/Iso80000/Part6.lean

complete

```
def PropertyKindCalculus.Iso80000.Part6.powerSpecies (id : String)
  (p : PropertyKindCalculus.ExaminationPrinciple) :
  PropertyKindCalculus.DimensionedKind
```

Proof for Definition 18.3.1

uses 0

powerSpecies id p := { kind := { id, scale := .ratio, examPrinciple := some p.id }, dim := EDim.power }. The general power carries no examination principle; the direct-parent edges of the family are collected in an Edge relation, each species an edge to power (mirroring the standard's listing of the AC quantities beneath power).

Effort

small

Tags

proved

Active and reactive power are distinct kinds — by defining construction, not by fiat. They share dimension $M \cdot L^2 \cdot T^{-3}$ and are both "powers", yet are different kinds because they isolate different components of the periodic process (the time-averaged real part vs the imaginary part of the complex power), proved through `distinct_of_examPrinciple` (§7.5). This is the AC-power analogue of `width ≠ distance`, `static ≠ kinetic friction force`, and `Helmholtz ≠ Gibbs energy`. Uses [the examination principle](#).

Lean code for Theorem18.3.2.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part6.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part6.activePower_ne_reactivePower :
  PropertyKindCalculus.Iso80000.Part6.activePower.kind ≠
    PropertyKindCalculus.Iso80000.Part6.reactivePower.kind
```

Proof for Theorem18.3.2

uses0

`activePower_ne_reactivePower` : `activePower.kind ≠ reactivePower.kind`, from `KindOfProperty.distinct_of_examPrinciple` applied to the differing principles `time-averaged-real` and `reactive-imaginary` (a decide on the `Option String` links). The companion `apparentPower_ne_complexPower` separates two further species, and `activePower_examinedBy` checks the `kind-to-principle` link. Axiom-free.

Effort

small

Tags

proved

Comparability is preserved. Active and reactive power, though distinct kinds, remain mutually comparable — they share the super-kind power, so combining them (into the complex power) is possible but only via an explicit up-cast, never silently. The companion `activePower_specializes_power` records that active power specializes power. Uses [specialization](#).

Lean code for Theorem18.3.3.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part6.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part6.activePower_reactivePower_compa
PropertyKindCalculus.MutuallyComparable
PropertyKindCalculus.Iso80000.Part6.Edge
PropertyKindCalculus.Iso80000.Part6.activePower.kind
PropertyKindCalculus.Iso80000.Part6.reactivePower.kind

```

Proof for Theorem 18.3.3

uses 0

```

activePower_reactivePower_comparable : MutuallyComparable Edge activePower.kind
reactivePower.kind, with witness power.kind, each species reaching it by one edge. Axiom-free.

```

18.4. Dimension does not classify; the kind does — the power and impedance cases

Part 6 holds the sharpest dimension collision in the series so far. *Active power* (6-56), *reactive power* (6-60), and *apparent power* (6-57) are all $M \cdot L^2 \cdot T^{-3}$, yet they are distinct kinds — and the collision is sharper than Part 5's joule-per-kelvin, because there even the unit string agreed, while here the standard itself spends three strings (w, var, VA) on the one dimension. *Resistance* (6-46) and *reactance* (6-51.4) collide the same way at the ohm — the real and imaginary parts of an impedance — and the relative permittivity (6-15), relative permeability (6-27), the susceptibilities, the coupling and leakage factors, and the quality, loss, and power factors fill out the dimension-one family.

Theorem **18.4.1.** **group** **statement"**>uses by 0

Effort **Tags**

small

proved

Active power is not reactive power, though both are $M \cdot L^2 \cdot T^{-3}$. There exist distinct IEC 80000-6 kinds with the same dimension — active and reactive power witness it, alongside apparent, complex, and non-active power. No dimension-only type system can separate them; the kind layer does. This is the electromagnetic case that a dimension does not classify a quantity, sharper than Part 5's. Uses [the dimension map](#).

Lean code for Theorem **18.4.1.**

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part6.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part6.iec80000_6_dim_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim

```

Proof for Theorem 18.4.1

uses 0

iec80000_6_dim_collision : $\exists a\ b : \text{DimensionedKind}, a.\text{kind} \neq b.\text{kind} \wedge a.\text{dim} = b.\text{dim}$,
witnessed by $\langle \text{activePower}, \text{reactivePower}, \dots \rangle$ With $\text{activePower_ne_reactivePower}$ and reflexivity.
The companion $\text{resistance_ne_reactance}$ is the ohm case (real vs imaginary part of an impedance).
Axiom-free.

Theorem 18.4.2. group statement" >uses 1 by 0

Effort
small

Tags
proved

The active-power watt is not the reactive-power var. The units of active and reactive power carry the same dimension $M \cdot L^2 \cdot T^{-3}$, yet are not commensurable — and the standard keeps them apart not only by kind but by string, w versus var , a sharper form of the rule that neither a dimension nor a unit string determines a kind. Uses [metrological units](#).

Lean code for Theorem 18.4.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part6.lean

complete

theorem PropertyKindCalculus.Iso80000.Part6.wattActive_varReactive_not_comm
 ¬PropertyKindCalculus.Iso80000.Part6.wattActive.Commensurable
 PropertyKindCalculus.Iso80000.Part6.varReactive

Proof for Theorem 18.4.2

uses 0

wattActive_varReactive_not_commensurable : $\neg \text{wattActive.Commensurable varReactive}$ by
unfolding to the underlying kinds and decide (active power's kind differs from the reactive-power kind;
the units are written "W" and "var").

Theorem 18.4.3. group statement" >uses 1 by 0

Effort
small

Tags
proved

The dimension-1 disambiguation, on the standard. There exist distinct IEC 80000-6 kinds with the same dimension one — relative permittivity and relative permeability witness it, alongside the susceptibilities, the coupling and leakage factors, and the quality, loss, and power factors. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem 18.4.3.

1 theorem

```
theorem · defined in PropertyKindCalculus/Iso80000/Part6.lean

theorem PropertyKindCalculus.Iso80000.Part6.iec80000_6_dim_one_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 18.4.3 uses 0

iec80000_6_dim_one_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1, witnessed by (relativePermittivity, relativePermeability, ...) with relativePermittivity_ne_relativePermeability (a decide) and reflexivity. Axiom-free.

18.5. The algebraic Remarks as kind-laws: electromagnetism built by the kind algebra

Many Part-6 definitions state a quantity's defining relation *algebraically* — the constitutive laws of circuit theory. Resistance is voltage per current (Ohm's law, 6-46); conductance is 1/R (6-47); capacitance is charge per voltage (6-13); power is voltage times current (6-45); resistivity is 1/σ (6-44); admittance is 1/Z (6-52.1); permeance is 1/R_m (6-40); the power factor is active over apparent power (6-58). Electric current itself is charge per time (6-1), crossing to ISO 80000-3 for its time factor. Each is formalized as an R12 kind-law over the catalogued kinds.

Theorem 18.5.1. group statement">us

Effort
small

Tags
proved

Resistance is voltage / current — a verified construction (Ohm's law). A resistance built as the quotient of a voltage by an electric current carries its quotient certificate by construction, over the $\mathbb{R}$ carrier, and canonicity makes the certificate determine the quantity. Uses [the quotient kind-law](#).

Lean code for Theorem 18.5.1.

1 theorem

```
theorem · defined in PropertyKindCalculus/Iso80000/Part6/DefiningRelations.lean

theorem PropertyKindCalculus.Iso80000.Part6.DefiningRelations.resistanceOf_
  (u :
    PropertyKindCalculus.Quantity
```

```

PropertyKindCalculus.Iso80000.Part6.voltage.kind ℝ)
(i :
  PropertyKindCalculus.Quantity
  PropertyKindCalculus.Iso80000.Part6.electricCurrent.kind ℝ) :
PropertyKindCalculus.Quantity.IsQuotient

PropertyKindCalculus.Iso80000.Part6.DefiningRelations.resistance_quot_volt
age_current
  (PropertyKindCalculus.Iso80000.Part6.DefiningRelations.resistanceOf
    u i)
  u i

```

Proof for Theorem 18.5.1

uses 0

resistanceOf u i := Quantity.div resistance_quot_voltage_current u i, and
 resistanceOf_isQuotient is rfl; resistance_certificate_canonical is eq_div_of_isQuotient. The
 kind-law resistance_quot_voltage_current : QuotientKind voltage.kind electricCurrent.kind
 resistance.kind is (rfl, rfl, rfl), and resistance_dim_from_voltage_current checks $M \cdot L^2 \cdot T^{-1} \cdot C^{-2}$
 follows from the relation. The same families give powerOf (= voltage × current, a ProductKind),
 conductanceOf (= 1/R), and the cross-part electricCurrent_quot_charge_duration (= charge / time, the
 time crossing to ISO 80000-3).

Theorem 18.5.2.

group

statement">us

Effort

Tags

small

proved

The power factor is dimension one because it is a ratio of two powers. From the relation $\lambda = P/S$,
 the powers cancel and the dimension is computed to be one — not stipulated, the electromagnetic
 analogue of the plane angle being a ratio of two lengths and the thermodynamic efficiency a ratio of
 two powers. Yet the power factor remains a distinct kind from every other dimension-one quantity (the
 collision capstone above). Uses [the dimension map](#).

theorem · defined in PropertyKindCalculus/Iso80000/Part6/DefiningRelations.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part6.DefiningRelations.powerFactor_dim
  PropertyKindCalculus.Iso80000.Part6.powerFactor.dim =
    PropertyKindCalculus.Iso80000.Part6.activePower.dim /
    PropertyKindCalculus.Iso80000.Part6.apparentPower.dim
```

Proof for Theorem 18.5.2

uses 0

powerFactor_dim_from_powers : powerFactor.dim = activePower.dim / apparentPower.dim, by reducing both sides via div_self ($a / a = 1$). The kind-law powerFactor_quot_active_apparent is (rfl, rfl, rfl). The companion power_dim_from_voltage_current checks $M \cdot L^2 \cdot T^{-3}$ follows from $P = U \cdot I$ (the charge generator cancelling between voltage and current).

18.6. Item index — IEC 80000-6

Every catalogued item of IEC 80000-6, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLio dimension this work assigns it (over the charge generator c). Each item number links to the formalized result it participates in — the scale-type distinction, the AC-power lattice, a dimension collision, a defining relation, or the catalogue itself. Symbols and unit strings are **citation locators**; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
6-1	electric current	I	A	$T^{-1} \cdot C$
6-2.1	electric charge	Q	C	C
6-2.2	elementary charge	e	C	C
6-3	electric charge density	ρ	C/m ³	$L^{-3} \cdot C$
6-4	surface density of electric charge	σ	C/m ²	$L^{-2} \cdot C$
6-5	linear density of electric charge	τ	C/m	$L^{-1} \cdot C$
6-6	electric dipole moment	p	C·m	$L \cdot C$
6-7	electric polarization	P	C/m ²	$L^{-2} \cdot C$
6-8	electric current density	J	A/m ²	$L^{-2} \cdot T^{-1} \cdot C$
6-9	linear electric current density	J_S	A/m	$L^{-1} \cdot T^{-1} \cdot C$
6-10	electric field strength	E	V/m	$M \cdot L \cdot T^{-2} \cdot C^{-1}$
6-11.1	electric potential	V	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
6-11.2	electric potential difference	V_ab	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
6-11.3	voltage	U	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
6-11.4	induced voltage	U_i	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
6-12	electric flux density	D	C/m ²	$L^{-2} \cdot C$
6-13	capacitance	C	F	$M^{-1} \cdot L^{-2} \cdot T^2 \cdot C^2$
6-14.1	electric constant	ϵ_0	F/m	$M^{-1} \cdot L^{-3} \cdot T^2 \cdot C^2$
6-14.2	permittivity	ϵ	F/m	$M^{-1} \cdot L^{-3} \cdot T^2 \cdot C^2$
6-15	relative permittivity	ϵ_r	1	1
6-16	electric susceptibility	χ	1	1
6-17	electric flux	Ψ	C	C
6-18	displacement current density	J_D	A/m ²	$L^{-2} \cdot T^{-1} \cdot C$
6-19.1	displacement current	I_D	A	$T^{-1} \cdot C$
6-19.2	total current	I_tot	A	$T^{-1} \cdot C$
6-20	total current density	J_tot	A/m ²	$L^{-2} \cdot T^{-1} \cdot C$
6-21	magnetic flux density	B	T	$M \cdot T^{-1} \cdot C^{-1}$
6-22.1	magnetic flux	Φ	Wb	$M \cdot L^2 \cdot T^{-1} \cdot C^{-1}$

Item	Quantity	Symbol	Unit	Dimension
6-22.2	protoflux	Ψ_p	Wb	$M \cdot L^2 \cdot T^{-1} \cdot C^{-1}$
6-22.3	linked magnetic flux	Φ_l	Wb	$M \cdot L^2 \cdot T^{-1} \cdot C^{-1}$
6-22.4	total magnetic flux	Ψ	Wb	$M \cdot L^2 \cdot T^{-1} \cdot C^{-1}$
6-23	magnetic moment	m	$A \cdot m^2$	$L^2 \cdot T^{-1} \cdot C$
6-24	magnetization	M	A/m	$L^{-1} \cdot T^{-1} \cdot C$
6-25	magnetic field strength	H	A/m	$L^{-1} \cdot T^{-1} \cdot C$
6-26.1	magnetic constant	μ_0	H/m	$M \cdot L \cdot C^{-2}$
6-26.2	permeability	μ	H/m	$M \cdot L \cdot C^{-2}$
6-27	relative permeability	μ_r	1	1
6-28	magnetic susceptibility	κ	1	1
6-29	magnetic polarization	J_m	T	$M \cdot T^{-1} \cdot C^{-1}$
6-30	magnetic dipole moment	j_m	Wb $\cdot$ m	$M \cdot L^3 \cdot T^{-1} \cdot C^{-1}$
6-31	coercivity	H_c	A/m	$L^{-1} \cdot T^{-1} \cdot C$
6-32	magnetic vector potential	A	Wb/m	$M \cdot L \cdot T^{-1} \cdot C^{-1}$
6-33	electromagnetic energy density	w	J/m ³	$M \cdot L^{-1} \cdot T^{-2}$
6-34	Poynting vector	S	W/m ²	$M \cdot T^{-3}$
6-35.1	phase speed of electromagnetic waves	c	m/s	$L \cdot T^{-1}$
6-35.2	speed of light in vacuum	c_0	m/s	$L \cdot T^{-1}$
6-36	source voltage	U_s	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
6-37.1	magnetic potential	V_m	A	$T^{-1} \cdot C$
6-37.2	magnetic tension	U_m	A	$T^{-1} \cdot C$
6-37.3	magnetomotive force	F_m	A	$T^{-1} \cdot C$
6-38	number of turns in a winding	N	1	1
6-39	reluctance	R_m	H ⁻¹	$M^{-1} \cdot L^{-2} \cdot C^2$
6-40	permeance	Λ	H	$M \cdot L^2 \cdot C^{-2}$
6-41.1	inductance	L	H	$M \cdot L^2 \cdot C^{-2}$
6-41.2	mutual inductance	L_{mn}	H	$M \cdot L^2 \cdot C^{-2}$
6-42.1	coupling factor	k	1	1

Item	Quantity	Symbol	Unit	Dimension
6-42.2	leakage factor	σ	1	1
6-43	conductivity	σ	S/m	$M^{-1} \cdot L^{-3} \cdot T \cdot C^2$
6-44	resistivity	ρ	$\Omega \cdot m$	$M \cdot L^3 \cdot T^{-1} \cdot C^{-2}$
6-45	power	P	W	$M \cdot L^2 \cdot T^{-3}$
6-46	resistance	R	Ω	$M \cdot L^2 \cdot T^{-1} \cdot C^{-2}$
6-47	conductance	G	S	$M^{-1} \cdot L^{-2} \cdot T \cdot C^2$
6-48	phase difference	ϕ	rad	1
6-49	electric current phasor	I	A	$T^{-1} \cdot C$
6-50	voltage phasor	U	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
6-51.1	impedance	Z	Ω	$M \cdot L^2 \cdot T^{-1} \cdot C^{-2}$
6-51.2	impedance of vacuum	Z_0	V/A	$M \cdot L^2 \cdot T^{-1} \cdot C^{-2}$
6-51.3	resistance	R	Ω	$M \cdot L^2 \cdot T^{-1} \cdot C^{-2}$
6-51.4	reactance	X	Ω	$M \cdot L^2 \cdot T^{-1} \cdot C^{-2}$
6-51.5	apparent impedance	Z	Ω	$M \cdot L^2 \cdot T^{-1} \cdot C^{-2}$
6-52.1	admittance	Y	S	$M^{-1} \cdot L^{-2} \cdot T \cdot C^2$
6-52.2	admittance of vacuum	Y_0	A/V	$M^{-1} \cdot L^{-2} \cdot T \cdot C^2$
6-52.3	conductance	G	S	$M^{-1} \cdot L^{-2} \cdot T \cdot C^2$
6-52.4	susceptance	B	S	$M^{-1} \cdot L^{-2} \cdot T \cdot C^2$
6-52.5	apparent admittance	Y	S	$M^{-1} \cdot L^{-2} \cdot T \cdot C^2$
6-53	quality factor	Q	1	1
6-54	loss factor	d	1	1
6-55	loss angle	δ	rad	1
6-56	active power	P	W	$M \cdot L^2 \cdot T^{-3}$
6-57	apparent power	S	VA	$M \cdot L^2 \cdot T^{-3}$
6-58	power factor	λ	1	1
6-59	complex power	S	VA	$M \cdot L^2 \cdot T^{-3}$
6-60	reactive power	Q	var	$M \cdot L^2 \cdot T^{-3}$
6-61	non-active power	Q'	VA	$M \cdot L^2 \cdot T^{-3}$

Item	Quantity	Symbol	Unit	Dimension
6-62	active energy	W	J	$M \cdot L^2 \cdot T^{-2}$

19. ISO 80000-7 — Light and radiation

The fifth part specified in full is ISO 80000-7, *Light and radiation*. Every item — all of 7-1.1 ... 7-37, including every sub-suffixed item — is catalogued: the speed of light and refractive index, radiant energy and its densities, radiant flux, intensity, and radiance, irradiance, exitance, and exposure, the luminous efficiency and efficacy, the luminous quantities, the photon quantities, colorimetry, colour temperature, the optical material properties, and the attenuation coefficients — each carrying its exact source as data: the part, the printed item designation, the principal quantity symbol, and the coherent SI unit symbol. Only *citation locators* are recorded; no normative content from the licensed standard is reproduced. The defining *mathematics* of selected remarks is specified in the sibling module `Part7.DefiningRelations`.

Light and radiation is where *dimension does not classify the kind* reaches its widest form, and where the catalogue meets two SI base quantities PhysLib's five-generator `Dimension` does not carry — luminous intensity and amount of substance. Both are handled by the Finkelstein–Whitehead *scale-spanning* reduction (the *Scale-spanning units* chapter, requirement *R13*), not by a new generator: the candela is the dimension of *power*, the steradian and the mole are dimension one. So the luminous quantities share the dimensions of their radiant partners — luminous flux $\equiv$ radiant flux, luminance $\equiv$ radiance, illuminance $\equiv$ irradiance — while staying distinct kinds, and a molar quantity drops the mole.

19.1. Quantity-kinds and units of ISO 80000-7

Definition

19.1.1.

group

statement">use

A catalogued kind pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "7-6.1"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream.

Lean code for Definition

19.1.1.

1 definition

structure (5 fields) · defined in PropertyKindCalculus/Iso80000/Catalogue.lean

complete

structure PropertyKindCalculus.Iso80000.CataloguedKind : Type

Proof for Definition 19.1.1

uses 0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with cite rendering e.g. ISO 80000-7, Second edition, 2019-08 item 7-6.1. The catalogue lists all 66 items in item order — the speed of light (7-1.1), radiant flux (7-4.1), luminous intensity (7-14), the photon quantities (7-19.1 ... 7-25), the molar absorption coefficient (7-37), ... — each paired with its coherent SI unit symbol.

Theorem

19.1.2.

group

uses 2

used by 0



Effort

small

Tags

proved

The candela reduces to power (item 7-14). Luminous intensity carries the dimension of power, $M \cdot L^2 \cdot T^{-3}$ — the scale-spanning reduction (R13) made a checked computation: PhysLib has no luminous-intensity generator, and none is needed, the spectral luminous efficiency $v(\lambda)$ that relates the candela to the watt being dimensionless. Uses [the dimension map](#) and [the scale-spanning units](#).

Lean code for Theorem 19.1.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part7.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part7.luminousIntensity_dim_eq_power
  PropertyKindCalculus.Iso80000.Part7.luminousIntensity.dim =
    PropertyKindCalculus.Dim.power
```

Proof for Theorem 19.1.2

uses 0

`luminousIntensity_dim_eq_power` : `luminousIntensity.dim = Dim.power`, by reflexivity over `Dim.luminousIntensity := Dim.power`. The companions `radiantFlux_dim_{mass,length,time}` record $M \cdot L^2 \cdot T^{-3}$ for radiant flux, `radiantEnergy_dim_time = -2` the joule, `photonFlux_dim_time = -1` the photon count rate, and `molarAbsorptionCoefficient_dim_length = 2` the molar coefficient as an area (the mole reduced).

19.2. The radiant / luminous / photon trios

ISO 80000-7 lists most measurands three times — a *radiant* (energetic, subscript e) version, a *luminous* (photopic, subscript v) version, and a *photon* (count, subscript p) version — and pairs them explicitly ("the corresponding photometric quantity is ..."). They are not the same kind of quantity: radiant flux is the total radiant power, luminous flux its $v(\lambda)$ -weighted response in the human eye, photon flux a count rate. A dimension-only model sees one type, "a real number of watts", for radiant and luminous flux, and cannot say why a count rate is a different thing again.

This is requirement R2 once more — the pattern Part 3 set on the length family, Part 4 on the force family, Part 5 on the thermodynamic potentials, Part 6 on the AC power quantities — now on the radiation trios. Each member is a distinct *kind* individuated *not by fiat* but by a *radiation mode* (Dybkaer's examination principle, §7.5). And the trios split two ways: the luminous mode is a *dimensionless reweighting* (luminous flux $\equiv$ radiant flux in dimension), while the photon mode is a genuine *re-dimensioning* (photon flux is T^{-1} , not $M \cdot L^2 \cdot T^{-3}$).

Definition19.2.1.

groupuses 2used by 0

A *radiation-mode kind* is a ratio-scale [quantity](#)-kind carrying an [examination principle](#) — this work's terse descriptor of which aspect of the radiation it isolates: the total energetics (subscript e), the $v(\lambda)$ -weighted photopic response (subscript v), or a photon count (subscript p). The radiant, luminous, and photon members of each measurand's trio differ in this principle.

Lean code for Definition19.2.1.

1 definition

def · defined in PropertyKindCalculus/Iso800000/Part7.lean

complete

def PropertyKindCalculus.Iso800000.Part7.modeKind (id : String) (m : PropertyKindCalculus.ExaminationPrinciple) (dim : Dimension) : PropertyKindCalculus.DimensionedKind

Proof for Definition 19.2.1

uses 0

modeKind id m dim := { kind := { id, scale := .ratio, examPrinciple := some m.id }, dim }, with the three modes RadiationMode.energetic, .photopic, .photonic. The flux trio is radiantFlux / luminousFlux / photonFlux, and likewise for energy, intensity, radiance, irradiance, exitance, and exposure.

Theorem19.2.2.

groupstatement">uses 1by 0

EffortTags

smallproved

Radiant and luminous flux are distinct kinds — by radiation mode, not by fiat. They share dimension $M \cdot L^2 \cdot T^{-3}$ and are both "flux", yet are different kinds because one is the total radiant power and the other its $v(\lambda)$ -weighted photopic response — proved through `distinct_of_examPrinciple` (§7.5). The photon member is distinct again, and even re-dimensions (T^{-1}). This is the AC-power-family pattern of Part 6 on the radiation trics. Uses [the examination principle](#).

Lean code for Theorem19.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso800000/Part7.lean

complete

theorem PropertyKindCalculus.Iso800000.Part7.radiantFlux_ne_luminousFlux : PropertyKindCalculus.Iso800000.Part7.radiantFlux.kind ≠ PropertyKindCalculus.Iso800000.Part7.luminousFlux.kind

Proof for Theorem 19.2.2

uses 0

radiantFlux_ne_luminousFlux : radiantFlux.kind $\neq$ luminousFlux.kind, from
KindOfProperty.distinct_of_examPrinciple on the differing modes radiant-energetic and luminous-
photopic-Vlambda. The companions radiantFlux_ne_photonFlux (the photon mode) and
radiance_ne_luminance cover further trios; radiantFlux_examinedBy and luminousFlux_examinedBy
check the kind-to-mode links. Axiom-free.

Theorem 19.2.3.

group

uses 2

used by 0

✓

Effort

Tags

small

proved

The luminous efficacy is dimension one because it is a ratio of two fluxes (item 7-11.1). From $K = \phi_v / \phi_e$, with luminous and radiant flux of equal dimension (the candela reducing to power), the dimension cancels and the efficacy is computed to be one — the photometric analogue of the power factor (Part 6) and the plane angle (Part 3). The scale coefficient $K_{cd} = 683 \text{ lm/W}$ is the candela's scale-spanning constant (R13). Uses [the dimension map](#) and [the quotient kind-law](#).

Lean code for Theorem 19.2.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part7/
DefiningRelations.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part7.DefiningRelations.luminousEfficacyDim :  
  PropertyKindCalculus.Iso80000.Part7.luminousEfficacy.dim =  
    PropertyKindCalculus.Iso80000.Part7.luminousFlux.dim /  
    PropertyKindCalculus.Iso80000.Part7.radiantFlux.dim
```

Proof for Theorem 19.2.3

uses 0

luminousEfficacy_dim_from_fluxes : luminousEfficacy.dim = luminousFlux.dim /
radiantFlux.dim, by reducing the right side via div_self ($a / a = 1$). The kind-law
luminousEfficacy_quot_fluxes is (rfl, rfl, rfl); the efficacy nonetheless remains a distinct kind
from every other dimension-one quantity (the collision capstone below).

19.3. Dimension does not classify; the kind does — the widest collisions in the series

ISO 80000-7 holds the widest dimension collisions of the catalogue. The candela reducing to power and the steradian to one, *radiant flux*, *luminous flux*, *radiant intensity*, and *luminous intensity* all share $M \cdot L^2 \cdot T^{-3}$; radiance, luminance, irradiance, illuminance, and the exitances all share $M \cdot T^{-3}$; and over two

dozen kinds — refractive index, the efficiencies and efficacies, emissivity, the absorptances, reflectances, and transmittances, the optical densities, the radiance and luminance factors, the chromaticity coordinates, the colour-matching functions, the photon number — share dimension one.

Theorem19.3.1.

group

statement">use

Effort

small

Tags

proved

Radiant flux is not luminous flux, though both are $M \cdot L^2 \cdot T^{-3}$. There exist distinct ISO 80000-7 kinds with the same dimension — radiant and luminous flux witness it, alongside radiant and luminous intensity. The candela reduces to power, so no dimension-only type system can separate them; the kind layer — and the unit, *w* versus *lm* — does. This is a collision between a base-quantity unit and a derived one, the scale-spanning case (R13) on the standard. Uses [the dimension map](#).

Lean code for Theorem19.3.1.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part7.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part7.iso80000_7_dim_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim
```

Proof for Theorem 19.3.1

uses 0

iso80000_7_dim_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim, witnessed by {radiantFlux, luminousFlux, ...}. The companion radiantFlux_dim_eq_radiantIntensity_dim is the *steradian* collision (flux versus intensity, the steradian being dimension one), and wattRadiant_lumen_not_commensurable / wattRadiant_wattPerSteradian_not_commensurable record the unit-level facts. Axiom-free.

Theorem19.3.2.

group

statement">use

Effort

small

Tags

proved

The dimension-1 disambiguation, on the standard — the widest in the series. There exist distinct ISO 80000-7 kinds with the same dimension one — refractive index and emissivity witness it, alongside the efficiencies, efficacies, absorptances, reflectances, transmittances, optical densities, the radiance and luminance factors, the reflectance factor, the chromaticity coordinates, the colour-matching functions, and the photon number. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem 19.3.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part7.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part7.iso80000_7_dim_one_collision :  
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 19.3.2

uses 0

```
iso80000_7_dim_one_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim ∧  
a.dim = 1, witnessed by ⟨refractiveIndex, emissivity, ...⟩ with refractiveIndex_ne_emissivity (a  
decide) and reflexivity. Axiom-free.
```

19.4. The algebraic Remarks as kind-laws: radiometry and photometry built by the kind algebra

Many Part-7 definitions state a quantity's defining relation *algebraically* — the constitutive laws of radiometry and photometry. Radiant flux is the time-derivative of radiant energy (7-4.1); photon flux of photon number (7-20); irradiance is the area-density of radiant flux (7-7.1); radiant intensity its solid-angle-density (7-5.1); the luminous efficacy is luminous flux over radiant flux (7-11.1). Each is specified as an R12 kind-law, several crossing to ISO 80000-3 for time, area, and solid angle.

Theorem 19.4.1. statement" > uses by 0

Effort small

Tags proved

Irradiance is radiant flux / area — a verified construction (item 7-7.1). An irradiance built as the quotient of a radiant flux by an area carries its quotient certificate by construction, over the $\mathbb{R}$ carrier, and canonicity makes the certificate determine the quantity. The companion `radiantIntensity_dim_from_flux_solidAngle` shows radiant intensity keeps the dimension of flux because the solid angle it divides by is dimension one — the steradian dropping, as a checked computation. Uses [the quotient kind-law](#).

Lean code for Theorem 19.4.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part7/DefiningRelations.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part7.DefiningRelations.irradianceOf_
  ( $\varphi$  :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part7.radiantFlux.kind  $\mathbb{R}$ )
  ( $a$  :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part3.area.kind  $\mathbb{R}$ ) :
  PropertyKindCalculus.Quantity.IsQuotient

PropertyKindCalculus.Iso80000.Part7.DefiningRelations.irradiance_quot_flux
_area
  (PropertyKindCalculus.Iso80000.Part7.DefiningRelations.irradianceOf
     $\varphi$   $a$ )
   $\varphi$   $a$ 

```

Proof for Theorem 19.4.1

uses 0

irradianceOf φ a := Quantity.div irradiance_quot_flux_area φ a , and irradianceOf_isQuotient is rfl; irradiance_certificate_canonical is eq_div_of_isQuotient. The same family gives luminousEfficacyOf (= Φ_v/Φ_e), and the cross-part radiantFlux_quot_energy_duration (= radiant energy / time) and photonFlux_quot_number_duration (= photon number / time), the time crossing to ISO 80000-3.

19.5. Item index — ISO 80000-7

Every catalogued item of ISO 80000-7, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLib dimension this work assigns it (over the radiation reductions). Each item number links to the formalized result it participates in — a trio distinction, a dimension collision, a defining relation, or the catalogue itself. Symbols and unit strings are *citation locators*; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
7-1.1	speed of light in a medium	c	m/s	$L \cdot T^{-1}$
7-1.2	refractive index	n	1	1
7-2.1	radiant energy	Q _e	J	$M \cdot L^2 \cdot T^{-2}$
7-2.2	spectral radiant energy	Q _{eλ}	J/nm	$M \cdot L \cdot T^{-2}$
7-3.1	radiant energy density	w	J/m ³	$M \cdot L^{-1} \cdot T^{-2}$
7-3.2	spectral radiant energy density in terms of wavelength	w _λ	J/(m ³ · nm)	$M \cdot L^{-2} \cdot T^{-2}$
7-3.3	spectral radiant energy density in terms of wavenumber	w _{ν̃}	J/m ²	$M \cdot T^{-2}$
7-4.1	radiant flux	Φ _e	W	$M \cdot L^2 \cdot T^{-3}$
7-4.2	spectral radiant flux	Φ _{eλ}	W/nm	$M \cdot L \cdot T^{-3}$
7-5.1	radiant intensity	I _e	W/sr	$M \cdot L^2 \cdot T^{-3}$
7-5.2	spectral radiant intensity	I _{eλ}	W/(sr · nm)	$M \cdot L \cdot T^{-3}$
7-6.1	radiance	L _e	W/(sr · m ²)	$M \cdot T^{-3}$
7-6.2	spectral radiance	L _{eλ}	W/(sr · m ² · nm)	$M \cdot L^{-1} \cdot T^{-3}$
7-7.1	irradiance	E _e	W/m ²	$M \cdot T^{-3}$
7-7.2	spectral irradiance	E _{eλ}	W/(m ² · nm)	$M \cdot L^{-1} \cdot T^{-3}$
7-8.1	radiant exitance	M _e	W/m ²	$M \cdot T^{-3}$
7-8.2	spectral radiant exitance	M _{eλ}	W/(m ² · nm)	$M \cdot L^{-1} \cdot T^{-3}$
7-9.1	radiant exposure	H _e	J/m ²	$M \cdot T^{-2}$
7-9.2	spectral radiant exposure	H _{eλ}	J/(m ² · nm)	$M \cdot L^{-2} \cdot T^{-2}$
7-10.1	luminous efficiency	V	1	1
7-10.2	spectral luminous efficiency	V(λ)	1	1
7-11.1	luminous efficacy of radiation	K	lm/W	1
7-11.2	spectral luminous efficacy	K(λ)	lm/W	1
7-11.3	maximum luminous efficacy	K _m	lm/W	1
7-11.4	luminous efficacy of a source	η _v	lm/W	1
7-12	luminous energy	Q _v	lm · s	$M \cdot L^2 \cdot T^{-2}$

Item	Quantity	Symbol	Unit	Dimension
7-13	luminous flux	Φ_v	lm	$M \cdot L^2 \cdot T^{-3}$
7-14	luminous intensity	I_v	cd	$M \cdot L^2 \cdot T^{-3}$
7-15	luminance	L_v	cd/m ²	$M \cdot T^{-3}$
7-16	illuminance	E_v	lx	$M \cdot T^{-3}$
7-17	luminous exitance	M_v	lm/m ²	$M \cdot T^{-3}$
7-18	luminous exposure	H_v	lx·s	$M \cdot T^{-2}$
7-19.1	photon number	N_p	1	1
7-19.2	photon energy	Q_p	J	$M \cdot L^2 \cdot T^{-2}$
7-20	photon flux	Φ_p	s ⁻¹	T ⁻¹
7-21	photon intensity	I_p	s ⁻¹ ·sr ⁻¹	T ⁻¹
7-22	photon radiance	L_p	m ⁻² ·s ⁻¹ ·sr ⁻¹	L ⁻² ·T ⁻¹
7-23	photon irradiance	E_p	m ⁻² ·s ⁻¹	L ⁻² ·T ⁻¹
7-24	photon exitance	M_p	m ⁻² ·s ⁻¹	L ⁻² ·T ⁻¹
7-25	photon exposure	H_p	m ⁻²	L ⁻²
7-26.1	tristimulus values for the CIE 1931 standard observer	X, Y, Z	1	1
7-26.2	tristimulus values for the CIE 1964 standard observer	X ₁₀ , Y ₁₀ , Z ₁₀	1	1
7-27.1	CIE colour-matching functions for the CIE 1931 standard observer	$\bar{x}$, $\bar{y}$, $\bar{z}$	1	1
7-27.2	CIE colour-matching functions for the CIE 1964 standard observer	$\bar{x}_{10}$, $\bar{y}_{10}$, $\bar{z}_{10}$	1	1
7-28.1	chromaticity coordinates in the CIE 1931 standard system	x, y, z	1	1
7-28.2	chromaticity coordinates in the CIE 1964 standard system	x ₁₀ , y ₁₀ , z ₁₀	1	1
7-29.1	colour temperature	T _c	K	θ
7-29.2	correlated colour temperature	T _{cp}	K	θ
7-30.1	emissivity	ε	1	1
7-30.2	emissivity at a specified wavelength	ε(λ)	1	1

Item	Quantity	Symbol	Unit	Dimension
7-31.1	absorptance	α	1	1
7-31.2	luminous absorptance	α_v	1	1
7-31.3	reflectance	ρ	1	1
7-31.4	luminous reflectance	ρ_v	1	1
7-31.5	transmittance	τ	1	1
7-31.6	luminous transmittance	τ_v	1	1
7-32.1	transmittance optical density	D	1	1
7-32.2	Napierian absorbance	A _n	1	1
7-33.1	radiance factor	β_e	1	1
7-33.2	luminance factor	β_v	1	1
7-34	reflectance factor	R	1	1
7-35.1	linear attenuation coefficient	μ	m ⁻¹	L ⁻¹
7-35.2	linear absorption coefficient	a _l	m ⁻¹	L ⁻¹
7-36.1	mass attenuation coefficient	μ_m	kg ⁻¹ ·m ²	M ⁻¹ ·L ²
7-36.2	mass absorption coefficient	α_m	kg ⁻¹ ·m ²	M ⁻¹ ·L ²
7-37	molar absorption coefficient	χ	m ² /mol	L ²

20. ISO 80000-8 — Acoustics

The sixth part specified in full is ISO 80000-8, *Acoustics*. Every item — all of 8-1 ... 8-17, including the sub-suffixed pressure pair (8-2.1, 8-2.2) — is catalogued: the logarithmic frequency range, static and sound pressure, the sound-particle kinematics (displacement, velocity, acceleration), volume flow rate, the sound-energy cluster (energy density, energy, power, intensity), sound exposure, the two impedances, the three logarithmic levels, and the reverberation time — each carrying its exact source as data: the part, the printed item designation, the principal quantity symbol, and the coherent SI unit symbol. Only *citation locators* are recorded; no normative content from the licensed standard is reproduced. The defining *mathematics* of selected remarks is specified in the sibling module `Part8.DefiningRelations`.

Acoustics brings two faces of the catalogue's central thesis, and — unusually — its boundary. The *logarithmic levels* (sound pressure level, sound power level, sound exposure level) are each $10 \cdot \lg(\text{quantity} / \text{reference})$, ratios on a logarithmic scale reported in the decibel: *dimension one*, three distinct kinds the dimension functor collapses to a single point. A genuine *dimension collision* has sound pressure and sound energy density carry the same dimension $M \cdot L^{-1} \cdot T^{-2}$ — force per area equals energy per volume — while staying distinct kinds. And the *boundary*: the two homonymous *impedances* are told apart by their dimension, the case where the necessary discriminator is, this time, also sufficient.

20.1. Quantity-kinds and units of ISO 80000-8

Definition 20.1.1. statement" > uses 1 by 0

A *catalogued kind* pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "8-2.2"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream.

Lean code for Definition 20.1.1.

1 definition

```
structure (5 fields) · defined in PropertyKindCalculus/Iso80000/Catalogue.lean
structure PropertyKindCalculus.Iso80000.CataloguedKind : Type
```

Proof for Definition 20.1.1 uses 0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with `cite` rendering e.g. ISO 80000-8, Second edition, 2020-02 item 8-2.2. The catalogue lists all 18 items in item order — the logarithmic frequency range (8-1), static and sound pressure (8-2.1, 8-2.2), the kinematics (8-3 ... 8-5), the sound-

energy cluster (8-7 ... 8-10), the impedances (8-12, 8-13), the levels (8-14 ... 8-16), ... — each paired with its coherent SI unit symbol.

20.2. The logarithmic levels — the acoustic face of dimension one

ISO 80000-8 lists three *levels*: the sound pressure level (8-14), the sound power level (8-15), and the sound exposure level (8-16). Each is $10 \cdot \lg(\text{quantity} / \text{reference})$ — a ratio of a quantity to a stated reference, taken on a logarithmic scale, and reported in the decibel. A level is *dimension one*: the logarithm of a ratio of like quantities. So the dimension functor sends all three to the single point 1, and a dimension-only model sees one type, "a real number of decibels", for all of them.

What keeps them apart is *which quantity each is the level of* — the pressure, the power, the exposure. This is requirement *R1* once more (the dimension does not classify the kind), in the form the part makes vivid: a shared unit symbol dB no more makes the sound pressure level the sound power level than the shared symbol 1 makes a Reynolds number a Froude number (ISO 80000-11).

Theorem **20.2.1.** **group** **statement">us**

Effort

small

Tags

proved

The decibel of sound pressure level and the decibel of sound power level are not commensurable, though both are dB and dimension one (items 8-14, 8-15). Same symbol, same dimension, different kind — the level of pressure is not the level of power. This is the ISO 80000-11 "1" pattern (Reynolds versus Euler) on the acoustic levels. Uses [the metrological unit](#).

Lean code for Theorem **20.2.1.**

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part8.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part8.decibelPressure_decibelPower_notCommensurable :
  ¬PropertyKindCalculus.Iso80000.Part8.decibelPressure.Commensurable
    PropertyKindCalculus.Iso80000.Part8.decibelPower
```

Proof for Theorem 20.2.1

uses 0

decibelPressure_decibelPower_not_commensurable, by decide after unfolding MetrologicalUnit.Commensurable on the two dimension-one kinds. The companion soundPressureLevel_ne_soundPowerLevel is the underlying kind distinction, and iso80000_8_dim_one_collision packages the dimension-one capstone. Axiom-free.

Effort

small

Tags

proved

The dimension-1 disambiguation, on the standard. There exist distinct ISO 80000-8 kinds with the same dimension one — the sound pressure level and the sound power level witness it, alongside the sound exposure level and the logarithmic frequency range. The dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem20.2.2.

1 theorem

theorem

· defined in

PropertyKindCalculus/Iso80000/Part8.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part8.iso80000_8_dim_one_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 20.2.2

uses 0

iso80000_8_dim_one_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1, witnessed by {soundPressureLevel, soundPowerLevel, ...} with the kind distinction a decide and reflexivity. Axiom-free.

20.3. A dimension collision, and the boundary where the dimension does classify

ISO 80000-8 shows both sides of the thesis. Sound pressure (8-2.2, Pa) and sound energy density (8-7, J/m³) carry the *same* dimension $M \cdot L^{-1} \cdot T^{-2}$ — pressure being force per area, energy density energy per volume, and the two equal — yet are different kinds: a collision the dimension cannot resolve.

But the part also marks the *boundary*. It lists two quantities both called an *impedance*: the characteristic impedance of a medium (8-12, Pa·s/m, $M \cdot L^{-2} \cdot T^{-1}$) and the acoustic impedance (8-13, Pa·s/m³, $M \cdot L^{-4} \cdot T^{-1}$). Here the dimension *does* tell the homonyms apart, because one is pressure over particle velocity and the other pressure over volume flow rate — the extra L^2 shows. Dimension is a *necessary* discriminator of kind, never a *sufficient* one; the impedances are the case where, this once, it is also sufficient, and naming them makes the limit of the dimension test explicit.

Effort

small

Tags

proved

Sound pressure is not sound energy density, though both are $M \cdot L^{-1} \cdot T^{-2}$ (items 8-2.2, 8-7). There exist distinct ISO 80000-8 kinds with the same dimension — sound pressure and sound energy density witness it. Force per area equals energy per volume, so no dimension-only type system can separate them; the kind layer — and the unit, Pa versus J/m³ — does. Uses [the dimension map](#).

Lean code for Theorem 20.3.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part8.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part8.iso80000_8_dim_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim
```

Proof for Theorem 20.3.1

uses 0

iso80000_8_dim_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim, witnessed by (soundPressure, soundEnergyDensity, ...). The dimension equality soundPressure_dim_eq_energyDensity_dim is a Dimension.ext computation (force/area = energy/volume); pascalSound_joulePerCubicMetre_not_commensurable records the unit-level fact. The contrasting acousticImpedance_dim_from_pressure_volumeFlow shows where the dimension instead discriminates. Axiom-free.

20.4. The algebraic Remarks as kind-laws: linear acoustics built by the kind algebra

Several Part-8 definitions state a quantity's defining relation *algebraically* — the constitutive laws of linear acoustics. Sound particle velocity is the time-derivative of displacement (8-4), acceleration of velocity (8-5); sound intensity is the *product* of sound pressure and particle velocity (8-10, $I = p \cdot u$); the two impedances are quotients of sound pressure (8-12, 8-13). Each is specified as an R12 kind-law, the kinematics crossing to ISO 80000-3 for time.

Theorem 20.4.1.

statement" > uses 1 by 0

Effort

small

Tags

proved

Sound intensity is sound pressure $\times$ particle velocity — a verified construction (item 8-10). A sound intensity built as the product of a sound pressure and a particle velocity carries its product certificate by construction, over the $\mathbb{R}$ carrier. The companion `soundIntensity_dim_from_pressure_velocity` shows the intensity is $M \cdot T^{-3}$ because the length in pressure and velocity cancels — the acoustic intensity law as a checked computation. Uses [the quotient kind-law](#).

Lean code for Theorem 20.4.1.

1 theorem

complete

theorem · defined in PropertyKindCalculus/Iso80000/Part8/DefiningRelations.lean

```

theorem PropertyKindCalculus.Iso80000.Part8.DefiningRelations.soundIntensity
  (p :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part8.soundPressure.kind  $\mathbb{R}$ )
  (u :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part8.particleVelocity.kind  $\mathbb{R}$ ) :
    PropertyKindCalculus.Quantity.IsProduct

PropertyKindCalculus.Iso80000.Part8.DefiningRelations.soundIntensity_prod_
pressure_velocity

(PropertyKindCalculus.Iso80000.Part8.DefiningRelations.soundIntensityOf
  p u)
p u

```

Proof for Theorem 20.4.1

uses 0

`soundIntensityOf p u` := `Quantity.mul soundIntensity_prod_pressure_velocity p u`, and `soundIntensityOf_isProduct` is `rfl`. The same family gives the quotient constructors `characteristicImpedanceOf (= p/u)` and the cross-part kinematic laws `particleVelocity_quot_displacement_duration (= displacement / time)`, the time crossing to ISO 80000-3.

20.5. Item index — ISO 80000-8

Every catalogued item of ISO 80000-8, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLib dimension this work assigns it. Each item number links to the formalized result it participates in — a dimension collision, a dimension-one level, a defining relation, or the catalogue itself. Symbols and unit strings are *citation locators*; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
8-1	logarithmic frequency range	G	oct	1
8-2.1	static pressure	p_s	Pa	$M \cdot L^{-1} \cdot T^{-2}$
8-2.2	sound pressure	p	Pa	$M \cdot L^{-1} \cdot T^{-2}$
8-3	sound particle displacement	δ	m	L
8-4	sound particle velocity	u	m/s	$L \cdot T^{-1}$
8-5	sound particle acceleration	a	m/s ²	$L \cdot T^{-2}$
8-6	volume flow rate	q_V	m ³ /s	$L^3 \cdot T^{-1}$
8-7	sound energy density	w	J/m ³	$M \cdot L^{-1} \cdot T^{-2}$
8-8	sound energy	Q	J	$M \cdot L^2 \cdot T^{-2}$
8-9	sound power	P	W	$M \cdot L^2 \cdot T^{-3}$
8-10	sound intensity	I	W/m ²	$M \cdot T^{-3}$
8-11	sound exposure	E	Pa ² · s	$M^2 \cdot L^{-2} \cdot T^{-3}$
8-12	characteristic impedance of a medium	Z_c	Pa · s/m	$M \cdot L^{-2} \cdot T^{-1}$
8-13	acoustic impedance	Z_a	Pa · s/m ³	$M \cdot L^{-4} \cdot T^{-1}$
8-14	sound pressure level	L_p	dB	1
8-15	sound power level	L_W	dB	1
8-16	sound exposure level	L_E	dB	1
8-17	reverberation time	T	s	T

21. ISO 80000-9 — Physical chemistry and molecular physics

The seventh part specified in full is ISO 80000-9, *Physical chemistry and molecular physics*. Every item — all of 9-1 ... 9-49, sixty-two including every sub-suffixed item — is catalogued: counting entities and the mole, the molar quantities, the concentrations and fractions, the chemical potential and the activity ladder, the reaction and equilibrium quantities, statistical mechanics, and the transport and electrochemical coefficients — each carrying its exact source as data. Only *citation locators* are recorded; no normative content from the licensed standard is reproduced. The defining *mathematics* of selected remarks is specified in the sibling module `Part9.DefiningRelations`.

Physical chemistry is the home of the *mole*, and so the sharpest test in the series of the Finkelstein–Whitehead *scale-spanning* reduction (the *Scale-spanning units* chapter, requirement *R13*): the mole is a human-selected dimensionless count of entities, N_A of them, so $\text{Dim.amountOfSubstance} = 1$, and every quantity *per mole* drops the mole. One reduction reorganizes the whole part. Amount of substance joins the dimensionless family. Molar mass *is a mass*, molar volume *a volume*, the amount-of-substance concentration *a number density*, molality *an inverse mass*. The seven J/mol quantities collide on one dimension; the three $\text{J}/(\text{mol} \cdot \text{K})$ quantities on another.

21.1. Quantity-kinds and units of ISO 80000-9

Definition 21.1.1.

statement" > uses 1 by 0

A *catalogued kind* pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "9-12.1"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream.

Lean code for Definition 21.1.1.

1 definition

```
structure (5 fields) · defined in PropertyKindCalculus/Iso80000/Catalogue.lean
```

complete

```
structure PropertyKindCalculus.Iso80000.CataloguedKind : Type
```

Proof for Definition 21.1.1

uses 0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with cite rendering e.g. ISO 80000-9, Second edition, 2019-08 item 9-12.1. The catalogue lists all 62 items in item order — the mole (9-2), the molar quantities (9-4 ... 9-8), the concentrations (9-9.1 ... 9-12.2), the activities (9-18 ... 9-27.3),

the partition functions (9-35.1 ... 9-35.4), the conductivities (9-44, 9-45), ... — each paired with its coherent SI unit symbol.

21.2. The mole reduces to dimension one — R13 at its sharpest

ISO 80000-9 is built on the *mole*. In the scale-spanning reading (R13), the mole is not a base dimension but a human-selected dimensionless *count*: N_A entities. So amount of substance is *dimension one*, and the $/\text{mol}$ in every molar quantity is invisible to the dimension functor. Molar mass (kg/mol) carries the dimension of *mass*; molar volume (m^3/mol) the dimension of *volume*; the amount-of-substance concentration (mol/m^3) the dimension of a *number density* L^{-3} — the very same dimension as a particle concentration; molality (mol/kg) the dimension of an *inverse mass*.

And amount of substance itself joins the dimensionless family, distinct in kind from the number of entities it counts ($N = n \cdot N_A$) — both dimension one, two kinds. This is requirement *R13* in the place that makes it unavoidable: a mol is no more a base unit than a cd (ISO 80000-7) or a Sh (IEC 80000-13).

Theorem **21.2.1.** **group** **uses 2** **used by 0** ✓

Effort
small

Tags

capstone

proved

Molar mass is a mass — the mole reduces (item 9-4). With the mole dimension one, kg/mol carries the dimension of mass, M . The companion facts record the rest of the reduction: the amount-of-substance concentration is a number density (equal to the particle concentration), molality is an inverse mass, and amount of substance is itself dimension one. This is the scale-spanning reduction (R13) at the place the mole lives. Uses [the dimension map](#) and [the scale-spanning units](#).

Lean code for Theorem 21.2.1.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part9.lean

complete

theorem PropertyKindCalculus.Iso80000.Part9.molarMass_dim_eq_mass :
PropertyKindCalculus.Iso80000.Part9.molarMass.dim =
PropertyKindCalculus.Dim.mass

Proof for Theorem 21.2.1

uses 0

molarMass_dim_eq_mass : molarMass.dim = Dim.mass, by reflexivity (the kind is declared at Dim.mass).
The companions amountOfSubstance_dim_eq_one,
amountConcentration_dim_eq_particleConcentration, and molality_dim_eq_inverseMass record the

rest; the defining-relation versions (molarMass_dim_from_mass_amount, amountConcentration_dim_from_amount_volume) derive the reduction *from* $M = m/n$ and $c = n/V$ with $\dim n = 1$. Axiom-free.

Theorem 21.2.2.

group

statement">uses

Effort

small

Tags

proved

The dimension-1 disambiguation, on the standard — with the mole inside it. There exist distinct ISO 80000-9 kinds with the same dimension one — amount of substance (the mole, reduced by R13) and the number of entities witness it, alongside the fractions, the activities, and the partition functions. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem 21.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part9.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part9.iso80000_9_dim_one_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 21.2.2

uses 0

iso80000_9_dim_one_collision, witnessed by (amountOfSubstance, numberOfEntities, ...) with the kind distinction a decide and reflexivity. Axiom-free.

21.3. The seven-fold J/mol collision

The mole reducing to dimension one, the seven energies-per-mole — molar internal energy, molar enthalpy, molar Helmholtz and Gibbs energies (9-6.1 ... 9-6.4), the chemical potential (9-17), the standard chemical potential (9-21), and the affinity of a reaction (9-30) — all carry the *same* dimension $M \cdot L^2 \cdot T^{-2}$ and the *same* unit string J/mol , yet are seven distinct kinds. A dimension-only model sees one type, "a real number of joules per mole", for all seven; the kind layer keeps the chemical potential from being read as an affinity. Likewise the molar heat capacity (9-7), molar entropy (9-8), and molar gas constant (9-37.1) all share $\text{J}/(\text{mol} \cdot \text{K})$.

Theorem 21.3.1.

statement">uses by 0

Effort

small

Tags

proved

Molar internal energy is not molar Gibbs energy, though both are J/mol and $\text{M} \cdot \text{L}^2 \cdot \text{T}^{-2}$ (items 9-6.1, 9-6.4). There exist distinct ISO 80000-9 kinds with the same dimension — seven J/mol quantities witness it. The mole reduces to dimension one, so no dimension-only type system can separate them; the kind layer — and the unit — does. Uses [the dimension map](#).

Lean code for Theorem 21.3.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part9.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part9.iso80000_9_dim_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim
```

Proof for Theorem 21.3.1

uses 0

iso80000_9_dim_collision, witnessed by (molarInternalEnergy, molarGibbsEnergy, ...). The unit fact jPerMol_internalEnergy_gibbs_not_commensurable is a decide; molarHeatCapacity_dim_eq_molarEntropy_dim records the parallel $\text{J}/(\text{mol} \cdot \text{K})$ collision, and molPerKg_molality_ionicStrength_not_commensurable the mol/kg one. Axiom-free.

21.4. The algebraic Remarks as kind-laws: physical chemistry built from the other parts

The molar quantities of ISO 80000-9 are defined as quotients *by the amount of substance* — molar mass is m/n (9-4), molar volume V/n (9-5), the amount concentration n/V (9-12.1), molality n/m (9-15), the molar energies U/n (9-6.1) — composing Part-9 kinds out of ISO 80000-3 (volume), -4 (mass), and -5 (energy) kinds and the mole. Each is specified as an R12 kind-law.

Theorem 21.4.1.

statement" uses 1 by 0

Effort

Tags

small

proved

Molar mass is mass / amount of substance — a verified construction (item 9-4). A molar mass built as the quotient of a mass by an amount of substance carries its quotient certificate by construction, over the $\mathbb{R}$ carrier, and canonicity makes the certificate determine the quantity. The companion molarMass_dim_from_mass_amount shows the result is a mass because the mole is dimension one. Uses [the quotient kind-law](#).

theorem · defined in PropertyKindCalculus/Iso80000/Part9/DefiningRelations.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part9.DefiningRelations.molarMassOf_i
  (m :
    PropertyKindCalculus.Quantity
    PropertyKindCalculus.Iso80000.Part4.mass.kind ℝ)
  (n :
    PropertyKindCalculus.Quantity
    PropertyKindCalculus.Iso80000.Part9.amountOfSubstance.kind ℝ) :
  PropertyKindCalculus.Quantity.IsQuotient

PropertyKindCalculus.Iso80000.Part9.DefiningRelations.molarMass_quot_mass_
amount
  (PropertyKindCalculus.Iso80000.Part9.DefiningRelations.molarMassOf m
    n)
  m n

```

Proof for Theorem 21.4.1

uses 0

molarMassOf m n := Quantity.div molarMass_quot_mass_amount m n, and molarMassOf_isQuotient is rfl; molarMass_certificate_canonical is eq_div_of_isQuotient. The same family gives amountConcentrationOf (= n/V) and the cross-part kind-laws molarVolume_quot_volume_amount, molarInternalEnergy_quot_energy_amount, and molality_quot_amount_mass, crossing to ISO 80000-3/-4/-5.

21.5. Item index — ISO 80000-9

Every catalogued item of ISO 80000-9, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLib dimension this work assigns it (the mole reduced). Each item number links to the formalized result it participates in. Symbols and unit strings are *citation locators*; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
9-1	number of entities	N	1	1
9-2	amount of substance	n	mol	1
9-3	relative atomic mass	A _r	1	1
9-4	molar mass	M	kg/mol	M
9-5	molar volume	V _m	m ³ /mol	L ³
9-6.1	molar internal energy	U _m	J/mol	M · L ² · T ⁻²
9-6.2	molar enthalpy	H _m	J/mol	M · L ² · T ⁻²
9-6.3	molar Helmholtz energy	F _m	J/mol	M · L ² · T ⁻²
9-6.4	molar Gibbs energy	G _m	J/mol	M · L ² · T ⁻²
9-7	molar heat capacity	C _m	J/(mol · K)	M · L ² · T ⁻² · Θ ⁻¹
9-8	molar entropy	S _m	J/(mol · K)	M · L ² · T ⁻² · Θ ⁻¹
9-9.1	particle concentration	n	m ⁻³	L ⁻³
9-9.2	molecular concentration	C	m ⁻³	L ⁻³
9-10	mass concentration	γ	kg/m ³	M · L ⁻³
9-11	mass fraction	w	1	1
9-12.1	amount-of-substance concentration	c	mol/m ³	L ⁻³
9-12.2	standard amount-of-substance concentration	c [°]	mol/m ³	L ⁻³
9-13	mole fraction	x	1	1
9-14	volume fraction	φ	1	1
9-15	molality	b	mol/kg	M ⁻¹
9-16	latent heat of phase transition	C _{pt}	J	M · L ² · T ⁻²
9-17	chemical potential	μ	J/mol	M · L ² · T ⁻²
9-18	absolute activity	λ	1	1
9-19	partial pressure	p	Pa	M · L ⁻¹ · T ⁻²
9-20	fugacity	p̂	Pa	M · L ⁻¹ · T ⁻²
9-21	standard chemical potential	μ [°]	J/mol	M · L ² · T ⁻²
9-22	activity factor	f	1	1
9-23	standard absolute activity in a mixture	λ [°]	1	1

Item	Quantity	Symbol	Unit	Dimension
9-24	activity of solute	a	1	1
9-25	activity coefficient	γ	1	1
9-26	standard absolute activity in a solution	λ°	1	1
9-27.1	activity of solvent	a_A	1	1
9-27.2	osmotic coefficient of solvent	ϕ	1	1
9-27.3	standard absolute activity of solvent	λ_A°	1	1
9-28	osmotic pressure	Π	Pa	$M \cdot L^{-1} \cdot T^{-2}$
9-29	stoichiometric number	ν	1	1
9-30	affinity of a chemical reaction	A	J/mol	$M \cdot L^2 \cdot T^{-2}$
9-31	extent of reaction	ξ	mol	1
9-32	standard equilibrium constant	K°	1	1
9-33	equilibrium constant (pressure basis)	K_p	$\text{Pa}^{(\sum \nu)}$	1
9-34	equilibrium constant (concentration basis)	K_c	$(\text{mol}/\text{m}^3)^{(\sum \nu)}$	1
9-35.1	microcanonical partition function	Ω	1	1
9-35.2	canonical partition function	Z	1	1
9-35.3	grand-canonical partition function	Ξ	1	1
9-35.4	molecular partition function	q	1	1
9-36.1	statistical weight of subsystem	g	1	1
9-36.2	degeneracy	g	1	1
9-37.1	molar gas constant	R	J/(mol·K)	$M \cdot L^2 \cdot T^{-2} \cdot \theta^{-1}$
9-37.2	specific gas constant	R_s	J/(kg·K)	$L^2 \cdot T^{-2} \cdot \theta^{-1}$
9-38	mean free path	l	m	L
9-39	diffusion coefficient	D	m ² /s	$L^2 \cdot T^{-1}$
9-40.1	thermal diffusion ratio	k_T	1	1
9-40.2	thermal diffusion factor	α_T	1	1
9-41	thermal diffusion coefficient	D_T	m ² /s	$L^2 \cdot T^{-1}$
9-42	ionic strength	I	mol/kg	M^{-1}
9-43	degree of dissociation	α	1	1

Item	Quantity	Symbol	Unit	Dimension
9-44	electrolytic conductivity	κ	S/m	$M^{-1} \cdot L^{-3} \cdot T \cdot C^2$
9-45	molar conductivity	Λ_m	$S \cdot m^2/mol$	$M^{-1} \cdot L^{-1} \cdot T \cdot C^2$
9-46	transport number of the ion	t	1	1
9-47	angle of optical rotation	α	rad	1
9-48	molar optical rotatory power	α_m	$rad \cdot m^2/mol$	L^2
9-49	specific optical rotatory power	α_m	$rad \cdot m^2/kg$	$M^{-1} \cdot L^2$

22. ISO 80000-10 — Atomic and nuclear physics

The eighth part specified in full is ISO 80000-10, *Atomic and nuclear physics*. Every item — all of 10-1.1 ... 10-89, one hundred and twenty-five including every sub-suffixed item — is catalogued: the nucleon numbers and masses, the atomic constants and magnetic moments, the quantum numbers, the precession frequencies, the decay and activity quantities, the cross sections, the fluences, the attenuation and stopping powers, the transport-in-matter and reactor quantities, and the dosimetry family — each carrying its exact source as data. Only *citation locators* are recorded; no normative content from the licensed standard is reproduced. The defining *mathematics* of selected remarks is specified in the sibling module `Part10.DefiningRelations`.

Atomic and nuclear physics holds the series' clearest external confirmation of the *dimension does not classify the kind* thesis — the standard itself coins *two unit names for one dimension*. The absorbed dose, specific energy imparted, and kerma (all the gray) and the dose equivalent (the sievert) all carry dimension $L^2 \cdot T^{-2}$ (energy per mass), and the standard separates them by unit name *because the kinds differ*: the gray for the physical doses, the sievert for the biologically-weighted one. This is the entropy-versus-heat-capacity J/K collision of ISO 80000-5, made sharper. A second case is the becquerel, the special name SI reserves for s^{-1} as *the unit of activity*, a kind the dimension cannot tell from the decay constant.

22.1. Quantity-kinds and units of ISO 80000-10

Definition

22.1.1.

statement" > uses 1 by 0

A *catalogued kind* pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "10-81.1"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream.

Lean code for Definition

22.1.1.

1 definition

structure (5 fields) · defined in PropertyKindCalculus/Iso80000/Catalogue.lean

complete

structure PropertyKindCalculus.Iso80000.CataloguedKind : Type

Proof for Definition 22.1.1

uses 0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with cite rendering e.g. ISO 80000-10, Second edition, 2019-08 item 10-81.1. The catalogue lists all 125 items in item order — the nucleon numbers (10-1.1 ... 10-1.3), the activity (10-27), the cross sections (10-38.1 ... 10-42.2), the dosimetry family (10-80.1 ... 10-89), ... — each paired with its coherent SI unit symbol.

22.2. The gray/sievert collision — the standard's own disambiguation

Four quantities of ISO 80000-10 carry the dimension $L^2 \cdot T^{-2}$ (energy per mass): the absorbed dose (10-81.1), the specific energy imparted (10-81.2), the kerma (10-86.1) — all in *grays* — and the dose equivalent (10-83.1) — in *sieverts*. The gray and the sievert are the *same* coherent unit, J/kg; the standard gives them different special names *precisely because the kinds differ*: the physical energy deposited per mass, versus that energy weighted for biological effect.

This is requirement *R1* (the dimension does not classify the kind) confirmed by the standard's own hand. A dimension-only model sees one type, "a real number of joules per kilogram", for absorbed dose and dose equivalent — and would let a gray be silently read as a sievert, the exact error the two unit names exist to prevent. It is the entropy-versus-heat-capacity J/K collision of ISO 80000-5, made sharper: here the collision is between two *named* units, not two uses of one.

Theorem22.2.1groupstatement"uses by 0

Effort
small

Tags

capstone

proved

Absorbed dose is not dose equivalent, though both are J/kg and $L^2 \cdot T^{-2}$ (items 10-81.1, 10-83.1). There exist distinct ISO 80000-10 kinds with the same dimension — absorbed dose and dose equivalent witness it, alongside the specific energy imparted and the kerma. The standard assigns two unit names (*gray*, *sievert*) to one dimension; no dimension-only type system can separate them, the kind layer — and the unit — does. Uses [the dimension map](#).

Lean code for Theorem22.2.1

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part10.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part10.iso80000_10_dim_collision :
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim
```

Proof for Theorem 22.2.1uses 0

iso80000_10_dim_collision, witnessed by (absorbedDose, doseEquivalent, ...). The unit fact gray_sievert_not_commensurable is a decide. Axiom-free.

Theorem22.2.2groupstatement"uses by 0

Effort	Tags
small	proved

The becquerel of activity and the reciprocal second of the decay constant are not commensurable, though both are T^{-1} (items 10-27, 10-24). The becquerel is the special name SI reserves for s^{-1} as the unit of activity — a kind distinction the dimension cannot see, shared also with the particle emission rate and the Larmor and cyclotron frequencies. Uses [the metrological unit](#).

Lean code for Theorem 22.2.2.

1 theorem

theorem

· defined in

PropertyKindCalculus/Iso80000/Part10.lean

complete

theorem

PropertyKindCalculus.Iso80000.Part10.becquerel_perSecond_not_commen

¬PropertyKindCalculus.Iso80000.Part10.becquerel.Commensurable

PropertyKindCalculus.Iso80000.Part10.perSecondDecay

Proof for Theorem 22.2.2

uses 0

becquerel_perSecond_not_commensurable,

by decide after unfolding MetrologicalUnit.Commensurable.

The underlying kind distinction is activity_ne_decayConstant, and

activity_dim_eq_decayConstant_dim records the shared dimension. Axiom-free.

22.3. The dimension-one family — the widest in the physical parts

ISO 80000-10 carries the largest dimension-one family of the physical parts: the atomic, neutron, and nucleon numbers; the eight quantum numbers (10-13.1 ... 10-13.8); the two g-factors; the relative mass excess and defect; the packing and binding fractions; the internal conversion and quality factors; the total ionization; and the reactor factors (resonance escape, fast fission, thermal utilization, non-leakage, multiplication). All share the dimension one; what tells them apart is the kind.

Theorem 22.3.1.

group statement">uses 1 by 0

Effort	Tags
small	proved

The dimension-1 disambiguation, on the standard — the widest in the physical parts. There exist distinct ISO 80000-10 kinds with the same dimension one — the atomic and neutron numbers

witness it, alongside the quantum numbers, the g-factors, and the reactor factors. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem 22.3.1.

1 theorem

complete

```
theorem · defined in PropertyKindCalculus/Iso80000/Part10.lean

theorem PropertyKindCalculus.Iso80000.Part10.iso80000_10_dim_one_collision
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 22.3.1

uses 0

iso80000_10_dim_one_collision, witnessed by {atomicNumber, neutronNumber, ...} with the kind distinction a decide and reflexivity. Axiom-free.

22.4. The algebraic Remarks as kind-laws: dosimetry built by the kind algebra

Several Part-10 definitions state a quantity's defining relation *algebraically*. Dose equivalent is the *product* of absorbed dose with the dimensionless quality factor (10-83.1, $H = D \cdot Q$); mean life is the *reciprocal* of the decay constant (10-25, $\tau = 1/\lambda$); specific activity is activity per mass (10-28); the absorbed-dose rate is absorbed dose per time (10-84); the mass attenuation coefficient is linear attenuation per mass density (10-50). Each is specified as an R12 kind-law, several crossing to ISO 80000-3 and -4.

Theorem 22.4.1.

statement" > uses by 0

Effort	Tags
small	proved

Dose equivalent is absorbed dose × quality factor — a verified construction (item 10-83.1). A dose equivalent built as the product of an absorbed dose and the dimensionless quality factor carries its product certificate by construction, over the $\mathbb{R}$ carrier. The companion `doseEquivalent_dim_from_dose_quality` shows it keeps the dimension of absorbed dose $L^2 \cdot T^{-2}$ because the quality factor is dimension one — the gray/sievert collision as a checked computation. Uses [the quotient kind-law](#).

theorem · defined in PropertyKindCalculus/Iso80000/Part10/DefiningRelations.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part10.DefiningRelations.doseEquivalent
  (d :
    PropertyKindCalculus.Quantity
    PropertyKindCalculus.Iso80000.Part10.absorbedDose.kind ℝ)
  (q :
    PropertyKindCalculus.Quantity
    PropertyKindCalculus.Iso80000.Part10.qualityFactor.kind ℝ) :
  PropertyKindCalculus.Quantity.IsProduct

PropertyKindCalculus.Iso80000.Part10.DefiningRelations.doseEquivalent_prod
_dose_quality

(PropertyKindCalculus.Iso80000.Part10.DefiningRelations.doseEquivalentOf
  d q)
  d q

```

Proof for Theorem 22.4.1

uses 0

doseEquivalentOf d q := Quantity.mul doseEquivalent_prod_dose_quality d q, and doseEquivalentOf_isProduct is rfl. The same families give specificActivityOf (= A/m) and the reciprocal/quotient kind-laws meanLife_recip_decayConstant, absorbedDoseRate_quot_dose_duration, and massAttenuation_quot_linear_density, crossing to ISO 80000-3 and -4.

22.5. Item index — ISO 80000-10

Every catalogued item of ISO 80000-10, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the PhysLib dimension this work assigns it. Each item number links to the formalized result it participates in. Symbols and unit strings are *citation locators*; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
10-1.1	atomic number	Z	1	1
10-1.2	neutron number	N	1	1
10-1.3	nucleon number	A	1	1
10-2	rest mass	m	kg	M
10-3	rest energy	E_0	J	$M \cdot L^2 \cdot T^{-2}$
10-4.1	atomic mass	m_a	kg	M
10-4.2	nuclidic mass	m	kg	M
10-4.3	unified atomic mass constant	m_u	kg	M
10-5.1	elementary charge	e	C	C
10-5.2	charge number	c	1	1
10-6	Bohr radius	a_0	m	L
10-7	Rydberg constant	R_∞	m^{-1}	L^{-1}
10-8	Hartree energy	E_H	eV	$M \cdot L^2 \cdot T^{-2}$
10-9.1	magnetic dipole moment	μ	$m^2 \cdot A$	$L^2 \cdot T^{-1} \cdot C$
10-9.2	Bohr magneton	μ_B	$m^2 \cdot A$	$L^2 \cdot T^{-1} \cdot C$
10-9.3	nuclear magneton	μ_N	$m^2 \cdot A$	$L^2 \cdot T^{-1} \cdot C$
10-10	spin	s	J · s	$M \cdot L^2 \cdot T^{-1}$
10-11	total angular momentum	J	J · s	$M \cdot L^2 \cdot T^{-1}$
10-12.1	gyromagnetic ratio	γ	$A \cdot m^2 \cdot J^{-1} \cdot s^{-1}$	$M^{-1} \cdot C$
10-12.2	gyromagnetic ratio of the electron	γ_e	$A \cdot m^2 \cdot J^{-1} \cdot s^{-1}$	$M^{-1} \cdot C$
10-13.1	quantum number	N	1	1
10-13.2	principal quantum number	n	1	1
10-13.3	orbital angular momentum quantum number	l	1	1
10-13.4	magnetic quantum number	m	1	1
10-13.5	spin quantum number	s	1	1
10-13.6	total angular momentum quantum number	j	1	1
10-13.7	nuclear spin quantum number	I	1	1
10-13.8	hyperfine structure quantum number	F	1	1

Item	Quantity	Symbol	Unit	Dimension
10-14.1	Landé factor	g	1	1
10-14.2	g factor of nucleus	g	1	1
10-15.1	Larmor angular frequency	ω_L	s^{-1}	T^{-1}
10-15.2	Larmor frequency	ν_L	s^{-1}	T^{-1}
10-15.3	nuclear precession angular frequency	ω_N	s^{-1}	T^{-1}
10-16	cyclotron angular frequency	ω_c	s^{-1}	T^{-1}
10-17	gyroradius	r_g	m	L
10-18	nuclear quadrupole moment	Q	m^2	L^2
10-19.1	nuclear radius	R	m	L
10-19.2	electron radius	r_e	m	L
10-20	Compton wavelength	λ_C	m	L
10-21.1	mass excess	Δ	kg	M
10-21.2	mass defect	B	kg	M
10-22.1	relative mass excess	Δ_r	1	1
10-22.2	relative mass defect	B_r	1	1
10-23.1	packing fraction	f	1	1
10-23.2	binding fraction	b	1	1
10-24	decay constant	λ	s^{-1}	T^{-1}
10-25	mean duration of life	τ	s	T
10-26	level width	Γ	eV	$M \cdot L^2 \cdot T^{-2}$
10-27	activity	A	Bq	T^{-1}
10-28	specific activity	a	Bq/kg	$M^{-1} \cdot T^{-1}$
10-29	activity density	c_A	Bq/m ³	$L^{-3} \cdot T^{-1}$
10-30	surface-activity density	a_S	Bq/m ²	$L^{-2} \cdot T^{-1}$
10-31	half life	$T_{1/2}$	s	T
10-32	alpha disintegration energy	Q_α	eV	$M \cdot L^2 \cdot T^{-2}$
10-33	maximum beta-particle energy	E_β	eV	$M \cdot L^2 \cdot T^{-2}$
10-34	beta disintegration energy	Q_β	eV	$M \cdot L^2 \cdot T^{-2}$

Item	Quantity	Symbol	Unit	Dimension
10-35	internal conversion factor	α	1	1
10-36	particle emission rate	$\dot{N}$	s^{-1}	T^{-1}
10-37.1	reaction energy	Q	eV	$M \cdot L^2 \cdot T^{-2}$
10-37.2	resonance energy	E_r	eV	$M \cdot L^2 \cdot T^{-2}$
10-38.1	cross section	σ	m^2	L^2
10-38.2	total cross section	σ_{tot}	m^2	L^2
10-39	direction distribution of cross section	σ_{Ω}	$m^2 \cdot sr^{-1}$	L^2
10-40	energy distribution of cross section	σ_E	m^2/J	$M^{-1} \cdot T^2$
10-41	direction and energy distribution of cross section	$\sigma_{\Omega E}$	$m^2/(J \cdot sr)$	$M^{-1} \cdot T^2$
10-42.1	macroscopic cross section	Σ	m^{-1}	L^{-1}
10-42.2	macroscopic total cross section	Σ_{tot}	m^{-1}	L^{-1}
10-43	particle fluence	Φ	m^{-2}	L^{-2}
10-44	particle fluence rate	$\dot{\Phi}$	$m^{-2} \cdot s^{-1}$	$L^{-2} \cdot T^{-1}$
10-45	radiant energy	R	eV	$M \cdot L^2 \cdot T^{-2}$
10-46	energy fluence	Ψ	eV/m^2	$M \cdot T^{-2}$
10-47	energy fluence rate	$\dot{\Psi}$	W/m^2	$M \cdot T^{-3}$
10-48	particle current density	J	$m^{-2} \cdot s^{-1}$	$L^{-2} \cdot T^{-1}$
10-49	linear attenuation coefficient	μ	m^{-1}	L^{-1}
10-50	mass attenuation coefficient	μ_m	$kg^{-1} \cdot m^2$	$M^{-1} \cdot L^2$
10-51	molar attenuation coefficient	μ_c	$m^2 \cdot mol^{-1}$	L^2
10-52	atomic attenuation coefficient	μ_a	m^2	L^2
10-53	half-value thickness	$d_{\frac{1}{2}}$	m	L
10-54	total linear stopping power	S	eV/m	$M \cdot L \cdot T^{-2}$
10-55	total mass stopping power	S_m	$eV \cdot m^2/kg$	$L^4 \cdot T^{-2}$
10-56	mean linear range	R	m	L
10-57	mean mass range	R_{ρ}	$kg \cdot m^{-2}$	$M \cdot L^{-2}$
10-58	linear ionization	N_{il}	m^{-1}	L^{-1}
10-59	total ionization	N_i	1	1

Item	Quantity	Symbol	Unit	Dimension
10-60	average energy loss per elementary charge produced	W_i	eV	$M \cdot L^2 \cdot T^{-2}$
10-61	mobility	μ	$m^2 / (V \cdot s)$	$M^{-1} \cdot T \cdot C$
10-62.1	particle number density	n	m^{-3}	L^{-3}
10-62.2	ion number density	n^+	m^{-3}	L^{-3}
10-63	recombination coefficient	α	$m^3 \cdot s^{-1}$	$L^3 \cdot T^{-1}$
10-64	diffusion coefficient	D	$m^2 \cdot s^{-1}$	$L^2 \cdot T^{-1}$
10-65	diffusion coefficient for fluence rate	D_ϕ	m	L
10-66	particle source density	S	$m^{-3} \cdot s^{-1}$	$L^{-3} \cdot T^{-1}$
10-67	slowing-down density	q	$m^{-3} \cdot s^{-1}$	$L^{-3} \cdot T^{-1}$
10-68	resonance escape probability	p	1	1
10-69	lethargy	u	1	1
10-70	average logarithmic energy decrement	ζ	1	1
10-71	mean free path	λ	m	L
10-72.1	slowing-down area	L^2_s	m^2	L^2
10-72.2	diffusion area	L^2	m^2	L^2
10-72.3	migration area	M^2	m^2	L^2
10-73.1	slowing-down length	L_s	m	L
10-73.2	diffusion length	L	m	L
10-73.3	migration length	M	m	L
10-74.1	neutron yield per fission	ν	1	1
10-74.2	neutron yield per absorption	η	1	1
10-75	fast fission factor	ϵ	1	1
10-76	thermal utilization factor	f	1	1
10-77	non-leakage probability	Λ	1	1
10-78.1	multiplication factor	k	1	1
10-78.2	infinite multiplication factor	k_∞	1	1
10-79	reactor time constant	T	s	T

Item	Quantity	Symbol	Unit	Dimension
10-80.1	energy imparted	ϵ	eV	$M \cdot L^2 \cdot T^{-2}$
10-80.2	mean energy imparted	$\bar{\epsilon}$	eV	$M \cdot L^2 \cdot T^{-2}$
10-81.1	absorbed dose	D	Gy	$L^2 \cdot T^{-2}$
10-81.2	specific energy imparted	z	Gy	$L^2 \cdot T^{-2}$
10-82	quality factor	Q	1	1
10-83.1	dose equivalent	H	Sv	$L^2 \cdot T^{-2}$
10-83.2	dose equivalent rate	$\dot{H}$	Sv/s	$L^2 \cdot T^{-3}$
10-84	absorbed-dose rate	$\dot{D}$	Gy/s	$L^2 \cdot T^{-3}$
10-85	linear energy transfer	L_{Δ}	eV/m	$M \cdot L \cdot T^{-2}$
10-86.1	kerma	K	Gy	$L^2 \cdot T^{-2}$
10-86.2	kerma rate	$\dot{K}$	Gy/s	$L^2 \cdot T^{-3}$
10-87	mass energy-transfer coefficient	μ_{tr}/ρ	$kg^{-1} \cdot m^2$	$M^{-1} \cdot L^2$
10-88	exposure	X	C/kg	$M^{-1} \cdot C$
10-89	exposure rate	$\dot{X}$	C/(kg·s)	$M^{-1} \cdot T^{-1} \cdot C$

23. ISO 80000-11 — Characteristic numbers

The sixth part specified is ISO 80000-11, *Characteristic numbers* — and it is where the *dimension does not classify the kind* thesis (*R1*) becomes *total*. Every item is catalogued: all 115 of items 11-4.1 ... 11-9.2, across the six clauses — momentum transfer (the Reynolds, Euler, Froude, Mach, Knudsen, ... numbers), transfer of heat (Fourier, Péclet, Rayleigh, Nusselt, Biot, ...), transfer of matter in a binary mixture (the mass-transfer analogues), constants of matter (Prandtl, Schmidt, Lewis, ...), magnetohydrodynamics (the magnetic and electric numbers), and miscellaneous — each carrying its exact source as data: the part, the printed item designation, the principal quantity symbol, and the coherent SI unit symbol. Only *citation locators* are recorded; no normative content from the licensed standard is reproduced.

A characteristic number is a *dimensionless* ratio. So **every one of the 115 is dimension one**: the dimension functor sends the Reynolds number (inertial / viscous forces), the Mach number (flow speed / sound speed), and the Prandtl number (momentum / thermal diffusivity) all to the *single point* 1. A dimension-only model sees one type — "a real number" — for the entire part and cannot tell any two of its 115 members apart. What keeps them apart is their *measurement principle*: the physical ratio each one expresses, carried as an examination principle (requirement *R2*). This is the pattern Part 3 set on the length family, Part 4 on the force family, Part 7 on the radiation trios — here at its widest, over a part with *no* dimensional variation at all.

23.1. Quantity-kinds of ISO 80000-11 — one dimension, 115 kinds

Definition23.1.1.

groupstatement">uses1by0

A *catalogued kind* pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "11-4.1"), the principal quantity symbol (e.g. Re), and the coherent SI unit symbol (1 for every characteristic number). The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream.

Lean code for Definition23.1.1.

1 definition

structure (5 fields) · defined in PropertyKindCalculus/Iso80000/Catalogue.lean

complete

structure PropertyKindCalculus.Iso80000.CataloguedKind : Type

Proof for Definition 23.1.1

uses 0

Realized as structure CataloguedKind over a DimensionedKind, with cite rendering e.g. ISO 80000-11, Second edition, 2019-10 item 11-4.1. The catalogue lists all 115 items in item order across clauses 4 ... 9; catalogue_length : catalogue.length = 115 checks the count.

Definition 23.1.2.

group

uses 2

used by 0

✓

A *characteristic number* is a ratio-scale [quantity](#)-kind of *dimension one*, individuated by its [examination principle](#) — the *measurement principle*, formed as context: defining-ratio. The context is one of ISO 80000-11's six clauses (momentum transfer, heat transfer, mass transfer, constants of matter, magnetohydrodynamics, miscellaneous); it is part of the principle because the same ratio name recurs across clauses as a *different* kind.

Lean code for Definition 23.1.2.

1 definition

```
def · defined in PropertyKindCalculus/Iso80000/Part11.lean
```

complete

```
def PropertyKindCalculus.Iso80000.Part11.charNum
  (ctx : PropertyKindCalculus.Iso80000.Part11.Context)
  (id principle : String) : PropertyKindCalculus.DimensionedKind
```

Proof for Definition 23.1.2

uses 0

charNum ctx id principle := { kind := { id, scale := .ratio, examPrinciple := some (ctx.id +
+ ": " ++ principle) }, dim := Dim.one }, with the clause Context an enumeration (momentum, heat,
mass, constants, mhd, misc). Every kind in the part is built by this one constructor.

Theorem 23.1.3.

group

statement" > uses 1 by 0

✓

Effort Tags

small

proved

Every characteristic number is dimension one — the widest collapse in the series. All 115 catalogued kinds share the dimensionless dimension; the [dimension functor](#) collapses the entire part to one point. What distinguishes them is the measurement principle, never the dimension. Uses [the dimension map](#).

Lean code for Theorem 23.1.3.

1 theorem

```
theorem · defined in PropertyKindCalculus/Iso80000/Part11.lean
```

complete

```

theorem PropertyKindCalculus.Iso80000.Part11.catalogue_all_dim_one
  (c : PropertyKindCalculus.Iso80000.CataloguedKind) :
  c ∈ PropertyKindCalculus.Iso80000.Part11.catalogue → c.qk.dim = 1

```

Proof for Theorem 23.1.3

uses 0

catalogue_all_dim_one : $\forall c \in \text{catalogue}, c.\text{qk}.\text{dim} = 1$, by `fin_cases` over the catalogue then reflexivity (each entry is a `charNum` application, whose `dim` is `Dim.one` by construction — `charNum_dim_one`). Axiom-free.

23.2. The sub-suffixed homonyms — same name, distinct kind

ISO 80000-11 reuses one name across its transport-phenomena clauses, distinguishing the variants only by a symbol suffix and a different defining ratio. The Froude number appears as 11-4.3 (Fr , momentum) and 11-5.4 (Fr^* , heat); the Fourier, Péclet, Grashof, Nusselt, Stanton, Graetz and Biot numbers each as a heat (Fo , ...) and a mass (Fo^* , ...) member; the Bejan number *four* times; and the Stokes number *five* times (Stk , Stk_1 ... Stk_4). These are the sub-suffixed quantity kinds — and they are the sharpest form of *R2* in the whole series: a member shares its name *and* its dimension (one) with its siblings, so neither the name nor the dimension can tell them apart. Only the measurement principle can.

Theorem 23.2.1.

group

statement">uses 1 by 0

✓

Effort

Tags

small

proved

The two Froude numbers are distinct kinds — by measurement principle, not by name. Both are named "Froude number" and both are dimension one (11-4.3, Fr , momentum transfer; 11-5.4, Fr^ , heat transfer), yet they isolate different physical ratios — inertial/gravitational versus gravitational/thermodiffusion forces — so the calculus proves them distinct from their differing principles. The Fourier (11-5.1, 11-6.1) and Nusselt (11-5.5, 11-6.4) numbers recur likewise. Uses [the examination principle](#).*

Lean code for Theorem 23.2.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part11.lean

complete

```

theorem PropertyKindCalculus.Iso80000.Part11.froudeMomentum_ne_froudeHeat :
  PropertyKindCalculus.Iso80000.Part11.froudeMomentum.kind ≠
  PropertyKindCalculus.Iso80000.Part11.froudeHeat.kind

```

Proof for Theorem 23.2.1

uses 0

froudeMomentum_ne_froudeHeat : froudeMomentum.kind ≠ froudeHeat.kind, from
KindOfProperty.distinct_of_examPrinciple on the differing principles momentum-transfer: ... and
heat-transfer: ...; froudeMomentum_id_eq_froudeHeat checks the names are equal. The companions
fourierHeat_ne_fourierMass and nusseltHeat_ne_nusseltMass cover the other recurring pairs. Axiom-
free.

Theorem 23.2.2.

group

statement">uses 1 by 0

✓

Effort

Tags

small

proved

The five Stokes numbers are pairwise distinct kinds. All five (11-4.32 ... 11-4.36) carry the name "Stokes number" and dimension one, distinguished only by symbol suffix (Stk , Stk_1 ... Stk_4) and the measurement principle each isolates — friction/inertia for particles in flow, for vibrating particles, rotameter calibration, viscous/gravity for settling particles, drag/internal-friction. Uses [the examination principle](#).

Lean code for Theorem 23.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso800000/
Part11.lean

complete

```
theorem PropertyKindCalculus.Iso800000.Part11.stokes_pairwise_distinct :  
  PropertyKindCalculus.Iso800000.Part11.stokesPlasma.kind ≠  
    PropertyKindCalculus.Iso800000.Part11.stokesVibrating.kind ∧  
  PropertyKindCalculus.Iso800000.Part11.stokesVibrating.kind ≠  
    PropertyKindCalculus.Iso800000.Part11.stokesRotameter.kind ∧  
  PropertyKindCalculus.Iso800000.Part11.stokesRotameter.kind ≠  
    PropertyKindCalculus.Iso800000.Part11.stokesGravity.kind ∧  
  PropertyKindCalculus.Iso800000.Part11.stokesGravity.kind ≠  
    PropertyKindCalculus.Iso800000.Part11.stokesDrag.kind ∧  
  PropertyKindCalculus.Iso800000.Part11.stokesPlasma.kind ≠  
    PropertyKindCalculus.Iso800000.Part11.stokesDrag.kind
```

Proof for Theorem 23.2.2

uses 0

stokes_pairwise_distinct, a conjunction of five distinct_of_examPrinciple applications on the
differing principles. Axiom-free.

Theorem 23.2.3.

group

statement">uses 1 by 0

✓

Effort

small

Tags

proved

The four Bejan numbers are distinct kinds — even two sharing a transport context. The momentum (11-4.15), the two heat (11-5.9, 11-5.10) and the mass (11-6.16) Bejan numbers are four kinds. The two heat-transfer ones share name and context, and differ only in the measurement principle (mechanical-work/diffusion-losses versus heat-transfer efficiency by entropy generation) — the finest distinction the calculus draws in this part. Uses [the examination principle](#).

Lean code for Theorem 23.2.3.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part11.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part11.bejan_four_distinct :
  PropertyKindCalculus.Iso80000.Part11.bejanMomentum.kind ≠
    PropertyKindCalculus.Iso80000.Part11.bejanHeat.kind ∧
    PropertyKindCalculus.Iso80000.Part11.bejanHeat.kind ≠
      PropertyKindCalculus.Iso80000.Part11.bejanEntropy.kind ∧
      PropertyKindCalculus.Iso80000.Part11.bejanEntropy.kind ≠
        PropertyKindCalculus.Iso80000.Part11.bejanMass.kind ∧
        PropertyKindCalculus.Iso80000.Part11.bejanMomentum.kind ≠
          PropertyKindCalculus.Iso80000.Part11.bejanMass.kind
```

Proof for Theorem 23.2.3

uses 0

bejan_four_distinct, a conjunction of four distinct_of_examPrinciple applications. Axiom-free.

Theorem 23.2.4.

group

statement" > uses 1 by 0

Effort

small

Tags

proved

One unit symbol, many kinds. The Reynolds and Euler units are not commensurable, though both carry the coherent unit symbol 1 and dimension one. Same symbol, same dimension, different kind — so a value of one cannot be read as a value of the other. The kind layer keeps them apart where the unit symbol and the dimension cannot. Uses [the metrological unit](#).

Lean code for Theorem 23.2.4.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part11.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part11.reynolds_euler_not_commensurable :  
  ¬PropertyKindCalculus.Iso80000.Part11.reynoldsUnit.Commensurable  
    PropertyKindCalculus.Iso80000.Part11.eulerUnit
```

Proof for Theorem 23.2.4 uses 0

reynolds_euler_not_commensurable : ¬ reynoldsUnit.Commensurable eulerUnit, by unfolding
Commensurable (equality of the referenced kinds) and decide; reynoldsUnit_wellFormed checks the unit
is well-formed (a ratio-scale kind bears a unit). Axiom-free.

23.3. Dimension does not classify; the measurement principle does

Theorem 23.3.1. uses 2 used by 0 ✓ LEAN

Effort

small

Priority

high

Tags

capstone

proved

The dimension-one collision, on the standard — total, the widest in the series. There exist distinct ISO 80000-11 kinds with the same dimension one — the Reynolds and Euler numbers witness it, alongside all 113 others. No dimension-only type system can separate any two characteristic numbers; the kind layer, through the measurement principle (R2), holds the entire part apart. This is dimension does not classify the kind (R1) at its absolute limit. Uses [the dimension map](#) and [the examination principle](#).

Lean code for Theorem 23.3.1.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part11.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part11.iso80000_11_dim_one_collision  
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 23.3.1 uses 0

```
iso80000_11_dim_one_collision : ∃ a b : DimensionedKind, a.kind ≠ b.kind ∧ a.dim = b.dim ∧
a.dim = 1, witnessed by {reynolds, euler, reynolds_ne_euler, rfl, rfl}. Axiom-free.
```

23.4. A characteristic number is a kind, not a constant

The collapse to dimension one (the collision above) invites a tempting confusion: if every characteristic number is dimension one, isn't it just a *number* — the same sort of thing as π or e ? It is not, and saying why is the clearest statement of the two-index architecture: a quantity is a *kind* carried over a *numeric representation*.

A mathematical constant such as π is a fixed *magnitude in the carrier*: it lives *below* the kind layer, in the numbers the magnitudes are drawn from, and it is the same value in every context. A characteristic number is the opposite in every respect: it is a *kind of quantity*, dimension one, living *above* the carrier; its value is *variable* — whatever the defining ratio produces in a given situation ($\rho v L / \mu$ for the Reynolds number, flow speed over sound speed for the Mach number). The two never coincide:

	π (a constant)	Reynolds, Mach, ... (characteristic numbers)
<i>What it is</i>	a fixed magnitude in the carrier	a kind of quantity (dimension one)
<i>Layer</i>	below the kind layer, in the representation	above the carrier, in the kind layer
<i>Value</i>	the same number in every context	variable — whatever $\rho v L / \mu$, v / c , ... produce
<i>How formed</i>	shipped by the carrier	by the quotient calculus, the dimensions cancelling to 1
<i>Identity</i>	one value	each a distinct kind — Reynolds $\neq$ Mach, though both dimension one

This is why the function calculus supplies π as a carrier constant and $e = \exp 1$ as a *derived* one (the only mathematical constant a numeric carrier need supply directly is π), while the Reynolds and Mach numbers are *kinds* in this part, never carrier constants. The kind layer keeps the two categories apart exactly where a dimension-only model — seeing "dimension one" for all of them — cannot.

Definition 23.4.1. group statement">use

A *carrier constant* is a fixed magnitude supplied by the numeric carrier — π , and the derived $e = \exp 1$ — living *below* the kind layer, in the representation type. It is the same value in every context, and it is *not* a [quantity](#)-kind: it carries no kind index, no measurement principle, and is not formed from other quantities. This is the categorical opposite of a characteristic number.

Lean code for Definition 23.4.1.

1 definition

def · defined in PropertyKindCalculus/QuantityFunction.lean

complete

```
def PropertyKindCalculus.MathCarrier.e {R : Type}
  [PropertyKindCalculus.MathCarrier R] [One R] : R
```

Proof for Definition 23.4.1 uses 0

In the function calculus (PropertyKindCalculus.QuantityFunction), the carrier capability MathCarrier carries pi as its *only* primitive constant — the one transcendental constant a numeric carrier ships directly (matching TorchLean's MathFunctions). Every other constant is *derived*: MathCarrier.e := MathCarrier.exp 1, and likewise ln 2, sqrt 2, Adding them as carrier fields would be redundant, since the operations already determine them.

Theorem 23.4.2. group statement">use

Effort Tags

small proved

The Reynolds and Mach numbers are distinct kinds, both dimension one — so neither is a carrier constant. Two characteristic numbers forget to the same dimension one yet are different kinds, formed by the quotient calculus from different physical ratios. Unlike π (a fixed magnitude in the carrier), each is a variable quantity whose value depends on the flow. Uses [the dimension map](#).

Lean code for Theorem 23.4.2.

1 theorem

theorem · defined in PropertyKindCalculus/Function.lean

complete

```
theorem PropertyKindCalculus.reynolds_ne_mach_but_both_dimensionless :
  PropertyKindCalculus.reynoldsK.kind ≠
    PropertyKindCalculus.machK.kind ∧
  PropertyKindCalculus.reynoldsK.toDimension = 1 ∧
  PropertyKindCalculus.machK.toDimension = 1
```

Proof for Theorem 23.4.2 uses 0

reynolds_ne_mach_but_both_dimensionless : reynoldsK.kind ≠ machK.kind ∧ reynoldsK.toDimension = 1 ∧ machK.toDimension = 1, the kind inequality by decide on the differing identities, the two dimensions by reflexivity. The kinds reynoldsK/machK are declared in the function-

calculus layer (`PropertyKindCalculus.Function`) for this contrast; the catalogued reynolds/euler of this part carry the same point across all 115 members. Axiom-free.

23.5. Item index — ISO 80000-11

Every catalogued item of ISO 80000-11, indexed by its printed item number, with the characteristic number's name, the principal quantity symbol, and the *measurement principle* — this work's terse descriptor of the defining ratio, which is what individuates the kind. The coherent SI unit and the PhysLib dimension are 1 for *every* item, so they are omitted from the table; that uniformity is exactly the point. Each item number links to the formalized result it participates in — a homonym distinction, the dimension-one collision, or the dimension-one theorem itself. Names, symbols, and principle descriptors are *citation locators*; nothing normative is reproduced.

Item	Characteristic number	Symbol	Measurement principle (dimension one)
11-4.1	Reynolds number	Re	inertial forces / viscous forces
11-4.2	Euler number	Eu	pressure drop / kinetic energy per volume
11-4.3	Froude number	Fr	inertial forces / gravitational forces
11-4.4	Grashof number	Gr	thermal buoyancy forces / viscous forces
11-4.5	Weber number	We	inertial forces / surface-tension forces
11-4.6	Mach number	Ma	speed of flow / speed of sound
11-4.7	Knudsen number	Kn	mean free path / characteristic length
11-4.8	Strouhal number	Sr	characteristic frequency / characteristic speed
11-4.9	drag coefficient	c_D	drag force / inertial force
11-4.10	Bagnold number	Bg	drag and gravitational force / inertial force
11-4.11	Bagnold number	Ba ₂	drag force / viscous force, solid particles
11-4.12	lift coefficient	c_L	lift force / inertial force
11-4.13	thrust coefficient	c_t	thrust force / inertial force
11-4.14	Dean number	Dn	centrifugal force / inertial force, curved pipe
11-4.15	Bejan number	Be	mechanical work / frictional energy loss
11-4.16	Lagrange number	Lg	mechanical work / frictional energy loss, Lagrange form
11-4.17	Bingham number	Bm	yield stress / viscous stress
11-4.18	Hedström number	He	yield stress / viscous stress, at flow limit
11-4.19	Bodenstein number	Bd	convective / diffusive matter transfer
11-4.20	Rossby number	Ro	inertial forces / Coriolis forces
11-4.21	Ekman number	Ek	viscous forces / Coriolis forces
11-4.22	elasticity number	El	relaxation time / diffusion time
11-4.23	Darcy friction factor	f_D	pressure loss / pipe-wall friction
11-4.24	Fanning number	f_F	wall shear stress / dynamic pressure
11-4.25	Goertler number	Go	centrifugal effects / viscous effects, boundary layer
11-4.26	Hagen number	Hg	pressure-gradient force / viscous force
11-4.27	Laval number	La	flow speed / critical sound speed, nozzle throat

Item	Characteristic number	Symbol	Measurement principle (dimension one)
11-4.28	Poiseuille number	Po _i	pressure force / viscous force, pipe flow
11-4.29	power number	Ne	agitator power / inertial power
11-4.30	Richardson number	Ri	potential energy / kinetic energy
11-4.31	Reech number	Re _e	object speed / wave speed, submerged
11-4.32	Stokes number	St _k	friction force / inertial force, particles in flow
11-4.33	Stokes number	St _{k1}	friction force / inertial force, vibrating particles
11-4.34	Stokes number	St _{k2}	drag / inertia, rotameter calibration
11-4.35	Stokes number	St _{k3}	viscous force / gravity force, settling particles
11-4.36	Stokes number	St _{k4}	drag force / internal friction force, particles
11-4.37	Laplace number	La	capillary force / viscous force, free surface
11-4.38	Blake number	BL	inertial force / viscous force, porous medium
11-4.39	Sommerfeld number	So	viscous force / load force, lubrication
11-4.40	Taylor number	Ta	centrifugal force / viscous force, rotating flow
11-4.41	Galilei number	Ga	gravitational force / viscous force, fluid film
11-4.42	Womersley number	Wo	inertial forces / viscous forces, oscillating flow
11-5.1	Fourier number	Fo	heat conduction rate / thermal storage rate
11-5.2	Péclet number	Pe	convective / conductive heat transfer rate
11-5.3	Rayleigh number	Ra	thermal buoyancy forces / viscous forces, free convection
11-5.4	Froude number	Fr*	gravitational forces / thermodiffusion forces
11-5.5	Nusselt number	Nu	convective / conductive heat transfer at a surface
11-5.6	Biot number	Bi	internal / surface thermal resistance
11-5.7	Stanton number	St	wall heat transfer / fluid heat-capacity flow
11-5.8	j-factor	j	heat transfer / mass transfer, Colburn analogy
11-5.9	Bejan number	Be ₁	mechanical work / frictional and thermal-diffusion losses
11-5.10	Bejan number	Be _S	heat-transfer efficiency / entropy generation
11-5.11	Stefan number	Ste	sensible heat / latent heat, phase change

Item	Characteristic number	Symbol	Measurement principle (dimension one)
11-5.12	Brinkman number	Br	viscous heat production / wall heat conduction
11-5.13	Clausius number	Cl	kinetic energy transfer / thermal conduction
11-5.14	Carnot number	Ca	maximum thermodynamic (Carnot) efficiency
11-5.15	Eckert number	Ec	kinetic energy / enthalpy difference
11-5.16	Graetz number	Gz	convective / conductive heat, laminar pipe
11-5.17	heat transfer number	K_Q	heat flow / kinetic energy of flow
11-5.18	Pomerantsev number	Po	generated heat / conducted heat in a body
11-5.19	Boltzmann number	Bz	convective heat / radiant heat
11-5.20	Stark number	Sk	radiant heat / conductive heat
11-6.1	Fourier number	Fo*	diffusive mass transfer / storage rate
11-6.2	Péclet number	Pe*	advective / diffusive mass transfer rate
11-6.3	Grashof number	Gr*	buoyancy forces / viscous forces, natural convection
11-6.4	Nusselt number	Nu*	mass flux / molecular-diffusion flux
11-6.5	Stanton number	St*	perpendicular / parallel surface mass transfer
11-6.6	Graetz number	Gz*	advective / radial diffusive mass transfer
11-6.7	mass transfer factor	j*	interface mass transfer / parallel flux, Chilton-Colburn
11-6.8	Atwood number	At	density difference / density sum, two fluids
11-6.9	Biot number	Bi*	interface / interior mass transfer rate
11-6.10	Morton number	Mo	gravitational forces / viscous forces, bubbles
11-6.11	Bond number	Bo	gravitational and inertial force / capillary force
11-6.12	Archimedes number	Ar	buoyancy forces / viscous forces, density difference
11-6.13	expansion number	Ex	buoyancy force / inertial force, rising bubbles
11-6.14	Marangoni number	Mg	surface-tension convection / thermal diffusion
11-6.15	Lockhart-Martinelli parameter	Lp	two-phase mass-flow-rate ratio by density
11-6.16	Bejan number	Be*	mechanical work / frictional and diffusion losses
11-6.17	cavitation number	Ca	static-vapour head / dynamic head

Item	Characteristic number	Symbol	Measurement principle (dimension one)
11-6.18	absorption number	Ab	mass flow rate / surface area, gas absorption
11-6.19	capillary number	Ca	gravitational forces / capillary forces
11-6.20	dynamic capillary number	Ca*	viscous force / capillary force, interface
11-7.1	Prandtl number	Pr	kinematic viscosity / thermal diffusivity
11-7.2	Schmidt number	Sc	kinematic viscosity / diffusion coefficient
11-7.3	Lewis number	Le	thermal diffusivity / diffusion coefficient
11-7.4	Ohnesorge number	Oh	viscous force / root of inertia times capillary force
11-7.5	Cauchy number	Cy	inertia forces / compression forces
11-7.6	Hooke number	Ho ₂	inertia forces / linear-stress forces
11-7.7	Weissenberg number	Wi	shear rate times relaxation time
11-7.8	Deborah number	De	relaxation time / observation time
11-7.9	Lorentz number	Lo	electrical conductivity / thermal conductivity
11-7.10	compressibility number	Z	real-gas / ideal-gas compressibility factor
11-8.1	Reynolds magnetic number	Rm	inertial force / magneto-dynamic viscous force
11-8.2	Batchelor number	Bt	inertia / magneto-dynamic diffusion
11-8.3	Nusselt electric number	Ne _e	convective / diffusive ion current
11-8.4	Alfvén number	Al	flow speed / Alfvén wave speed
11-8.5	Hartmann number	Ha	magnetically induced stress / viscous force
11-8.6	Cowling number	Co	magnetic / kinematic energy density
11-8.7	Stuart electrical number	Se	electric / kinematic energy density
11-8.8	magnetic pressure number	N _{mp}	gas pressure / magnetic pressure
11-8.9	Chandrasekhar number	Q	Lorentz force / viscous force
11-8.10	Prandtl magnetic number	Pr _m	kinematic viscosity / magnetic viscosity
11-8.11	Roberts number	Ro _m	thermal diffusivity / magnetic viscosity
11-8.12	Stuart number	N _{St}	magnetic force / inertial force
11-8.13	magnetic number	N _{mg}	magnetic body force / viscous force
11-8.14	electric field parameter	E _f	Coulomb force / Lorentz force
11-8.15	Hall number	HL	gyrofrequency / collision frequency

Item	Characteristic number	Symbol	Measurement principle (dimension one)
11-8.16	Lundquist number	Lu	Alfvén speed / magneto-dynamic diffusion speed
11-8.17	Joule magnetic number	Jo _m	Joule heating / magnetic field energy
11-8.18	Grashof magnetic number	Gr _m	thermo-magnetic buoyancy / viscous force
11-8.19	Naze number	Na	Alfvén wave speed / sound speed
11-8.20	Reynolds electric number	Re _e	fluid speed / charged-particle drift speed
11-8.21	Ampère number	Am	electric surface current / magnetic field
11-9.1	Arrhenius number	α	chemical activation energy / thermal energy
11-9.2	Landau-Ginzburg number	κ	penetration depth / coherence length

24. ISO 80000-12 — Condensed matter physics

The ninth part specified in full is ISO 80000-12, *Condensed matter physics*. Every item — all of 12-1.1 ... 12-38.2, sixty including every sub-suffixed item — is catalogued: the lattice geometry, the reciprocal lattice and wavenumbers, the Debye and Grüneisen quantities, the transport and thermoelectric coefficients, the work functions and band energies, the carrier densities, and the magnetism and superconductivity quantities — each carrying its exact source as data. Only *citation locators* are recorded; no normative content from the licensed standard is reproduced. The defining *mathematics* of selected remarks is specified in the sibling module `Part12.DefiningRelations`.

Condensed matter physics is where *dimension does not classify the kind* stops being the exception and becomes the rule: nearly every quantity collides in dimension with several others. Five families dominate, and each is one big collision — *thirteen lengths* on L , *seven energies* on $M \cdot L^2 \cdot T^{-2}$, *five temperatures* on θ , *five carrier densities* on L^{-3} , and *five reciprocal lengths* on L^{-1} . The five named temperatures are especially striking: the Debye, Fermi, Curie, Néel, and superconduction-transition temperatures share one dimension *and* one unit (the kelvin), and only the kind tells them apart.

24.1. Quantity-kinds and units of ISO 80000-12

Definition

24.1.1.

statement" > uses 1 by 0

A catalogued kind pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "12-27.1"), the principal quantity symbol, and the coherent SI unit symbol. The citation travels with the kind as data, so the source of each definition can be rendered or audited downstream.

Lean code for Definition

24.1.1.

1 definition

structure (5 fields) · defined in PropertyKindCalculus/Iso80000/Catalogue.lean

complete

structure PropertyKindCalculus.Iso80000.CataloguedKind : Type

Proof for Definition 24.1.1

uses 0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with cite rendering e.g. ISO 80000-12, Second edition, 2019-08 item 12-27.1. The catalogue lists all 60 items in item order — the lattice geometry (12-1.1 ... 12-8), the band energies (12-24.1 ... 12-27.2), the carrier densities (12-29.1 ... 12-29.5), the superconductivity quantities (12-35.1 ... 12-38.2), ... — each paired with its coherent SI unit symbol.

24.2. Collisions everywhere — the kind classifies where the dimension cannot

ISO 80000-12 makes the catalogue's thesis the norm. Five families each collapse to one dimension:

- *Seven energies* — work function, ionization energy, electron affinity, Fermi energy, gap energy, exchange integral, superconductor energy gap — all $M \cdot L^2 \cdot T^{-2}$.
- *Five temperatures* — Debye, Fermi, Curie, Néel, superconduction-transition — all θ , and all in the kelvin.
- *Thirteen lengths* — lattice and position vectors, spacings, mean free paths, the penetration and coherence lengths — all L .
- *Five carrier densities* and *five reciprocal lengths* — all L^{-3} and L^{-1} .

A dimension-only model sees one type per family; the kind layer keeps the Fermi energy from the gap energy, the Curie temperature from the Néel temperature, the lattice spacing from the Burgers vector. The five *named* temperatures are the sharpest case: one dimension, one unit symbol, five kinds.

Theorem

24.2.1.

group

statement" >uses by 0

Effort
small

Tags

capstone

proved

Fermi energy is not gap energy, though both are J and $M \cdot L^2 \cdot T^{-2}$ (items 12-27.1, 12-27.2). There exist distinct ISO 80000-12 kinds with the same dimension — the seven energies witness it. No dimension-only type system can separate them; the kind layer — and the unit — does. Uses [the dimension map](#).

Lean code for Theorem

24.2.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part12.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part12.iso80000_12_dim_collision :  
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim
```

Proof for Theorem 24.2.1

uses 0

iso80000_12_dim_collision, witnessed by (fermiEnergy, gapEnergy, ...). The unit fact jouleFermi_jouleGap_not_commensurable is a decide; the companions

latticePlaneSpacing_dim_eq_burgersVector_dim (the length family) and curieTemperature_dim_eq_neelTemperature_dim (the temperature family) record the other collisions. Axiom-free.

Theorem

24.2.2.

group

statement">uses1by 0

Effort

Tags

small

proved

The kelvin of the Curie temperature and the kelvin of the Néel temperature are not commensurable, though both are 0 (items 12-35.1, 12-35.2). Five named temperatures — the Debye, Fermi, Curie, Néel, and superconduction-transition — share one dimension and one unit symbol, and only the kind tells them apart. Uses [the metrological unit](#).

Lean code for Theorem

24.2.2.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part12.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part12.kelvinCurie_kelvinNeel_not_com
  ¬PropertyKindCalculus.Iso80000.Part12.kelvinCurie.Commensurable
    PropertyKindCalculus.Iso80000.Part12.kelvinNeel
```

Proof for Theorem 24.2.2

uses 0

kelvinCurie_kelvinNeel_not_commensurable, by decide after unfolding MetrologicalUnit.Commensurable. Axiom-free.

Theorem

24.2.3.

group

statement">uses1by 0

Effort

Tags

small

proved

The dimension-1 disambiguation, on the standard. There exist distinct ISO 80000-12 kinds with the same dimension one — the Bragg angle and the structure factor witness it, alongside the order parameters, the atomic scattering factor, the Debye-Waller factor, the Grüneisen parameters, and the mobility ratio. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem

24.2.3.

1 theorem

theorem · defined in

PropertyKindCalculus/Iso80000/Part12.lean

complete

theorem PropertyKindCalculus.Iso80000.Part12.iso80000_12_dim_one_collision
 $\exists a\ b, a.\text{kind} \neq b.\text{kind} \wedge a.\text{dim} = b.\text{dim} \wedge a.\text{dim} = 1$

Proof for Theorem 24.2.3

uses 0

iso80000_12_dim_one_collision, witnessed by (braggAngle, structureFactor, ...) with the kind distinction a decide and reflexivity. Axiom-free.

24.3. The algebraic Remarks as kind-laws: thermoelectricity built by the kind algebra

Two Part-12 definitions state the thermoelectric coefficients *algebraically*: the Seebeck coefficient is the thermoelectric voltage per temperature (12-21, $S = dE/dT$), and the Peltier coefficient is the Seebeck coefficient times temperature (12-22, $\Pi = S \cdot T$, the Kelvin relation). Both are specified as R12 kind-laws, crossing to ISO 80000-5 for the thermodynamic temperature.

Theorem 24.3.1.

statement" > uses 1 by 0

Effort

small

Tags

proved

Peltier coefficient is Seebeck coefficient $\times$ temperature — a verified construction (item 12-22). A Peltier coefficient built as the product of a Seebeck coefficient and a thermodynamic temperature carries its product certificate by construction, over the $\mathbb{R}$ carrier. The companion `seebeck_dim_from_voltage_temperature` shows the Seebeck coefficient is V/K because it is a voltage per temperature. Uses [the quotient kind-law](#).

Lean code for Theorem 24.3.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part12/DefiningRelations.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part12.DefiningRelations.peltierOf_is
  (s :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part12.seebeckCoefficient.kind  $\mathbb{R}$ )
  (t :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part5.thermodynamicTemperature.kind
       $\mathbb{R}$ ) :
  PropertyKindCalculus.Quantity.IsProduct
```

```
PropertyKindCalculus.Iso80000.Part12.DefiningRelations.peltier_prod_seebec
k_temperature
  (PropertyKindCalculus.Iso80000.Part12.DefiningRelations.peltierOf s
    t)
  s t
```

Proof for Theorem 24.3.1

uses 0

`peltierOf s t` := `Quantity.mul peltier_prod_seebec_temperature s t`, and `peltierOf_isProduct` is `rfl`. The companion `seebeckOf` (= thermoelectric voltage / temperature) carries the quotient certificate, with `seebeck_certificate_canonical` its canonicity; both cross to ISO 80000-5.

24.4. Item index — ISO 80000-12

Every catalogued item of ISO 80000-12, indexed by its printed item number, with the principal quantity symbol, the coherent SI unit, and the `PhysLib` dimension this work assigns it. Each item number links to the formalized result it participates in. Symbols and unit strings are *citation locators*; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
12-1.1	lattice vector	$\mathbf{R}$	m	L
12-1.2	fundamental lattice vectors	$\mathbf{a}_1$	m	L
12-2.1	angular reciprocal lattice vector	$\mathbf{G}$	m^{-1}	L^{-1}
12-2.2	fundamental reciprocal lattice vectors	$\mathbf{b}_1$	m^{-1}	L^{-1}
12-3	lattice plane spacing	d	m	L
12-4	Bragg angle	θ	1	1
12-5.1	short-range order parameter	r	1	1
12-5.2	long-range order parameter	R	1	1
12-5.3	atomic scattering factor	f	1	1
12-5.4	structure factor	F	1	1
12-6	Burgers vector	$\mathbf{b}$	m	L
12-7.1	particle position vector	$\mathbf{r}$	m	L
12-7.2	equilibrium position vector	$\mathbf{R}_\theta$	m	L
12-7.3	displacement vector	$\mathbf{u}$	m	L
12-8	Debye-Waller factor	D	1	1
12-9.1	angular wavenumber	k	m^{-1}	L^{-1}
12-9.2	Fermi angular wavenumber	k_F	m^{-1}	L^{-1}
12-9.3	Debye angular wavenumber	q_D	m^{-1}	L^{-1}
12-10	Debye angular frequency	ω_D	s^{-1}	T^{-1}
12-11	Debye temperature	θ_D	K	θ
12-12	density of vibrational states	g	$\text{m}^{-3} \cdot \text{s}$	$\text{L}^{-3} \cdot \text{T}$
12-13	thermodynamic Grüneisen parameter	γ_G	1	1
12-14	Grüneisen parameter	γ	1	1
12-15.1	mean free path of phonons	l_p	m	L
12-15.2	mean free path of electrons	l_e	m	L
12-16	energy density of states	n_E	$\text{J}^{-1} \cdot \text{m}^{-3}$	$\text{M}^{-1} \cdot \text{L}^{-5} \cdot \text{T}^2$
12-17	residual resistivity	ρ_θ	$\Omega \cdot \text{m}$	$\text{M} \cdot \text{L}^3 \cdot \text{T}^{-1} \cdot \text{C}^{-2}$
12-18	Lorenz coefficient	L	V^2/K^2	$\text{M}^2 \cdot \text{L}^4 \cdot \text{T}^{-4} \cdot \text{C}^{-2} \cdot \theta^{-2}$

Item	Quantity	Symbol	Unit	Dimension
12-19	Hall coefficient	R _H	m ³ /C	$L^3 \cdot C^{-1}$
12-20	thermoelectric voltage	E _{ab}	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
12-21	Seebeck coefficient	S _{ab}	V/K	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1} \cdot \theta^{-1}$
12-22	Peltier coefficient	Π _{ab}	V	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1}$
12-23	Thomson coefficient	μ	V/K	$M \cdot L^2 \cdot T^{-2} \cdot C^{-1} \cdot \theta^{-1}$
12-24.1	work function	φ	J	$M \cdot L^2 \cdot T^{-2}$
12-24.2	ionization energy	E _i	J	$M \cdot L^2 \cdot T^{-2}$
12-25	electron affinity	χ	J	$M \cdot L^2 \cdot T^{-2}$
12-26	Richardson constant	A	A · m ⁻² · K ⁻²	$L^{-2} \cdot T^{-1} \cdot C \cdot \theta^{-2}$
12-27.1	Fermi energy	E _F	J	$M \cdot L^2 \cdot T^{-2}$
12-27.2	gap energy	E _g	J	$M \cdot L^2 \cdot T^{-2}$
12-28	Fermi temperature	T _F	K	θ
12-29.1	electron density	n	m ⁻³	L ⁻³
12-29.2	hole density	p	m ⁻³	L ⁻³
12-29.3	intrinsic carrier density	n _i	m ⁻³	L ⁻³
12-29.4	donor density	n _d	m ⁻³	L ⁻³
12-29.5	acceptor density	n _a	m ⁻³	L ⁻³
12-30	effective mass	m*	kg	M
12-31	mobility ratio	b	1	1
12-32.1	relaxation time	τ	s	T
12-32.2	carrier lifetime	τ	s	T
12-33	diffusion length	L	m	L
12-34	exchange integral	J	J	$M \cdot L^2 \cdot T^{-2}$
12-35.1	Curie temperature	T _C	K	θ
12-35.2	Néel temperature	T _N	K	θ
12-35.3	superconduction transition temperature	T _c	K	θ
12-36.1	thermodynamic critical magnetic flux density	B _c	T	$M \cdot T^{-1} \cdot C^{-1}$
12-36.2	lower critical magnetic flux density	B _{c1}	T	$M \cdot T^{-1} \cdot C^{-1}$

Item	Quantity	Symbol	Unit	Dimension
12-36.3	upper critical magnetic flux density	B_{c2}	T	$M \cdot T^{-1} \cdot C^{-1}$
12-37	superconductor energy gap	Δ	J	$M \cdot L^2 \cdot T^{-2}$
12-38.1	London penetration depth	λ_L	m	L
12-38.2	coherence length	ξ	m	L

25. IEC 80000-13 — Information science and technology

The tenth and last part specified is IEC 80000-13, *Information science and technology* — the second IEC-published part, after electromagnetism. Every item — all of 13-1 ... 13-42, forty-two in all (the part uses flat item numbers, no sub-suffixes) — is catalogued: the teletraffic quantities, storage and transfer, the powers and signal energy, and the coding- and information-theory quantities — each carrying its exact source as data. Only *citation locators* are recorded; no normative content from the licensed standard is reproduced. The defining *mathematics* of selected remarks is specified in the sibling module `Part13.DefiningRelations`.

Information science is where the *scale-spanning* reduction (the *Scale-spanning units* chapter, requirement *R13*) meets *human-selected counts of information*. Its characteristic quantities are all *dimension one* — and yet they carry special, mutually *incommensurable* units, and belong to distinct kinds. Information content and entropy are measured in the *shannon* (or hartley, or natural unit); traffic intensity in the *erlang*; storage capacity in the *bit*. All are dimension one; the dimension functor collapses them to a single point, and only the kind keeps the shannon of information from the erlang of traffic from the bit of storage. The shannon, erlang, and bit are scale-spanning units in exactly the sense of the candela and the mole: dimensionless references chosen by people.

25.1. Quantity-kinds and units of IEC 80000-13

Definition25.1.1.

statement">uses1by0

A *catalogued kind* pairs a dimensioned [quantity](#)-kind with its exact source in the series — the part, the printed item designation (e.g. "13-24"), the principal quantity symbol, and the unit symbol (for the information quantities, the special named unit — shannon, erlang, bit — rather than the bare 1). The citation travels with the kind as data.

Lean code for Definition25.1.1.

1 definition

structure (5 fields) · defined in PropertyKindCalculus/Iso80000/Catalogue.lean

complete

```
structure PropertyKindCalculus.Iso80000.CataloguedKind : Type
```

Proof for Definition 25.1.1

uses0

Realized as structure `CataloguedKind` over a `DimensionedKind`, with cite rendering e.g. IEC 80000-13, Edition 2.0, 2025-02 item 13-24. The catalogue lists all 42 items in item order — the

traffic quantities (13-1 ... 13-8), storage (13-9, 13-10), the information content and entropies (13-24 ... 13-37), the rates (13-38 ... 13-42), ... — each paired with its unit symbol.

25.2. Dimension one, three kind families, incommensurable special units (R13)

IEC 80000-13's characteristic quantities all live at dimension one — and this is exactly the place the catalogue's thesis turns into a **unit** question. Three families share dimension one but carry *different special units*, and belong to *different kinds*:

- *Information content and entropy* (13-24 ...) in the *shannon* Sh , *hartley* $Hart$, or *natural unit* nat — alternative units of one kind, related by the logarithm base.
- *Traffic intensity* (13-1 ...) in the *erlang* E .
- *Storage capacity* (13-9, 13-10) in the *bit* (or octet, or byte).

A dimension-only model sees one type, "a real number, dimension one", for all of them, and would let a bit of storage be silently read as an erlang of traffic. The shannon, erlang, and bit are *scale-spanning* units (R13), human-selected references for a dimensionless count — exactly like the candela (ISO 80000-7) and the mole (ISO 80000-9). The kind layer is what keeps them apart.

Theorem	25.2.1.	group	statement">uses by 0
Effort small	Tags capstone proved		

The shannon of information content and the erlang of traffic intensity are not commensurable, though both are dimension one (items 13-24, 13-1). Two human-selected dimensionless references for two different kinds — a bit of information is not an erlang of traffic. The companion shannon_bit_not_commensurable separates information from storage. This is the scale-spanning collision (R13) on information. Uses [the metrological unit](#).

Lean code for Theorem 25.2.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part13.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part13.shannon_erlang_not_commensurable
  →PropertyKindCalculus.Iso80000.Part13.shannon.Commensurable
    PropertyKindCalculus.Iso80000.Part13.erlang
```

Proof for Theorem 25.2.1 uses 0
shannon_erlang_not_commensurable and shannon_bit_not_commensurable, each by decide after unfolding MetrologicalUnit.Commensurable on the dimension-one kinds. Axiom-free.

Theorem 25.2.2. group statement" >uses 1 by 0

Effort
small

Tags
proved

The dimension-1 collision, on the standard. There exist distinct IEC 80000-13 kinds with the same dimension one — information content and traffic intensity witness it, alongside storage capacity, the entropies, and the probabilities. With distinct special units (shannon, erlang, bit), the dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem 25.2.2.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part13.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part13.iec80000_13_dim_one_collision
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim ∧ a.dim = 1
```

Proof for Theorem 25.2.2 uses 0
iec80000_13_dim_one_collision, witnessed by ⟨informationContent, trafficIntensity, ...⟩ with the kind distinction a decide and reflexivity. Axiom-free.

25.3. The information rates re-dimension to T^{-1}

Per-character entropies are dimension one, but the *per-second* rates re-dimension to T^{-1} : the average information rate, the average transinformation rate, the channel (time) capacity, the bit and transfer rates,

the call intensities. They collide there — distinct kinds, one dimension — just as the dimension-one quantities collide at the point one.

Theorem25.3.1.

group

statement">uses1by0

Effort

Tags

small

proved

The bit rate is not the call intensity, though both are T^{-1} (items 13-13, 13-7). There exist distinct IEC 80000-13 kinds with the same dimension T^{-1} — the bit rate and the call intensity witness it, alongside the transfer and modulation rates and the information rates. Dimension cannot separate them; the kind layer does. Uses [the dimension map](#).

Lean code for Theorem25.3.1.

1 theorem

theorem · defined in PropertyKindCalculus/Iso80000/Part13.lean

complete

```
theorem PropertyKindCalculus.Iso80000.Part13.iec80000_13_dim_collision :  
  ∃ a b, a.kind ≠ b.kind ∧ a.dim = b.dim
```

Proof for Theorem 25.3.1

uses 0

iec80000_13_dim_collision, witnessed by (binaryDigitRate, callIntensity, ...). The shared dimension is binaryDigitRate_dim_eq_callIntensity_dim; the kind distinction is a decide. Axiom-free.

25.4. The algebraic Remarks as kind-laws: digital transmission built by the kind algebra

Several Part-13 definitions state a quantity's defining relation *algebraically*. The signal energy per binary digit is the *product* of carrier power and the bit period (13-19, $E_{\text{bit}} = P_c \cdot T_{\text{bit}}$); the periods are *reciprocals* of the rates (13-12, 13-14). Each is specified as an R12 kind-law.

Theorem25.4.1.

statement">uses1by0

Effort

Tags

small

proved

Signal energy per bit is carrier power × bit period — a verified construction (item 13-19). A signal energy built as the product of a carrier power and a bit period carries its product certificate by

construction, over the $\mathbb{R}$ carrier. The companion `signalEnergy_dim_from_power_period` shows it is $M \cdot L^2 \cdot T^{-2}$ because it is a power times a time. Uses [the quotient kind-law](#).

Lean code for Theorem 25.4.1.

1 theorem

complete

theorem · defined in PropertyKindCalculus/Iso80000/Part13/DefiningRelations.lean

```
theorem PropertyKindCalculus.Iso80000.Part13.DefiningRelations.signalEnergy
  (p :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part13.carrierPower.kind  $\mathbb{R}$ )
  (t :
    PropertyKindCalculus.Quantity
      PropertyKindCalculus.Iso80000.Part13.periodOfBinaryDigits.kind
       $\mathbb{R}$ ) :
  PropertyKindCalculus.Quantity.IsProduct

PropertyKindCalculus.Iso80000.Part13.DefiningRelations.signalEnergy_prod_p
power_period
  (PropertyKindCalculus.Iso80000.Part13.DefiningRelations.signalEnergyOf
    p t)
  p t
```

Proof for Theorem 25.4.1

uses 0

`signalEnergyOf p t` := `Quantity.mul signalEnergy_prod_power_period p t`, and `signalEnergyOf_isProduct` is `rfl`. The reciprocal kind-laws `periodOfDataElements_recip_transferRate` and `periodOfBinaryDigits_recip_binaryDigitRate` give the periods, with `periodOfDataElements_dim_from_transferRate` the dimension. Axiom-free.

25.5. Item index — IEC 80000-13

Every catalogued item of IEC 80000-13, indexed by its printed item number, with the principal quantity symbol, the unit (the special named unit where the standard gives one), and the PhysLib dimension this work assigns it. Each item number links to the formalized result it participates in. Symbols and unit strings are *citation locators*; nothing normative is reproduced.

Item	Quantity	Symbol	Unit	Dimension
13-1	traffic intensity	A	E	1
13-2	traffic offered intensity	A_o	E	1
13-3	traffic carried intensity	Y	E	1
13-4	mean queue length	L	1	1
13-5	loss probability	B	1	1
13-6	waiting probability	W	1	1
13-7	call intensity	λ	s^{-1}	T^{-1}
13-8	completed call intensity	μ	s^{-1}	T^{-1}
13-9	storage capacity	M	bit	1
13-10	equivalent binary storage capacity	M_e	bit	1
13-11	transfer rate	r	s^{-1}	T^{-1}
13-12	period of data elements	T	s	T
13-13	binary digit rate	r_bit	bit/s	T^{-1}
13-14	period of binary digits	T_bit	s	T
13-15	equivalent binary digit rate	r_e	bit/s	T^{-1}
13-16	modulation rate	r_m	Bd	T^{-1}
13-17	quantizing distortion	T_Q	W	$M \cdot L^2 \cdot T^{-3}$
13-18	carrier power	P_c	W	$M \cdot L^2 \cdot T^{-3}$
13-19	signal energy per binary digit	E_bit	J	$M \cdot L^2 \cdot T^{-2}$
13-20	error probability	P	1	1
13-21	Hamming distance	d_n	1	1
13-22	clock frequency	f_cl	Hz	T^{-1}
13-23	decision content	D_a	1	1
13-24	information content	I(x)	Sh	1
13-25	entropy	H	Sh	1
13-26	maximum entropy	H_0	Sh	1
13-27	relative entropy	H_r	1	1
13-28	redundancy	R	Sh	1

Item	Quantity	Symbol	Unit	Dimension
13-29	relative redundancy	r	1	1
13-30	joint information content	$I(x, y)$	Sh	1
13-31	conditional information content	$I(x y)$	Sh	1
13-32	conditional entropy	$H(X Y)$	Sh	1
13-33	equivocation	$H_x(X Y)$	Sh	1
13-34	irrelevance	$H_y(Y X)$	Sh	1
13-35	transinformation content	$T(x, y)$	Sh	1
13-36	mean transinformation content	T	Sh	1
13-37	character mean entropy	H'	Sh	1
13-38	average information rate	H^*	Sh/s	T^{-1}
13-39	character mean transinformation content	T'	Sh	1
13-40	average transinformation rate	T^*	Sh/s	T^{-1}
13-41	channel capacity per character	C'	Sh	1
13-42	channel time capacity	C^*	Sh/s	T^{-1}

26. External Cross-References

This work formalizes Dybkær's **Ontology on Property** and aligns its concepts with the **International Vocabulary of Metrology** (VIM). The correspondence is recorded as typed, declaration-indexed metadata — two attributes, `@[dybkær ...]` and `@[vim4 ...]`, attached to the declarations of the core spine — so the two index tables below are **generated from the source**, not maintained by hand. Each row maps an external locus to the external term and to this work's declaration, linked to its definition in this blueprint. Only locators are recorded; no normative text from either copyrighted source is reproduced.

26.1. Dybkær — An Ontology on Property

Indexed by Dybkær section, with the term as printed and this work's corresponding declaration.

René Dybkær, An Ontology on Property for Physical, Chemical, and Biological Systems, 2009

Locus	Term	This work	Note
§ 3.3	system	System	
§ 6	quantities of the same kind	MutuallyComparable	
§ 6.19	kind-of-property	KindOfProperty	
§ 7	examination method	ExaminationMethod	Ch. 7
§ 7	examination procedure	ExaminationProcedure	Ch. 7
§ 7.5	examination principle	ExaminationPrinciple	
§ 9.15	property value	PropertyValue	
§ 10.14	property value scale	ValueScale	
§ 12	operator-based scale types	ScaleType	the four-way division, Fig. 12.21
§ 13.3.3	quantity (number times reference)	Quantity	
§ 13.3.5	reciprocal of a quantity	ReciprocalKind	
§ 13.3.5	quotient of two quantities	QuotientKind	
§ 13.3.5	product of two quantities	ProductKind	
§ 13.5	extensive kind	Extensive	
§ 18.12	metrological unit	MetrologicalUnit	
§ 20	component	Component	
§ 20	dedicated kind-of-property	DedicatedKind	the System — Component ; kind-of-property triple (IUPAC/IFCC)

26.2. VIM 4 2CD — International Vocabulary of Metrology

Indexed by VIM 4 2CD clause. The 2CD is a restricted committee draft; only clause numbers and terms are recorded.

The correspondence is mostly with this work's core spine — VIM's *quantity*, *ratio/intervall/ordinal quantity*, *measurement scale*, *value of a quantity*, and the *measurement principle/method/procedure* chain map directly to PKC declarations. Three VIM concepts reach into the further layers: *quantity dimension* (1.9)

and *quantity with the unit one* (1.10) are the Dimension layer's dim map and the dimension-one thesis; *quantity calculus* (1.29) is the Interaction algebra; the *international system of quantities* (1.8) is the ISO/IEC 80000 catalogue and the *coherent derived unit* (1.15) the unit each catalogued kind carries.

JCGM, International Vocabulary of Metrology — Basic and general concepts and associated terms (VIM), 4th edition, Committee Draft 2, 2023-07-31

Locus	Term	This work	Note
1.1	quantity	Quantity	
1.2	quantities of the same kind	MutuallyComparable	
1.3	ratio quantity	IsRational	
1.4	interval quantity	IsDifferential	
1.8	international system of quantities (ISQ)	catalogue	the ISO/IEC 80000 catalogue of quantity-kinds
1.9	quantity dimension	toDimension	the forgetful dim map into PhysLib
1.10	quantity with the unit one	dim_not_injective	the dimension-1 thesis: distinct kinds collapse to one dimension
1.12	measurement unit	MetrologicalUnit	
1.15	coherent derived unit	CataloguedKind	the coherentUnit of each kind
1.20–1.21	multiple / submultiple of a unit	SIPrefix	the decimal prefix factor
1.23	measurement scale	ValueScale	
1.24	value of a quantity	PropertyValue	
1.29	quantity calculus	KMul	kind interaction (KMul / KDiv)
1.30	quantity equation	ProductKind	realized as R12 kind-laws
1.33	ordinal quantity	IsOrdinal	
2.4	measurement principle	ExaminationPrinciple	
2.5	measurement method	ExaminationMethod	
2.7	measurement procedure	ExaminationProcedure	

Many VIM clauses have no PKC counterpart, for two reasons. Some are **metrological- procedure** concerns outside this ontology's scope — *true value* (1.26), *conventional value* (1.27), *measurement result* (2.10), *measured value* (2.11), *measurement model/function* (2.12–2.13), *influence quantity* (2.16), *correction* (2.17), and the *metrology/measurand/primary method* terms (2.2, 2.3, 2.6, 2.8, 2.9). Others are **not yet formalized**: the *base/derived quantity* and *base/derived/off-system unit* distinctions (1.6, 1.7, 1.13, 1.14, 1.18), *system of quantities/of units* as first-class objects (1.5, 1.16, 1.17, 1.19), *reference quantity* (1.11), the *unit/numerical-value equation* forms (1.31, 1.32), and *conversion factor between units* (1.22), whose round-trip is a planned capstone.

27. License

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